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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Class: II CSE-II SEM

Course Name: CN

Course Code:AS22- 05PC05

Academic Year: 2023-24

Section: A,B&C

LECTURE NOTES

UNIT -I

Introduction to Computer Networks:

- A **computer network** is a **digital telecommunications network** which allows nodes to share resources. In computer networks, computing devices exchange data with each other using connections (data links) between nodes. These data links are established over cable media such as wires or optic cables, or wireless media such as Wi-Fi.
- A **computer network** is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending or receiving data from the other node/device through the network.

Advantages of Computer Network:

There are many **advantages of computer network** and benefits of computer networking, some of them are File sharing, Resource sharing, better connectivity and communications is briefly explained below.

- **File sharing:** Fundamental goals of a computer network is to allow file sharing
- **Server-Client model:** Client-server model is a distributed application framework. The server is a master system which stores the data and provides the processing service. A client is a user system which accesses the data from the server and can perform processing in its system or in a server system.
- **Resource sharing:** All the resources such as printers, modems, scanners and fax machines etc. can be shared by all the system in a computer network.
- **Better connectivity and communications:** A computer network allow all the user or computers at a different location to communicate easily, widely used example is email, video conference etc.

Need of computer network:

Computer networks have become invaluable to organizations as well as individuals. Some of its main uses are **Internet access, eCommerce, Entertainment, VoIP** is briefly explained below.

Disadvantages of Computer Network:

There are many **disadvantages of computer network** of computer networking, some of them are Lack of data security and privacy, Independence, Robustness is briefly explained below.

- **Lack of data security and privacy** due to the huge number of the user accessing the internet from



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every corner of the earth, handling data is a challenging task and data can be stolen which violate



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the privacy of the user.

- **Presence of computer viruses and the malware:** A virus can easily spread through an interconnected workstation or over the internet which can damage or steal the data from the computer system in a computer network.
- **Lack of Independence:** In most cases, the client is dependent on the centralized server and client user lack any freedom whatsoever. the centralized server can make hinder in the decision making of the user system.
- **Lack of Robustness:** As a client is dependent on the centralized server, if the centralized server breaks down, the entire system of networks would be useless and also if any of the links between the end system fail then the network will standstill.
- **Need an efficient handler:** As a computer network consisting of many devices and software, so to handle the smooth functioning of the transmission of information, it requires high technical skills and knows- how its administration and its operations.

1. Message. The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
2. Sender. The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
3. Receiver. The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
4. Transmission medium. The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves
5. Protocol. A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

NETWORK HARDWARE:

A network is a set of devices (often referred to as *nodes*) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

Distributed Processing

Most networks use distributed processing, in which a task is divided among multiple computers. Instead of one single large machine being responsible for all aspects of a process, separate computers (usually a personal computer or workstation) handle a subset.

Network Criteria



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A network must be able to meet a certain number of criteria. The most important of these are performance, reliability, and security.

Performance:

Performance can be measured in many ways, including transit time and response time. Transit time is the amount of time required for a message to travel from one device to another. Response time is the elapsed time between an inquiry and a response. The performance of a network depends on a number of factors, including the number of users, the type of transmission medium, the capabilities of the connected hardware, and the efficiency of the software. Performance is often evaluated by two networking metrics: throughput and delay. We often need more throughput and less delay. However, these two criteria are often contradictory. If we try to send more data to the network, we may increase throughput but we increase the delay because of traffic congestion in the network.

Reliability:

In addition to accuracy of delivery, network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.

Security:

Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

Physical Structures:

Type of Connection

A network is two or more devices connected through links. A link is a communications pathway that transfers data from one device to another. For visualization purposes, it is simplest to imagine any link as a line drawn between two points. For communication to occur, two devices must be connected in some way to the same link at the same time. There are two possible types of connections: point-to-point and multipoint.

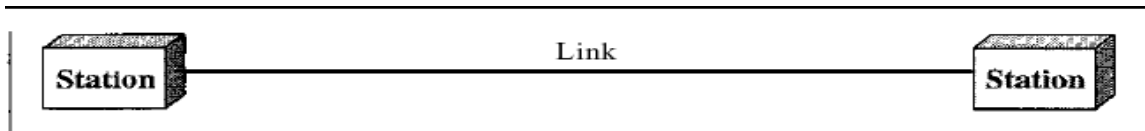
Point-to-Point

A point-to-point connection provides a dedicated link between two devices. The entire capacity of the link is reserved for transmission between those two devices. Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible. When you change television channels by infrared remote control, you are establishing a point-to-point connection between the remote control and the television's control system.

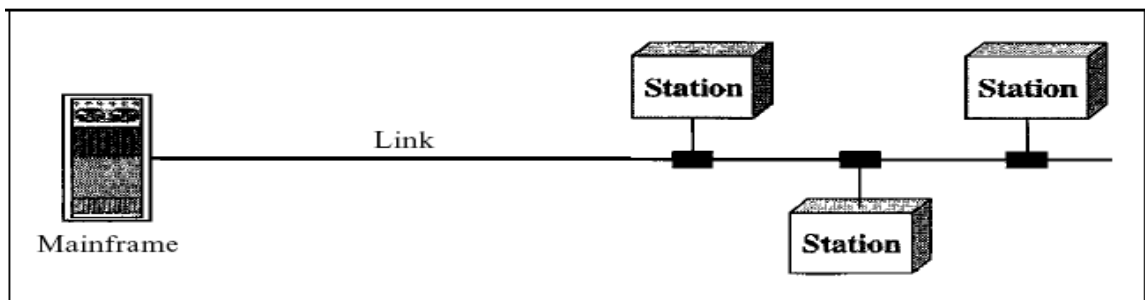
Multipoint

A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link. In a multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is a *spatially shared* connection. If users must take turns, it is a *timeshared* connection.

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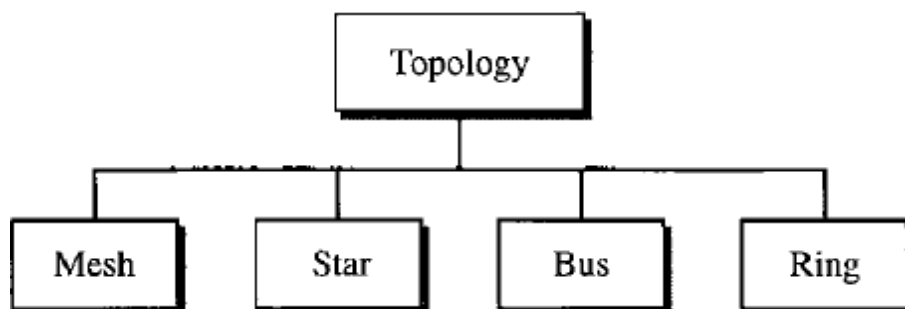
a. Point-to-point



b. Multipoint

Physical Topology

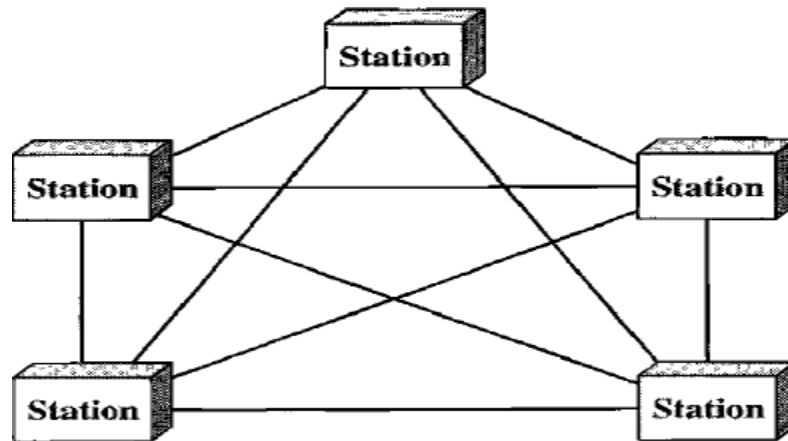
The term *physical topology* refers to the way in which a network is laid out physically. One or more devices connect to a link; two or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another. There are four basic topologies possible: mesh, star, bus, and ring.



Mesh: In a mesh topology, every device has a dedicated point-to-point link to every other device. The term *dedicated* means that the link carries traffic only between the two devices it connects. To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node. Node 1 must be connected to $n - 1$ nodes, node 2 must be connected to $n - 1$ nodes, and finally node n must be connected to $n - 1$ nodes. We need $n(n - 1)$ physical links. However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2. In other words, we can say that in a mesh topology, we need $n(n - 1) / 2$ duplex-mode links.

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To accommodate that many links, every device on the network must have $n - 1$ input/output (IO) ports to be connected to the other $n - 1$ stations.



Advantages:

1. The use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
2. A mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system. Third, there is the advantage of privacy or security. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages. Finally, point-to-point links make fault identification and fault isolation easy. Traffic can be routed to avoid links with suspected problems. This facility enables the network manager to discover the precise location of the fault and aids in finding its cause and solution.

Disadvantages:

1. Disadvantage of a mesh are related to the amount of cabling because every device must be connected to every other device, installation and reconnection are difficult.
2. Second, the sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate. Finally, the hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

For these reasons a mesh topology is usually implemented in a limited fashion, for example, as a backbone connecting the main computers of a hybrid network that can include several other topologies.

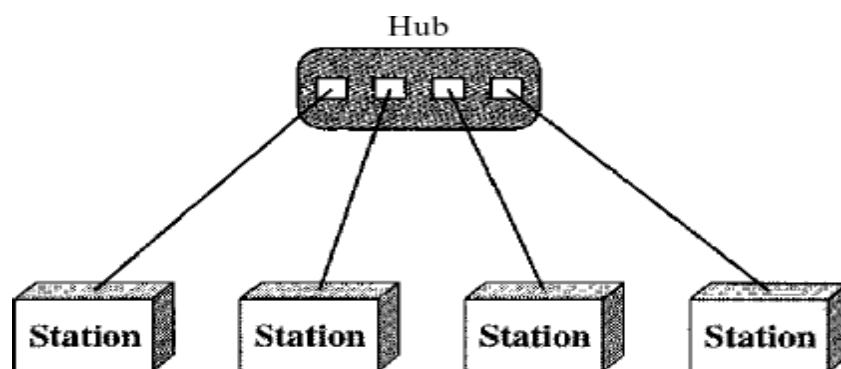
Star Topology:

In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. The devices are not directly linked to one another. Unlike a mesh topology, a star topology does not allow direct traffic between devices. The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.

A star topology is less expensive than a mesh topology. In a star, each device needs only one link and one I/O port to connect it to any number of others. This factor also makes it easy to install and reconfigure. Far less cabling needs to be housed, and additions, moves, and deletions involve only one connection: between that device and the hub.

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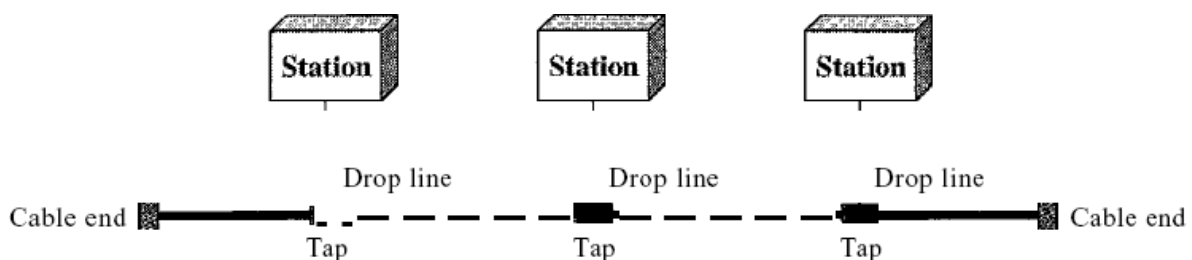
Other advantages include robustness. If one link fails, only that link is affected. All other links remain active. This factor also lends itself to easy fault identification and fault isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.



One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead. Although a star requires far less cable than a mesh, each node must be linked to a central hub. For this reason, often more cabling is required in a star than in some other topologies (such as ring or bus).

Bus Topology:

The preceding examples all describe point-to-point connections. A bus topology, on the other hand, is multipoint. One long cable acts as a backbone to link all the devices in a network



Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable. A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core. As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther. For this reason there is a limit on the number of taps a bus can support and on the distance between those taps.

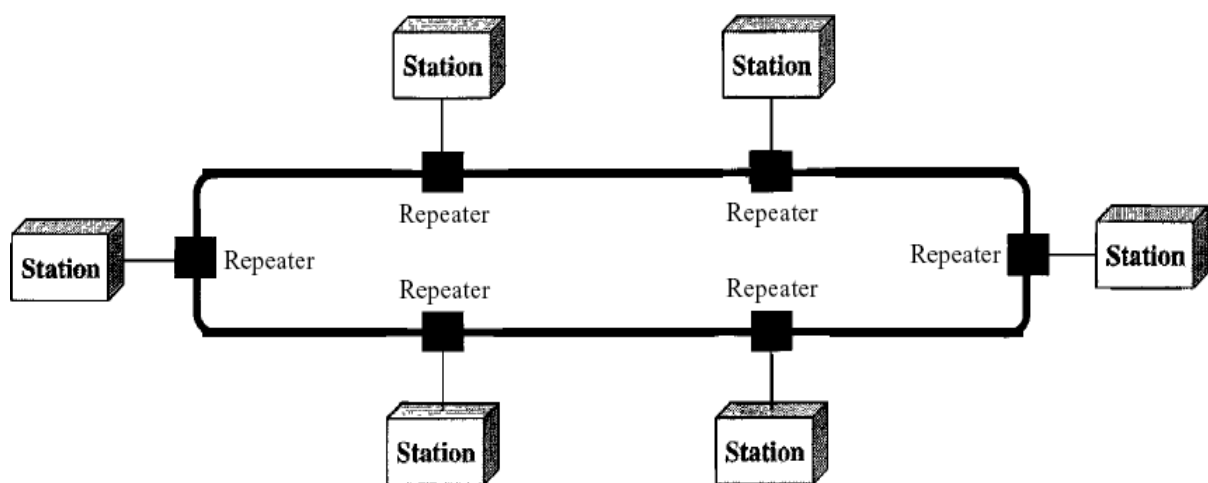
Advantages of a bus topology include ease of installation. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies.

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Disadvantages include difficult reconnection and fault isolation. A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices.

Bus topology was the one of the first topologies used in the design of early local area networks. Ethernet LANs can use a bus topology, but they are less popular.

Ring Topology



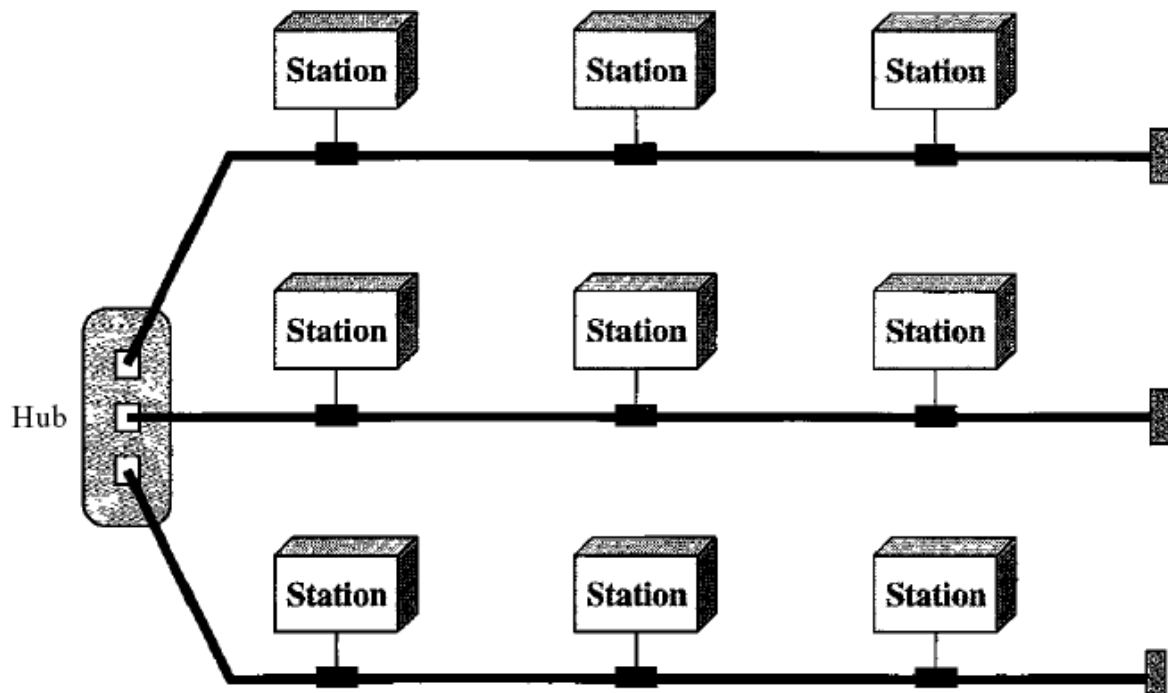
In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater.

A ring is relatively easy to install and reconfigure. Each device is linked to only its immediate neighbors (either physically or logically). To add or delete a device requires changing only two connections. The only constraints are media and traffic considerations (maximum ring length and number of devices). In addition, fault isolation is simplified. Generally in a ring, a signal is circulating at all times. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location.

However, unidirectional traffic can be a disadvantage. In a simple ring, a break in the ring (such as a disabled station) can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break. Ring topology was prevalent when IBM introduced its local-area network Token Ring. Today, the need for higher-speed LANs has made this

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topology less popular. Hybrid Topology A network can be hybrid. For example, we can have a main star topology with each branch connecting several stations in a bus topology as shown in Figure.



Categories of Networks:

1. Local Area Networks:

Local area networks, generally called LANs, are privately-owned networks within a single building or campus of up to a few kilometres in size. They are widely used to connect personal computers and workstations in company offices and factories to share resources (e.g., printers) and exchange information. LANs are distinguished from other kinds of networks by three **characteristics**:

- (1) Their size,
- (2) Their transmission technology, and
- (3) Their topology.

LANs are restricted in size, which means that the worst-case transmission time is bounded and known in advance. Knowing this bound makes it possible to use certain kinds of designs that would not otherwise be possible. It also simplifies network management. LANs may use a transmission technology consisting of a cable to which all the machines are attached, like the telephone company party lines once used in rural areas. Traditional LANs run at speeds of 10 Mbps to 100 Mbps, have low delay (micro-seconds or nanoseconds), and make very few errors. Newer LANs operate at up to 10 Gbps. Various topologies are possible for broadcast LANs. Figure 1 shows two of them. In a bus (i.e., a linear cable) network, at any instant at most one machine is the master and is allowed to transmit. All other machines are required to refrain from sending. An arbitration mechanism is needed to resolve conflicts when two or more machines want to transmit simultaneously. The arbitration



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mechanism may be centralized

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or distributed. **IEEE 802.3**, popularly called **Ethernet**, for example, is a bus-based broadcast network with decentralized control, usually operating at 10 Mbps to 10 Gbps. Computers on an Ethernet can transmit whenever they want to; if two or more packets collide, each computer just waits a random time and tries again later.

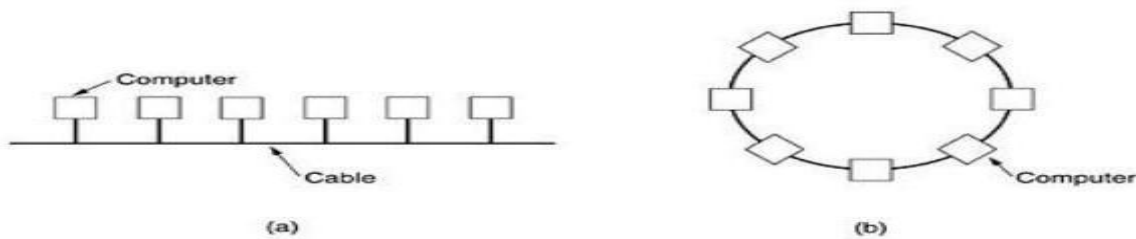


Fig.1: Two broadcast networks . (a) Bus. (b) Ring.

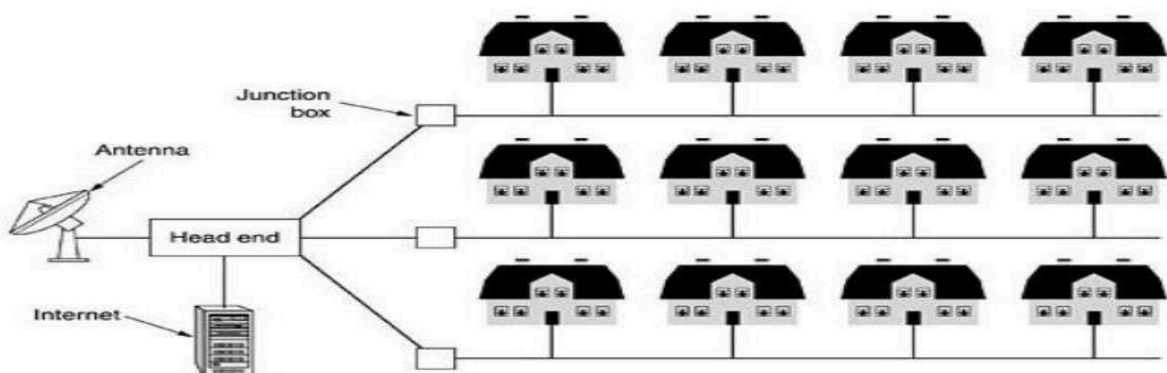
A second type of broadcast system is the ring. In a ring, each bit propagates around on its own, not waiting for the rest of the packet to which it belongs. Typically, each bit circumnavigates the entire ring in the time it takes to transmit a few bits, often before the complete packet has even been transmitted. As with all other broadcast systems, some rule is needed for arbitrating simultaneous accesses to the ring. Various methods, such as having the machines take turns, are in use. IEEE 802.5 (the IBM token ring), is a ring-based LAN operating at 4 and 16 Mbps.

2. Metropolitan Area Network (MAN):

A metropolitan area network, or MAN, covers a city.

The best-known example of a MAN is the cable television network available in many cities. This system grew from earlier community antenna systems used in areas with poor over-the-air television reception. In these early systems, a large antenna was placed on top of a nearby hill and signal was then piped to the subscribers' houses.

At first, these were locally-designed, ad hoc systems. Then companies began jumping into the business, getting contracts from city governments to wire up an entire city. The next step was television programming and even entire channels designed for cable only. Often these channels were highly specialized, such as all news, all sports, all cooking, all gardening, and so on. But from their inception until the late 1990s, they were intended for television reception only. **To a first approximation, a MAN might look something like the system shown in Fig. In this figure both television signals and Internet are fed into the centralized head end for subsequent distribution to people's homes. Cable television is not the only MAN. Recent developments in high-speed wireless Internet access resulted in another MAN, which has been standardized as IEEE 802.16.**





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3. Wide Area Network (WAN):

- A WAN (Wide Area Network) spans a large geographical area, often a country or continent.
- The WAN in Fig. 1-10 is a network that connects offices in Perth, Melbourne, and Brisbane. Each of these offices contains computers intended for running user (i.e., application) programs.
- We will follow traditional usage and call these machines hosts. The rest of the network that connects these hosts is then called the communication subnet, or just subnet for short.
- The job of the subnet is to carry messages from host to host, just as the telephone system carries words (just sounds) from speaker to listener. In most WANs, the subnet consists of two distinct components: transmission lines and switching elements. Transmission lines move bits between machines. They can be made of copper wire, optical fiber, or even radio links. Most companies do not have transmission lines lying about, so instead they lease the lines from a telecommunications company.
- Switching elements, or just switches, are specialized computers that connect two or more transmission lines. When data arrive on an incoming line, the switching element must choose an outgoing line on which to forward them. These switching computers have been called by various names in the past; the name router is now most commonly used.
- The WAN as we have described it looks similar to a large wired LAN, but there are some important differences that go beyond long wires. **Usually in a WAN, the hosts and subnet are owned and operated by different people.**
- Subnet Router Perth Brisbane Melbourne Transmission line Figure 1-10. WAN that connects three branch offices in Australia. example, the employees might be responsible for their own computers, while the company's IT department is in charge of the rest of the network.
- We will see clearer boundaries in the coming examples, in which the network provider or telephone company operates the subnet. Separation of the pure communication aspects of the network (the subnet) from the application aspects (the hosts) greatly simplifies the overall network design.
- A second difference is that the routers will usually connect different kinds of networking technology. The networks inside the offices may be switched Ethernet, for example, while the long-distance transmission lines may be SONET links. Some device needs to join them. The astute reader will notice that this goes beyond our definition of a network. This means that many WANs will in fact be internetworks, or composite networks that are made up of more than one network.
- A final difference is in what is connected to the subnet. This could be individual computers, as was the case for connecting to LANs, or it could be entire LANs. This is how larger networks are built from smaller ones. As far as the subnet is concerned, it does the same job. We are now in a position to look at two other varieties of WANs. First, rather than

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lease dedicated transmission lines, a company might connect its offices to the Internet. This allows connections to be made between the offices as virtual links that use the underlying capacity of the Internet.

- This arrangement, shown in Fig. 1-11, is called a VPN (Virtual Private Network). Compared to the dedicated arrangement, a VPN has the usual advantage of virtualization, which is that it provides flexible reuse of a resource (Internet connectivity). Consider how easy it is to add a fourth office to see this. A VPN also has the usual disadvantage of virtualization, which is a lack of control over the underlying resources. With a dedicated line, the capacity is clear.
- With a VPN your mileage may vary with your Internet service. Internet Perth Brisbane Melbourne Link via the internet Figure 1-11. WAN using a virtual private network.
- The second variation is that the subnet may be run by a different company. The subnet operator is known as a network service provider and the offices are its customers. This structure is shown in Fig. 1-12.
- subnet operator will connect to other customers too, as long as they can pay and it can provide service. Since it would be a disappointing network service if the customers could only send packets to each other, the subnet operator will also connect to other networks that are part of the Internet. Such a subnet operator is called an ISP (Internet Service Provider) and the subnet is an ISP network. Its customers who connect to the ISP receive Internet service. We can use the ISP network to preview some key issues that we will study in later chapters. In most WANs, the network contains many transmission lines, each connecting a pair of routers. If two routers that do not share a transmission line wish to communicate, they must do this indirectly, via other routers. **WAN using an ISP network. may be many paths in the network that connect these two routers. How the network makes the decision as to which path to use is called the routing algorithm. Many such algorithms exist. How each router makes the decision as to where to send a packet next is called the forwarding algorithm.**

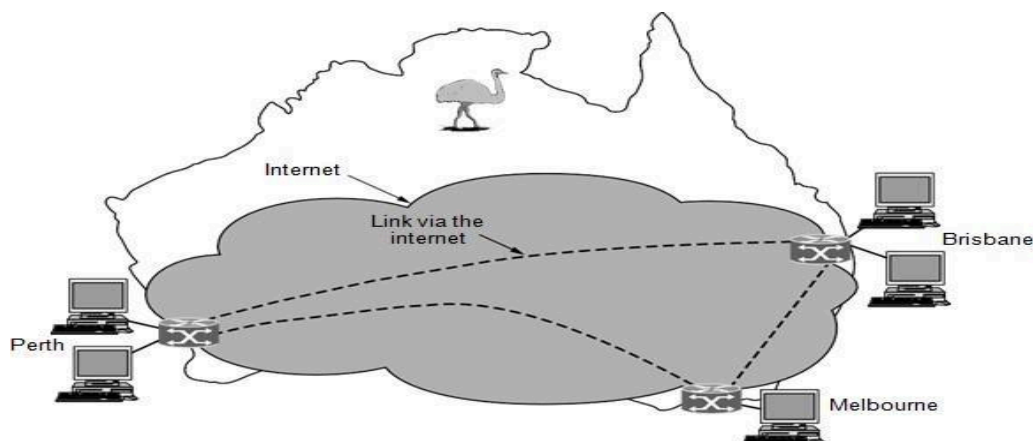


Figure 1-11. WAN using a virtual private network.

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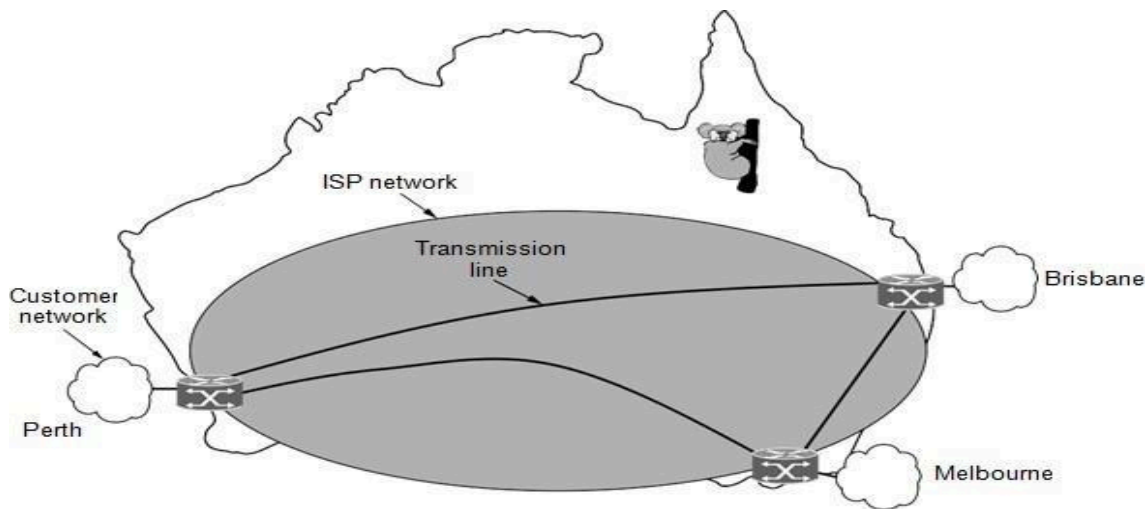


Figure 1-12. WAN using an ISP network.

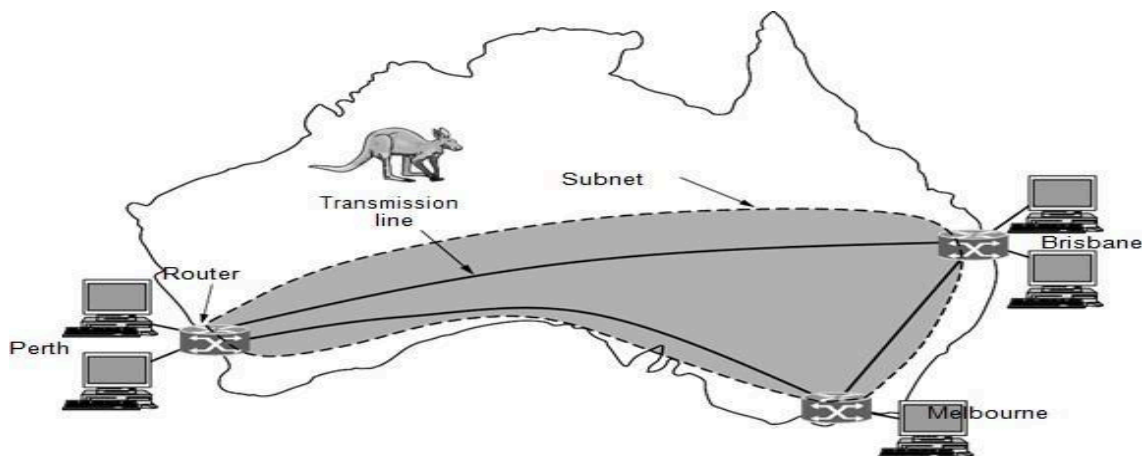


Figure 1-10. WAN that connects three branch offices in Australia.

4. THE INTERNET:

Many networks exist in the world, often with different hardware and software. People connected to one network often want to communicate with people attached to a different one. The fulfillment of this desire requires that different, and frequently incompatible, networks be connected. A collection of interconnected networks is called an **internetwork** or **internet**.

Subnets, networks, and internetworks are often confused. The term "subnet" makes the most sense in the context of a wide area network, where it refers to the collection of routers and communication lines owned by the network operator.

The general name for a machine that makes a connection between two or more networks and provides the necessary translation, both in terms of hardware and software, is a **gateway**. Gateways are distinguished by the layer at which they operate in the protocol hierarchy.

NETWORK SOFTWARE:

The first computer networks were designed with the hardware as the main concern and the software as an afterthought.

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Protocol Hierarchies:

To reduce their design complexity, most networks are organized as a stack of **layers** or **levels**, each one built upon the one below it. The number of layers, the name of each layer, the contents of each layer, and the function of each layer differ from network to network. The purpose of each layer is to offer certain services to the higher layers while shielding those layers from the details of how the offered services are implemented. In a sense, each layer is a kind of virtual machine, offering certain services to the layer above it.

When layer n on one machine carries on a conversation with layer n on another machine, the rules and conventions used in this conversation are collectively known as the layer n protocol. Basically, a **protocol** is an agreement between the communicating parties on how communications to proceed.

A five-layer network is illustrated in Fig. 1-13. The entities comprising the corresponding layers on different machines are called **peers**. The peers may be software processes, hardware devices, or even human beings. In other words, it is the peers that communicate by using the protocol to talk to each other. No data are directly transferred from layer n on one machine to layer n on another machine. Instead, each layer passes data and control information to the layer immediately below it, until the lowest layer is reached. Below layer 1 is the **physical medium** through which actual communication occurs. In Fig. 1-13, virtual communication is shown by dotted lines and physical communication by solid lines.

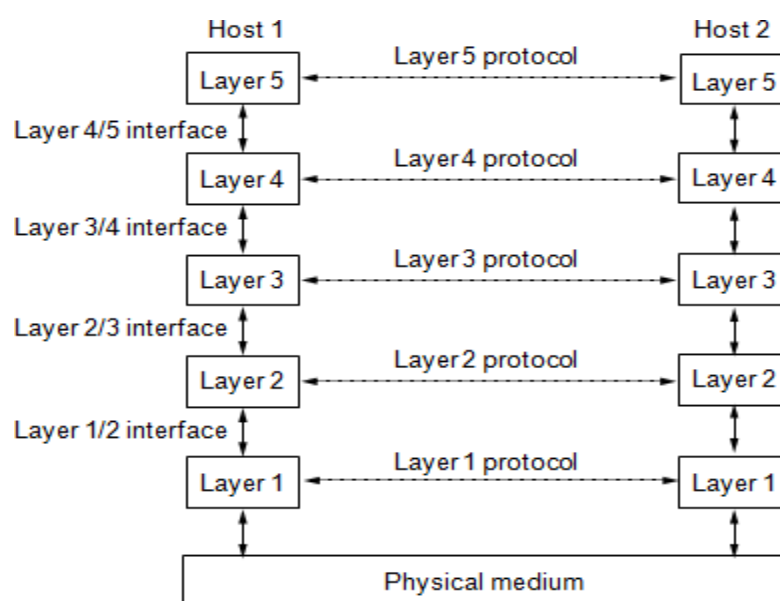


Figure 1-13. Layers, protocols, and interfaces.

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Between each pair of adjacent layers is an interface. The interface defines which primitive operations and services the lower layer makes available to the upper one. When network designers decide how many layers to include in a network and what each one should do, one of the most important considerations is defining clean interfaces between the layers.

A set of layers and protocols is called a network architecture. The specification of an architecture must contain enough information to allow an implementer to write the program or build the hardware for each layer so that it will correctly obey the appropriate protocol. It is not even necessary that the interfaces on all machines in a network be the same, provided that each machine can correctly use all the protocols. A list of the protocols used by a certain system, one protocol per layer, is called a **protocol stack**.

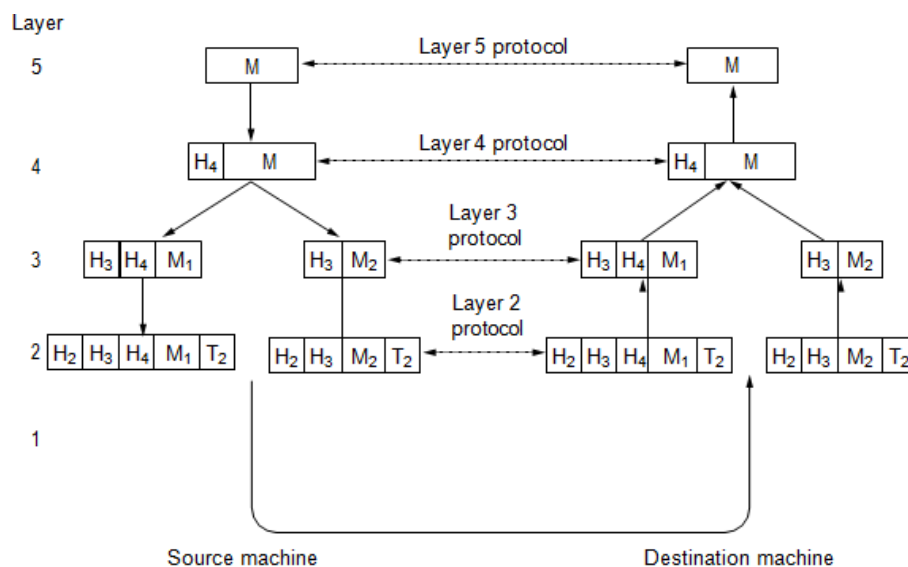


Figure 1-15. Example information flow supporting virtual communication in layer 5.

Now consider a more technical example: how to provide communication to the top layer of the five-layer network in Fig. 1-15. A message, M , is produced by an application process running in layer 5 and given to layer 4 for transmission. Layer 4 puts a **header** in front of the message to identify the message and passes the result to layer 3. The header includes control information, such as addresses, to allow layer 4 on the destination machine to deliver the message. Other examples of control information used in some layers are sequence numbers (in case the lower layer does not preserve message order), sizes, and times.

In many networks, no limit is placed on the size of messages transmitted in the layer 4 protocol but there is nearly always a limit imposed by the layer 3 protocol. Consequently, layer 3 must break up the incoming messages into smaller units, packets, prepending a layer 3 header to each packet. In this example, M is split into two parts, M_1 and M_2 , that will be transmitted separately.

Layer 3 decides which of the outgoing lines to use and passes the packets to layer 2. Layer 2 adds to each piece not only a header but also a trailer, and gives the resulting unit to layer 1 for physical transmission. At the receiving machine the message moves upward, from layer to layer, with headers being stripped off as it progresses. None of the headers for layers below n are passed up to layer n .



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Design Issues for the Layers:

Some of the key design issues that occur in computer networks will come up in layer after layer..

- **Reliability** is the design issue of making a network that operates correctly even though it is made up of a collection of components that are themselves unreliable. Think about the bits of a packet traveling through the network. There is a chance that some of these bits will be received damaged (inverted) due to fluke electrical noise, random wireless signals, hardware flaws, software bugs and so on. How is it possible that we find and fix these errors?

One mechanism for finding errors in received information uses codes for error detection. Information that is incorrectly received can then be retransmitted until it is received correctly. More powerful codes allow for error correction, where the correct message is recovered from the possibly incorrect bits that were originally received. Both of these mechanisms work by adding redundant information. They are used at low layers, to protect packets sent over individual links, and high layers, to check that the right contents were received.

- Another reliability issue is finding a working path through a network. Often there are multiple paths between a source and destination, and in a large network, there may be some links or routers that are broken. Suppose that the network is down in Germany. Packets sent from London to Rome via Germany will not get through, but we could instead send packets from London to Rome via Paris. **The network should automatically make this decision. This topic is called routing.**
- A second design issue concerns the evolution of the network. Over time, networks grow larger and new designs emerge that need to be connected to the existing network. We have recently seen the key structuring mechanism used to support change by dividing the overall problem and hiding implementation details: **protocol layering**. There are many other strategies as well.
- **Since there are many computers on the network, every layer needs a mechanism for identifying the senders and receivers that are involved in a particular message. This mechanism is called addressing or naming, in the low and high layers, respectively.**
- Designs that continue to work well when the network gets large are said to be **scalable**.
- **A third design issue is resource allocation. Networks provide a service to hosts from their underlying resources, such as the capacity of transmission lines.**
 - Sometimes the problem is that the network is oversubscribed because too many computers want to send too much traffic, and the network cannot deliver it all. This overloading of the network is called **congestion**.
 - The last major design issue is to secure the network by defending it against different kinds of threats. One of the threats we have mentioned previously is that of eavesdropping on communications. Mechanisms that provide **confidentiality** defend against this threat, and they are used in multiple layers.

Connection-Oriented Versus Connectionless Service:

- Layers can offer two different types of service to the layers above them:

Connection-oriented and connectionless.



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- **Connection-oriented** service is modeled after the telephone system.
- Similarly, to use a connection-oriented network service, the service user first establishes a connection, uses



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the connection, and then releases the connection.

- Essential aspect of a connection is that it acts like a tube: the sender pushes objects(bits) in at one end, and the receiver takes them out at the other end. In most cases the order is preserved so that the bits arrive in the order they were sent.

In contrast to connection-oriented service, **connectionless** service is modeled after the postal system. Each message (letter) carries the full destination address, and each one is routed through the intermediate nodes inside the system independent of all the subsequent messages.

- There are different names for messages in different contexts.
- **apacket** is a message at the network layer. When the intermediate nodes receive a message in full before sending it on to the next node, this is called **store-and-forward switching**.
- The alternative, in which the onward transmission of a message at a node starts before it is completely received by the node, is called **cut-through switching**. Normally, when two messages are sent to the same destination, the first one sent will be the first one to arrive. However, it is possible that the first one sent can be delayed so that the second one arrives first.
- Reliable connection-oriented service has two minor variations: message sequences and byte streams..

In other situations, the convenience of not having to establish a connection to send one message is desired, but reliability is essential. The **acknowledged datagram** service can be provided for these applications. It is like sending a registered letter and requesting a return receipt. When the receipt comes back, the sender is absolutely sure that the letter was delivered to the intended party and not lost along the way. Text messaging on mobile phones is an example.

Service	Example
Reliable message stream	Sequence of pages
Reliable byte stream	Movie download
Unreliable connection	Voice over IP
Unreliable datagram	Electronic junk mail
Acknowledged datagram	Text messaging
Request-reply	Database query

Still another service is the **request-reply** service. In this service the sender transmits a single datagram containing a request; the reply contains the answer. Request-reply is commonly used to implement communication



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in the client- servermodel: the client issues a request and the server responds to it. For example, a mobile phone client might send a query to a map server to retrieve the map data for the current location. Figure 1-16 summarizes the types of services discussed above.

Connection- less

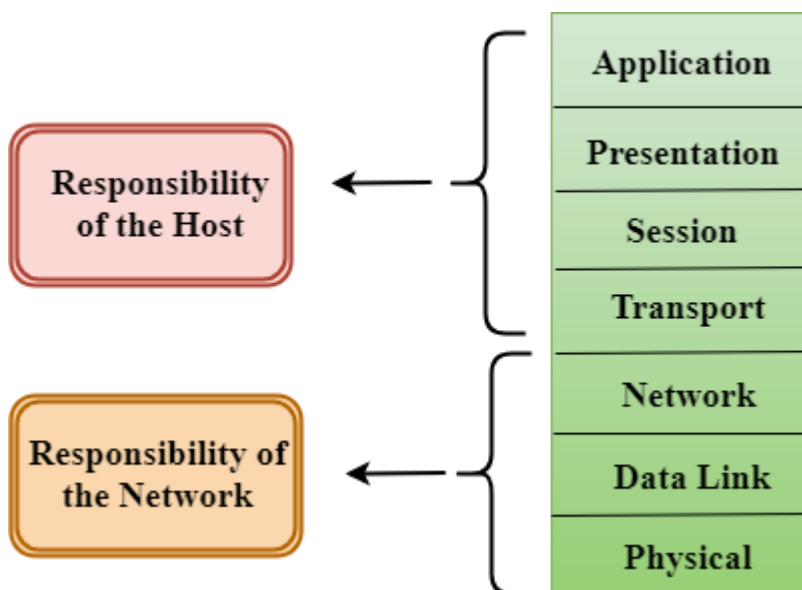
Figure 1-16. Six different types of service.

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OSI Reference Model:

- OSI stands for **Open System Interconnection** is a reference model that describes how information from a [software](#) application in one [computer](#) moves through a physical medium to the software application in another computer.
- OSI consists of seven layers, and each layer performs a particular network function.
- OSI model was developed by the International Organization for Standardization (ISO) in 1984, and it is now considered as an architectural model for the inter-computer communications.
- OSI model divides the whole task into seven smaller and manageable tasks. Each layer is assigned a particular task.
- Each layer is self-contained, so that task assigned to each layer can be performed independently.

Characteristics of OSI Model:



- The OSI model is divided into two layers: upper layers and lower layers.
- The upper layer of the OSI model mainly deals with the application related issues, and they are implemented only in the software. The application layer is closest to the end user. Both



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the end user and the application layer interact with the software applications. An upper layer refers to the layer just above another layer.

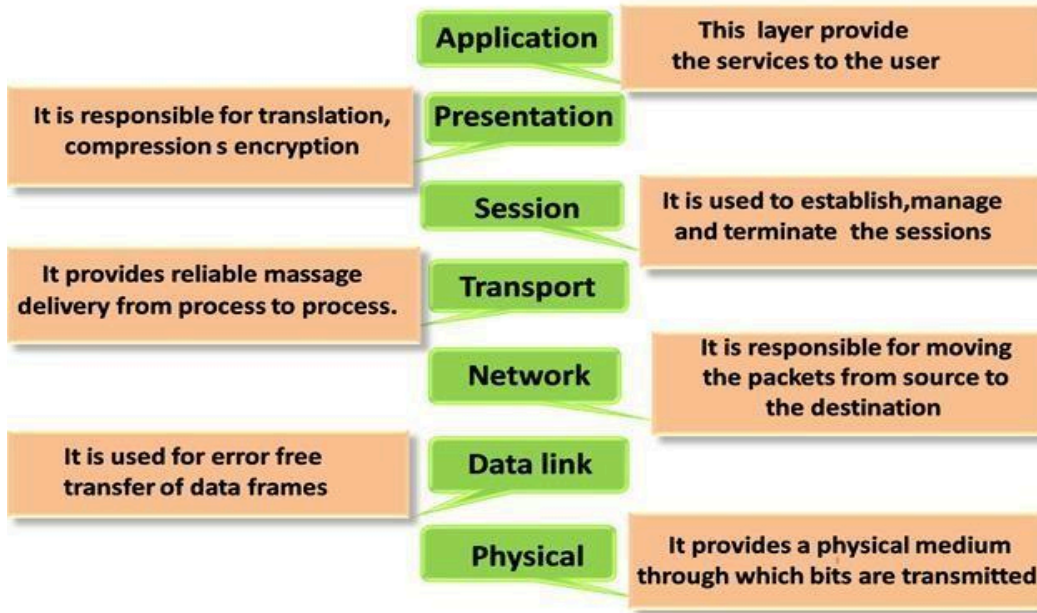
- The lower layer of the OSI model deals with the data transport issues. The data link layer and the physical layer are implemented in hardware and software. The physical layer is the lowest layer of the OSI model and is closest to the physical medium. The physical layer is mainly responsible for placing the information on the physical medium.

Functions of the OSI Layers

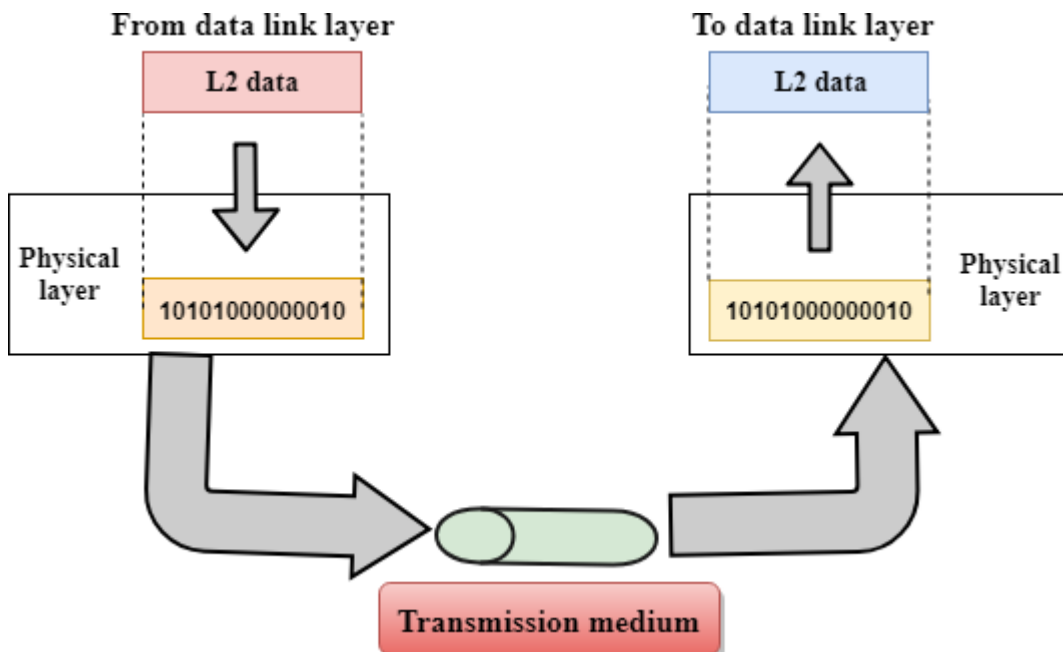
There are the seven OSI layers. Each layer has different functions. A list of seven layers are given below:

1. Physical Layer
2. Data-Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

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Physical layer



- The main functionality of the physical layer is to transmit the individual bits from one node to another node.

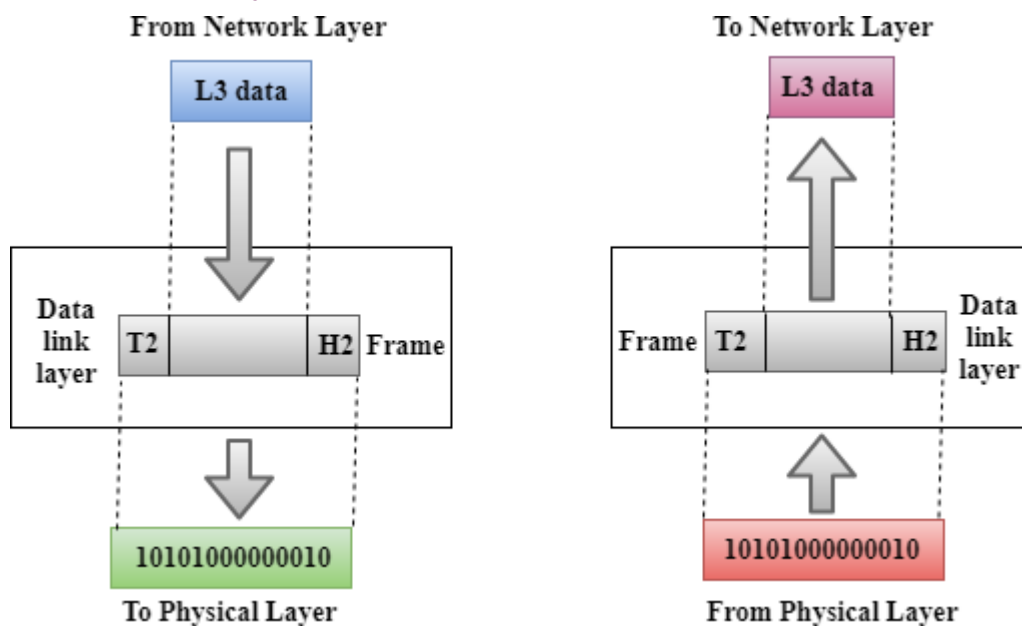
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- It is the lowest layer of the OSI model.
- It establishes, maintains and deactivates the physical connection.
- It specifies the mechanical, electrical and procedural network interface specifications.

Functions of a Physical layer:

- **Line Configuration:** It defines the way how two or more devices can be connected physically.
- **Data Transmission:** It defines the transmission mode whether it is simplex, half-duplex or full-duplex mode between the two devices on the network.
- **Topology:** It defines the way how network devices are arranged.
- **Signals:** It determines the type of the signal used for transmitting the information.

Data-Link Layer



- This layer is responsible for the error-free transfer of data frames.
- It defines the format of the data on the network.
- It provides a reliable and efficient communication between two or more devices.
- It is mainly responsible for the unique identification of each device that resides on a local network.

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- It contains two sub-layers:
 - **Logical Link Control Layer**
 - It is responsible for transferring the packets to the Network layer of the receiver that is receiving.
 - It identifies the address of the network layer protocol from the header.
 - It also provides flow control.
 - **Media Access Control Layer**
 - A Media access control layer is a link between the Logical Link Control layer and the network's physical layer.
 - It is used for transferring the packets over the network.

Functions of the Data-link layer

- **Framing:** The data link layer translates the physical's raw bit stream into packets known as Frames. The Data link layer adds the header and trailer to the frame. The header which is added to the frame contains the hardware destination and source address.

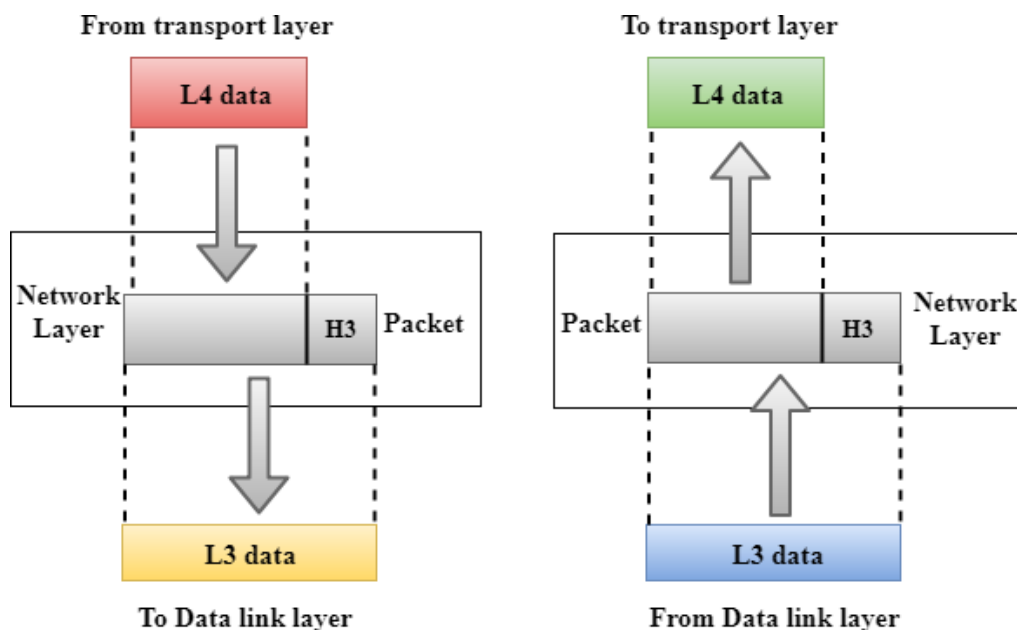
Header	Packet	Trailer
--------	--------	---------

- **Physical Addressing:** The Data link layer adds a header to the frame that contains a destination address. The frame is transmitted to the destination address mentioned in the header.
- **Flow Control:** Flow control is the main functionality of the Data-link layer. It is the technique through which the constant data rate is maintained on both the sides so that no data get corrupted. It ensures that the transmitting station such as a server with higher processing speed does not exceed the receiving station, with lower processing speed.
- **Error Control:** Error control is achieved by adding a calculated value CRC (Cyclic Redundancy Check) that is placed to the Data link layer's trailer which is added to the message frame before it is sent to the physical layer. If any error seems to occur, then the receiver sends the acknowledgment for the retransmission of the corrupted frames.

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- **Access Control:** When two or more devices are connected to the same communication channel, then the data link layer protocols are used to determine which device has control over the link at a given time.

Network Layer



- It is a layer 3 that manages device addressing, tracks the location of devices on the network.
- It determines the best path to move data from source to the destination based on the network conditions, the priority of service, and other factors.
- The Data link layer is responsible for routing and forwarding the packets.
- Routers are the layer 3 devices, they are specified in this layer and used to provide the routing services within an internetwork.
- The protocols used to route the network traffic are known as Network layer protocols. Examples of protocols are IP and Ipv6.

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Functions of Network Layer:

- **Internetworking:** An internetworking is the main responsibility of the network layer. It provides a logical connection between different devices.
- **Addressing**

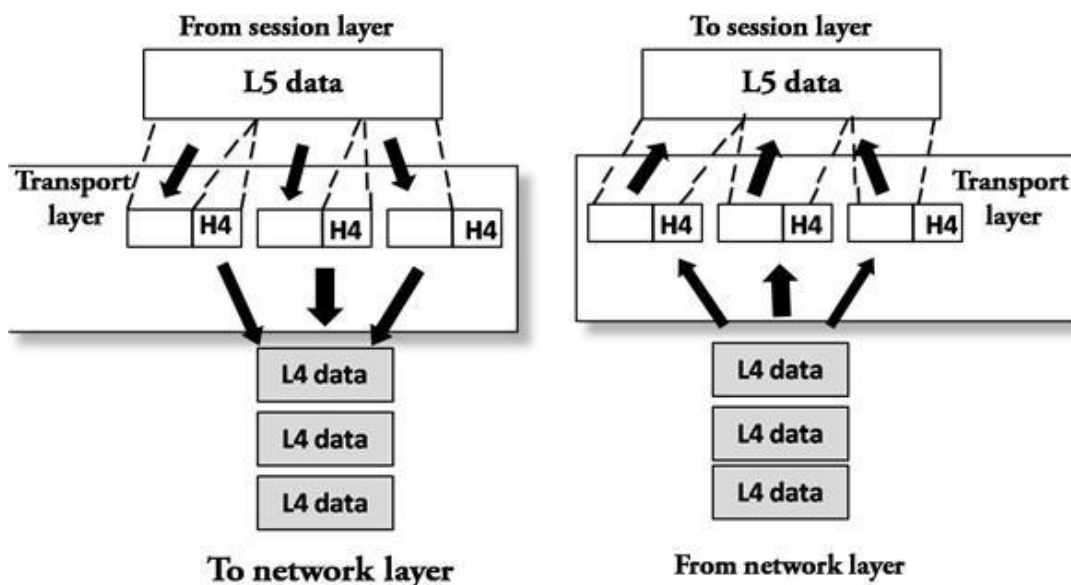
: A Network layer adds the source and destination address to the header of the frame. Addressing is used to identify the device on the internet.

- **Routing**

: Routing is the major component of the network layer, and it determines the best optimal path out of the multiple paths from source to the destination.

- **Packetizing:** A Network Layer receives the packets from the upper layer and converts them into packets. This process is known as Packetizing. It is achieved by internet protocol (IP).

Transport Layer



- The Transport layer is a Layer 4 ensures that messages are transmitted in the order in which they are sent and there is no duplication of data.



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- The main responsibility of the transport layer is to transfer the data completely.
- It receives the data from the upper layer and converts them into smaller units known as segments.
- This layer can be termed as an end-to-end layer as it provides a point-to-point connection between source and destination to deliver the data reliably.

The two protocols used in this layer are:

- **Transmission Control Protocol**
 - It is a standard protocol that allows the systems to communicate over the internet.
 - It establishes and maintains a connection between hosts.
 - When data is sent over the TCP connection, then the TCP protocol divides the data into smaller units known as segments. Each segment travels over the internet using multiple routes, and they arrive in different orders at the destination. The transmission control protocol reorders the packets in the correct order at the receiving end.
- **User Datagram Protocol**
 - User Datagram Protocol is a transport layer protocol.
 - It is an unreliable transport protocol as in this case receiver does not send any acknowledgment when the packet is received, the sender does not wait for any acknowledgment. Therefore, this makes a protocol unreliable.

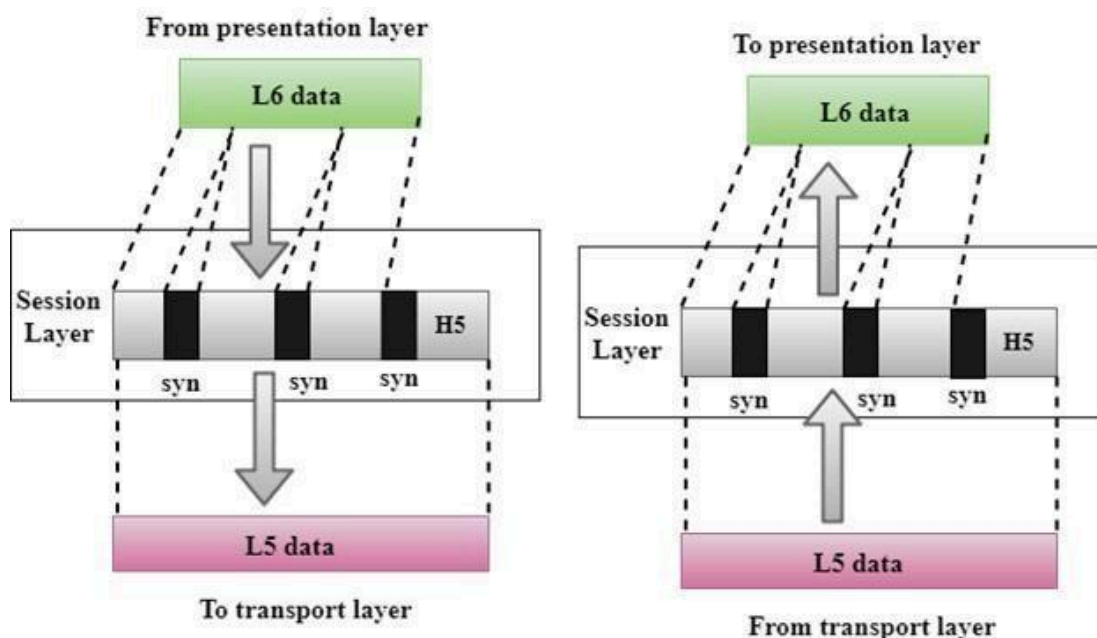
Functions of Transport Layer:

- **Service-point addressing:** Computers run several programs simultaneously due to this reason, the transmission of data from source to the destination not only from one computer to another computer but also from one process to another process. The transport layer adds the header that contains the address known as a service-point address or port address. The responsibility of the network layer is to transmit the data from one computer to another computer and the responsibility of the transport layer is to transmit the message to the correct process.

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- **Segmentation and reassembly:** When the transport layer receives the message from the upper layer, it divides the message into multiple segments, and each segment is assigned with a sequence number that uniquely identifies each segment. When the message has arrived at the destination, then the transport layer reassembles the message based on their sequence numbers.
- **Connection control:** Transport layer provides two services Connection-oriented service and connectionless service. A connectionless service treats each segment as an individual packet, and they all travel in different routes to reach the destination. A connection-oriented service makes a connection with the transport layer at the destination machine before delivering the packets. In connection-oriented service, all the packets travel in the single route.
- **Flow control:** The transport layer also responsible for flow control but it is performed end-to-end rather than across a single link.
- **Error control:** The transport layer is also responsible for Error control. Error control is performed end-to-end rather than across the single link. The sender transport layer ensures that message reach at the destination without any error.

Session Layer



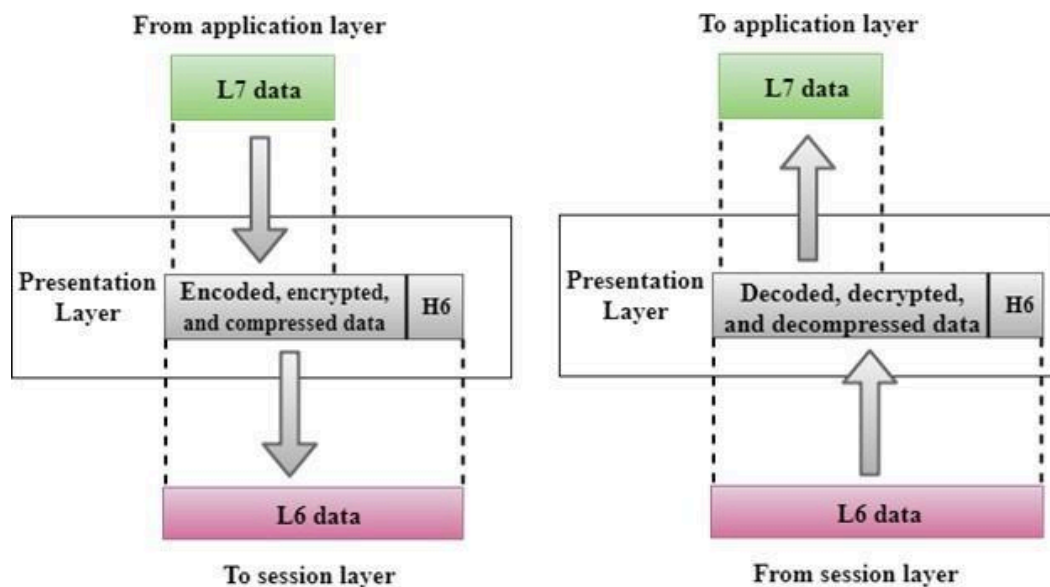
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- It is a layer 3 in the OSI model.
- The Session layer is used to establish, maintain and synchronizes the interaction between communicating devices.

Functions of Session layer:

- **Dialog control:** Session layer acts as a dialog controller that creates a dialog between two processes or we can say that it allows the communication between two processes which can be either half-duplex or full-duplex.
- **Synchronization:** Session layer adds some checkpoints when transmitting the data in a sequence. If some error occurs in the middle of the transmission of data, then the transmission will take place again from the checkpoint. This process is known as Synchronization and re-covery.

Presentation Layer



- A Presentation layer is mainly concerned with the syntax and semantics of the information exchanged between the two systems.
- It acts as a data translator for a network.

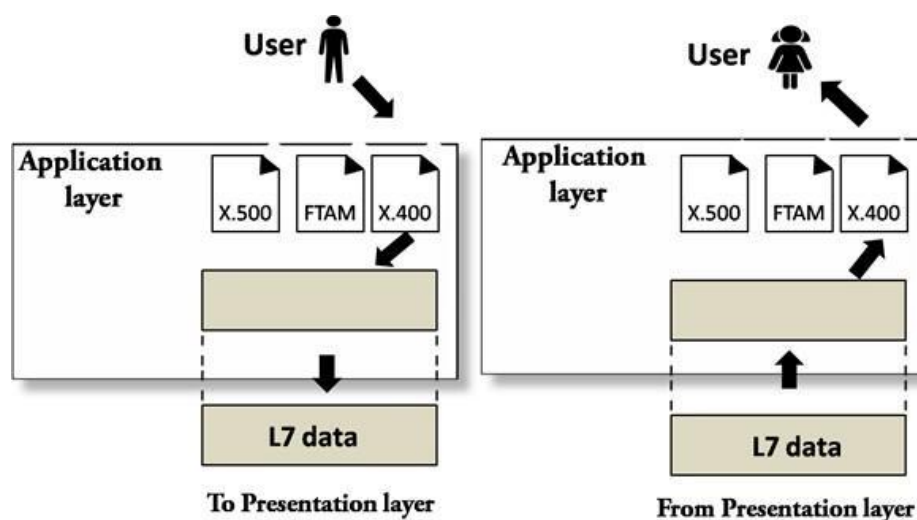
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- This layer is a part of the operating system that converts the data from one presentation format to another format.
- The Presentation layer is also known as the syntax layer.

Functions of Presentation layer:

- **Translation:** The processes in two systems exchange the information in the form of character strings, numbers and so on. Different computers use different encoding methods, the presentation layer handles the interoperability between the different encoding methods. It converts the data from sender-dependent format into a common format and changes the common format into receiver-dependent format at the receiving end.
- **Encryption:** Encryption is needed to maintain privacy. Encryption is a process of converting the sender-transmitted information into another form and sends the resulting message over the network.
- **Compression:** Data compression is a process of compressing the data, i.e., it reduces the number of bits to be transmitted. Data compression is very important in multimedia such as text, audio, video.

Application Layer





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- An application layer serves as a window for users and application processes to access network service.
- It handles issues such as network transparency, resource allocation, etc.
- An application layer is not an application, but it performs the application layer functions.
- This layer provides the network services to the end-users.

Functions of Application layer:

- **File transfer, access, and management (FTAM):** An application layer allows a user to access the files in a remote computer, to retrieve the files from a computer and to manage the files in a remote computer.
- **Mail services:** An application layer provides the facility for email forwarding and storage.
- **Directory services:** An application provides the distributed database sources and is used to provide that global information about various objects.

The TCP/IP Reference Model:

The TCP/IP reference model was developed prior to OSI model. The major design goals of this model were,

1. To connect multiple networks together so that they appear as a single network.
2. To survive after partial subnet hardware failures.
3. To provide a flexible architecture.

Unlike OSI reference model, TCP/IP reference model has only 4 layers. They are,

1. Host-to-Network Layer
2. Internet Layer
3. Transport Layer
4. Application Layer

Host-to-Network Layer:

The TCP/IP reference model does not really say much about what happens here, except to point out that the host has to connect to the network using some protocol so it can send IP packets to it. This protocol is not defined and varies from host to host and network to network.

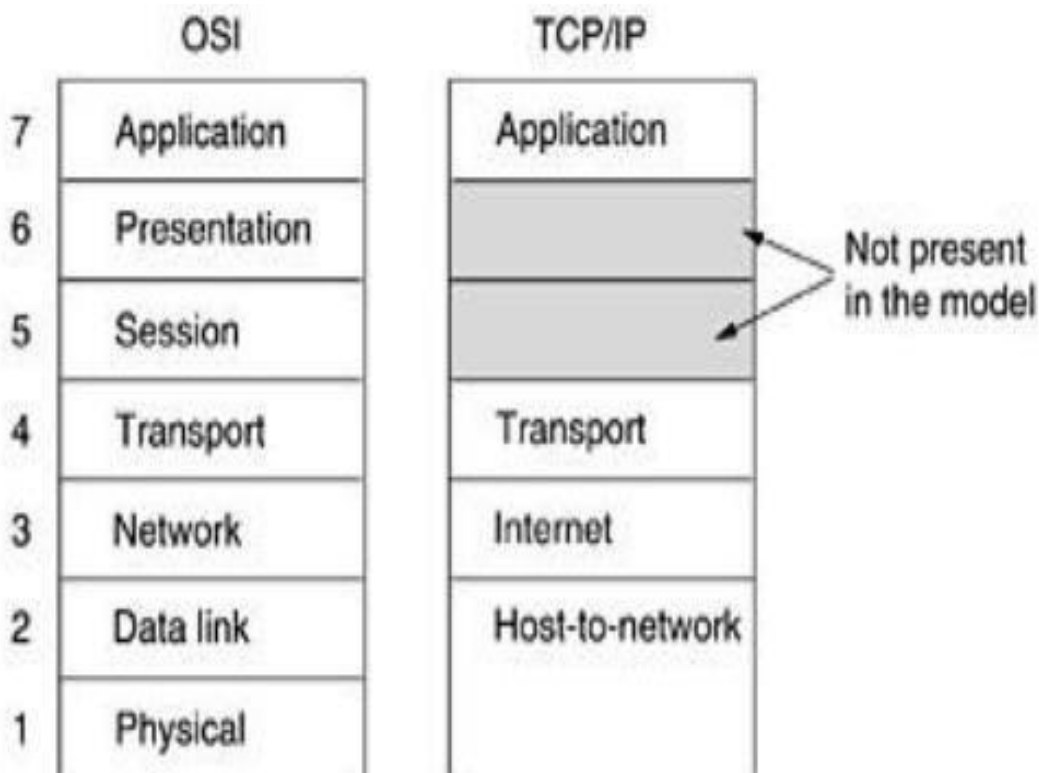
Internet Layer:

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This layer, called the internet layer, is the linchpin that holds the whole architecture together. Its job is to permit hosts to inject packets into any network and have them travel independently to the destination (potentially on a different network). They may even arrive in a different order than they were sent, in which case it is the job of higher layers to rearrange them, if in-order delivery is desired. Note that "internet" is used here in a generic sense, even though this layer is present in the Internet.

The internet layer defines an official packet format and protocol called IP (Internet Protocol). The job of the internet layer is to deliver IP packets where they are supposed to go. Packet routing is clearly the major issue here, as is avoiding congestion. For these reasons, it is reasonable to say that the TCP/IP internet layer is similar in functionality to the OSI network layer. Fig. shows this correspondence.

The Transport Layer: the layer above the internet layer in the TCP/IP model is now usually called the transport layer. It is designed to allow peer entities on the source and destination hosts to carry on a conversation, just as in the OSI transport layer. Two end-to-end transport protocols have been defined here. The first one, TCP (Transmission Control Protocol), is a reliable connection-oriented protocol that allows a byte stream originating on one machine to be delivered without error on any other machine in the internet. It fragments the incoming byte stream into discrete messages and passes each one on to the internet layer. At the destination,



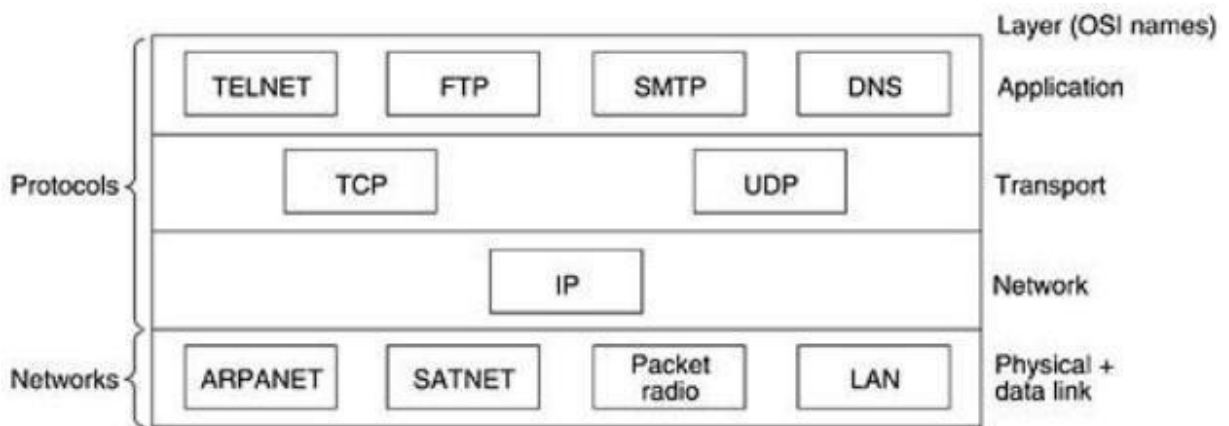
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the receiving TCP process reassembles the received messages into the output stream. TCP also handles flow control to make sure a fast sender cannot swamp a slow receiver with more messages than it can handle.

Fig.1: The TCP/IP reference model.

The second protocol in this layer, UDP (User Datagram Protocol), is an unreliable, connectionless protocol for applications that do not want TCP's sequencing or flow control and wish to provide their own. It is also widely used for one-shot, client-server-type request-reply queries and applications in which prompt delivery is more important than accurate delivery, such as transmitting speech or video. The relation of IP, TCP, and UDP is shown in Fig.2. Since the model was developed, IP has been implemented on many other networks.

Fig.2: Protocols and networks in the TCP/IP model initially.



The Application Layer:

The TCP/IP model does not have session or presentation layers. On top of the transport layer is the application layer. It contains all the higher-level protocols. The early ones included virtual terminal (TELNET), file transfer (FTP), and electronic mail (SMTP), as shown in Fig. The virtual terminal protocol allows a user on one machine to log onto a distant machine and work there. The file transfer protocol provides a way to move data efficiently from one machine to another. Electronic mail was originally just a kind of file transfer, but later a specialized protocol (SMTP) was developed for it. Many other protocols have been added to these over the years: the Domain Name System (DNS) for mapping host names onto their network addresses, NNTP, the protocol for moving USENET news articles around, and HTTP, the protocol for fetching pages on the World Wide Web, and many others.



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Comparison of the OSI and TCP/IP Reference Models:

The OSI and TCP/IP reference models have much in common. Both are based on the concept of a stack of independent protocols. Also, the functionality of the layers is roughly similar. For example, in both models the layers up through and including the transport layer are there to provide an end-to-end, network-independent transport service to processes wishing to communicate. These layers form the transport provider. Again in both models, the layers above transport are application-oriented users of the transport service. Despite these fundamental similarities, the two models also have many differences. Three concepts are central to the OSI model:

1. Services.
2. Interfaces.
3. Protocols.

Probably the biggest contribution of the OSI model is to make the distinction between these three concepts explicit. Each layer performs some services for the layer above it. The service definition tells what the layer does, not how entities above it access it or how the layer works. It defines the layer's semantics.

A layer's interface tells the processes above it how to access it. It specifies what the parameters are and what results to expect. It, too, says nothing about how the layer works inside.

Finally, the peer protocols used in a layer are the layer's own business. It can use any protocols it wants to, as long as it gets the job done (i.e., provides the offered services). It can also change them at will without affecting software in higher layers.

As a consequence, the protocols in the OSI model are better hidden than in the TCP/IP model and can be replaced relatively easily as the technology changes.

Another difference is in the area of connectionless versus connection-oriented communication.

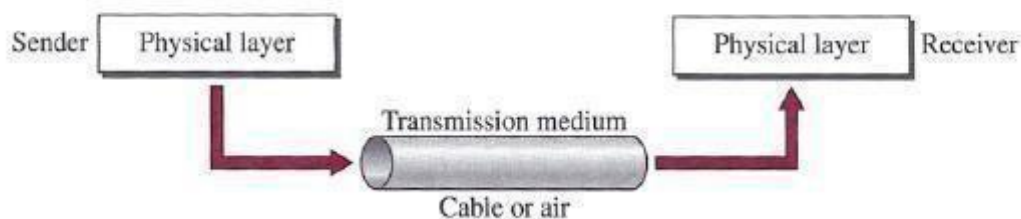
The OSI model supports both connectionless and connection-oriented communication in the network layer, but only connection-oriented communication in the transport layer, where it counts (because the transport service is visible to the users). The TCP/IP model has only one mode in the network layer (connectionless) but supports both modes in the transport layer, giving the users a choice. This choice is especially important for simple request-response protocols.

1.5. Guided transmission media

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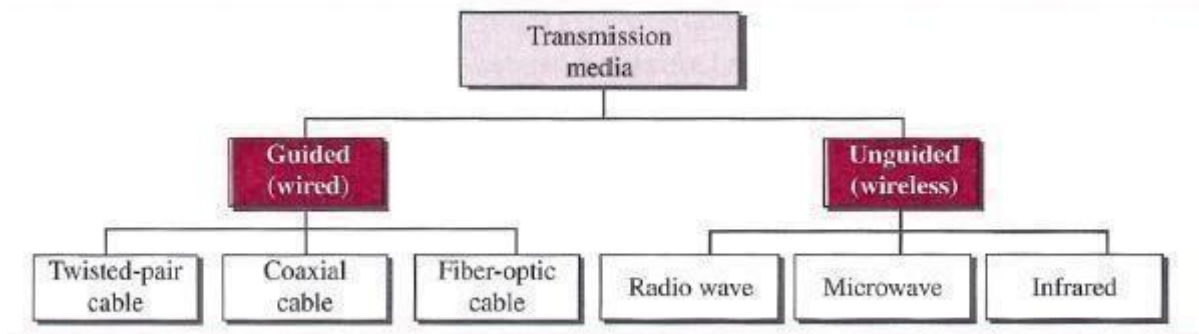
Transmission media are actually located below the physical layer and are directly controlled by the physical layer. We could say that transmission media belong to layer zero. Below figure shows the position of transmission media in relation to the physical layer.

Transmission medium and physical layer



In telecommunications, transmission media can be divided into two broad categories: guided and unguided. Guided media include twisted-pair cable, coaxial cable, and fiber-optic cable. Unguided medium is free space.

Classes of transmission media



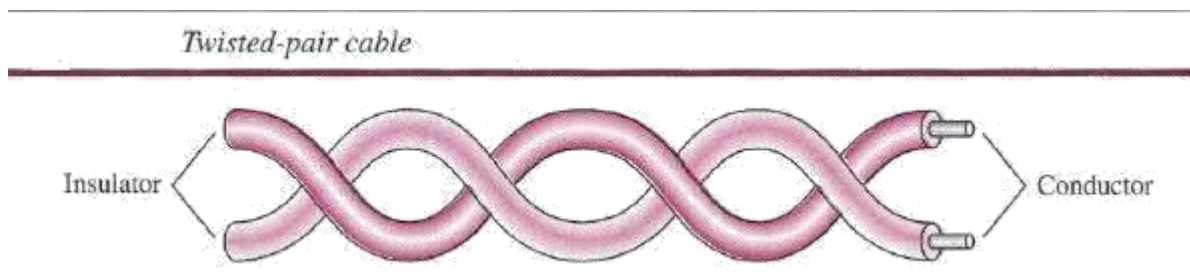
GUIDED MEDIA

Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable. A signal traveling along any of these media is directed and contained by the physical limits of the medium. Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current. Optical fiber is a cable that accepts and transports signals in the form of light.

Twisted-Pair Cable:

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A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together, as shown in following figure.



One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference. The receiver uses the difference between the two. In addition to the signal sent by the sender on one of the wires, interference (noise) and crosstalk may affect both wires and create unwanted signals. If the two wires are parallel, the effect of these unwanted signals is not the same in both wires because they are at different locations relative to the noise or crosstalk sources (e.g., one is closer and the other is farther). This results in a difference at the receiver. By twisting the pairs, a balance is maintained. For example, suppose in one twist, one wire is closer to the noise source and the other is farther; in the next twist, the reverse is true. Twisting makes it probable that both wires are equally affected by external influences (noise or crosstalk). This means that the receiver, which calculates the difference between the two, receives no unwanted signals. The unwanted signals are mostly canceled out. From the above discussion, it is clear that the number of twists per unit of length (e.g., inch) has some effect on the quality of the cable.

Performance

One way to measure the performance of twisted-pair cable is to compare attenuation versus frequency and distance. A twisted-pair cable can pass a wide range of frequencies. However, below figure shows that with increasing frequency, the attenuation, measured in decibels per kilometer (dB/km), sharply increases with frequencies above 100 kHz. Note that *gauge* is a measure of the thickness of the wire.

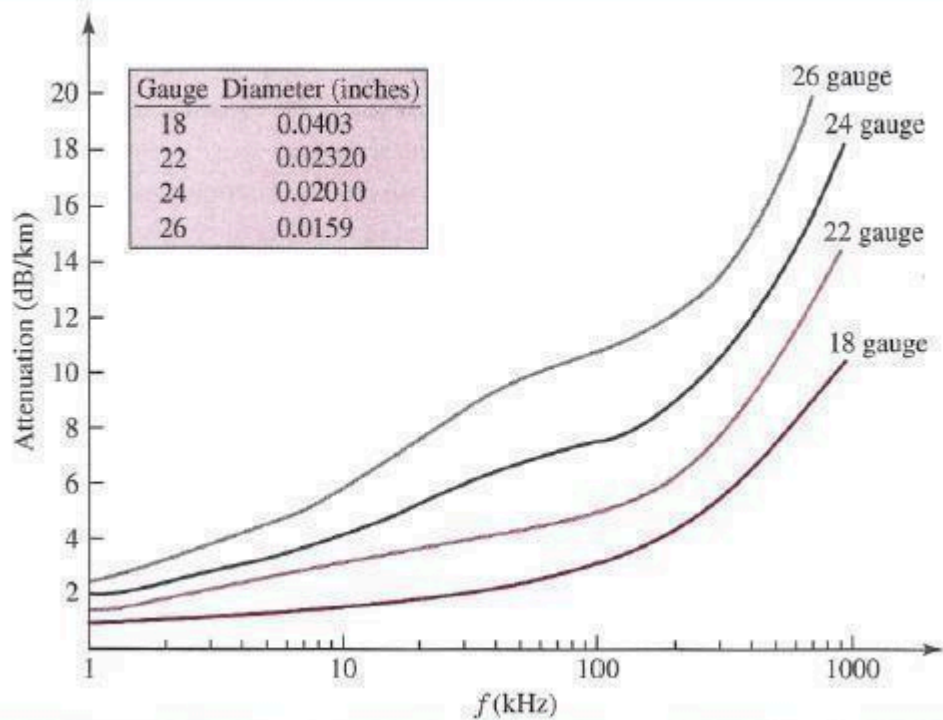
Applications

Twisted-pair cables are used in telephone lines to provide voice and data channels. The local loop-the line that connects subscribers to the central telephone office commonly consists of unshielded twisted-pair cables.

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The DSL lines that are used by the telephone companies to provide high-data-rate connections also use the high-bandwidth capability of unshielded twisted-pair cables.

UTP performance



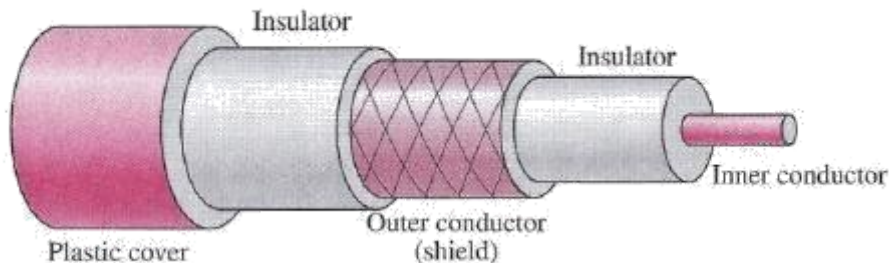
Local-area networks, such as IOBase-T and IOOBase-T, also use twisted-pair cables.

Coaxial Cable:

Coaxial cable (or *coax*) carries signals of higher frequency ranges than those in twisted pair cable, in part because the two media are constructed quite differently. Instead of having two wires, coax has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating shield, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two. The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit. This outer conductor is also enclosed in an insulating shield, and the whole cable is protected by a plastic cover.

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Coaxial cable



Performance

As we did with twisted-pair cable, we can measure the performance of a coaxial cable. We notice in Figure 7.9 that the attenuation is much higher in coaxial cable than in twisted-pair cable. In other words, although coaxial cable has a much higher bandwidth, the signal weakens rapidly and requires the frequent use of repeaters.

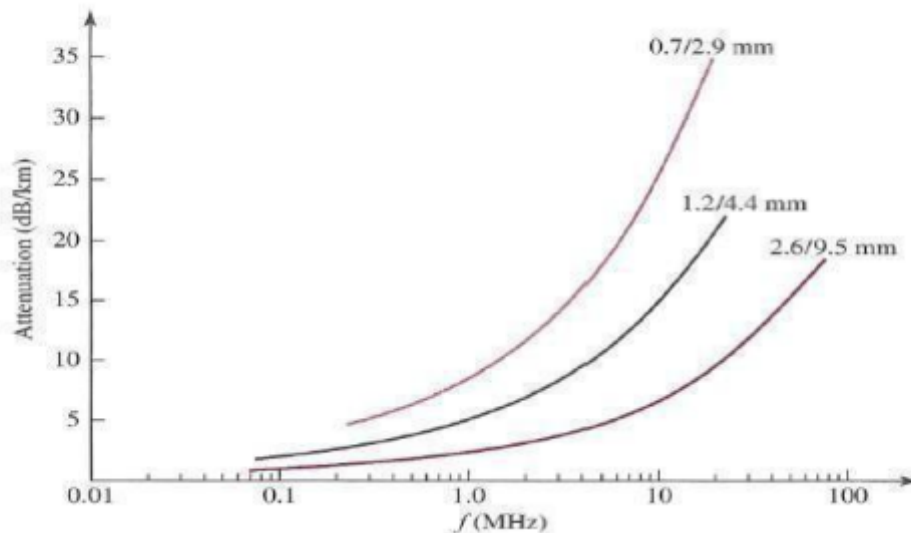
Applications

Coaxial cable was widely used in analog telephone networks where a single coaxial network could carry 10,000 voice signals. Later it was used in digital telephone networks where a single coaxial cable could carry digital data up to 600 Mbps. However, coaxial cable in telephone networks has largely been replaced today with fiber optic cable.

Cable TV networks also use coaxial cables. In the traditional cable TV network, the entire network used coaxial cable. Later, however, cable TV providers replaced most of the media with fiber-optic cable; hybrid networks use coaxial cable only at the network boundaries, near the consumer premises. Cable TV uses RG-59 coaxial cable. Another common application of coaxial cable is in traditional Ethernet LANs (see Because of its high bandwidth, and consequently high data rate, coaxial cable was chosen for digital transmission in early Ethernet LANs.

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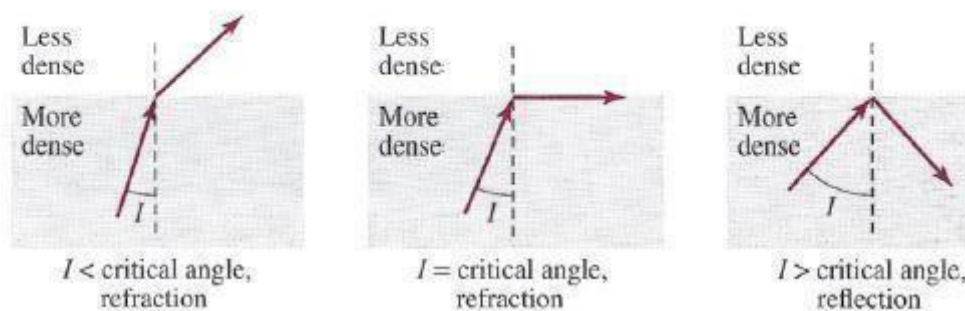
Coaxial cable performance



Fiber-Optic Cable:

A fiber-optic cable is made of glass or plastic and transmits signals in the form of light. To understand optical fiber, we first need to explore several aspects of the nature of light. Light travels in a straight line as long as it is moving through a single uniform substance. If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes direction. Below figure shows how a ray of light changes direction when going from a denser to a less dense substance. As

Bending of light ray



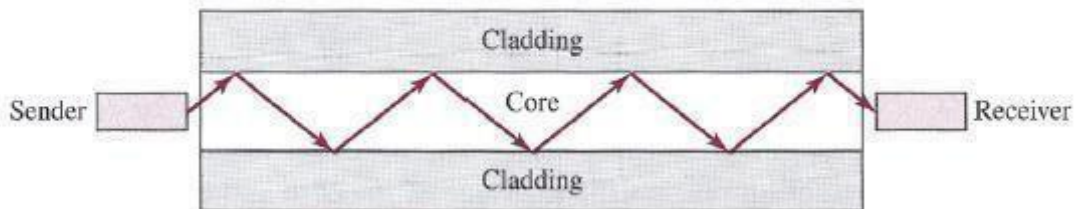
the figure shows, if the angle of

incidence I (the angle the ray makes with the line perpendicular to the interface between the two substances)

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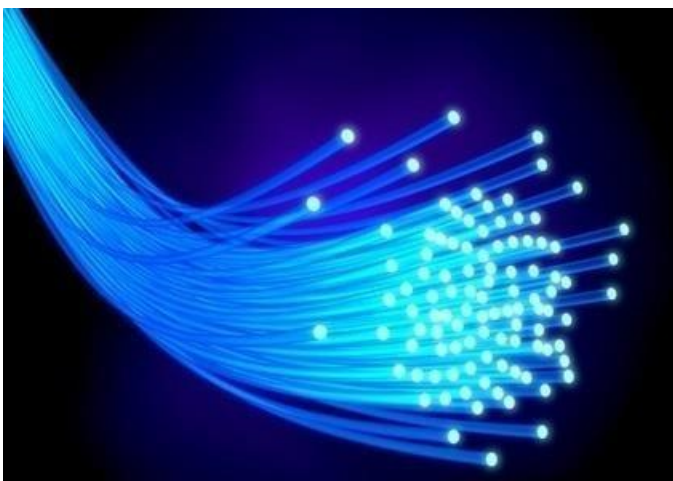
is less than the **critical angle**, the ray **refracts** and moves closer to the surface. If the angle of incidence is equal to the critical angle, the light bends along the interface. If the angle is greater than the critical angle, the ray **reflects** (makes a turn) and travels again in the denser substance. Note that the critical angle is a property of the substance, and its value differs from one substance to another. Optical fibers use reflection to guide light through a channel. A glass or plastic core is surrounded by a cladding of less dense glass or plastic. The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it. See below figure.

Optical fiber



Fiber Cable:

Optical fiber cables are transparent, flexible fibers made up of glass or plastic through which light waves can pass. A bunch of fiber optic cables is shown in the following diagram –



Structure of a Fiber – Optic Cable

A cross section of a fiber optic cable reveals three parts –

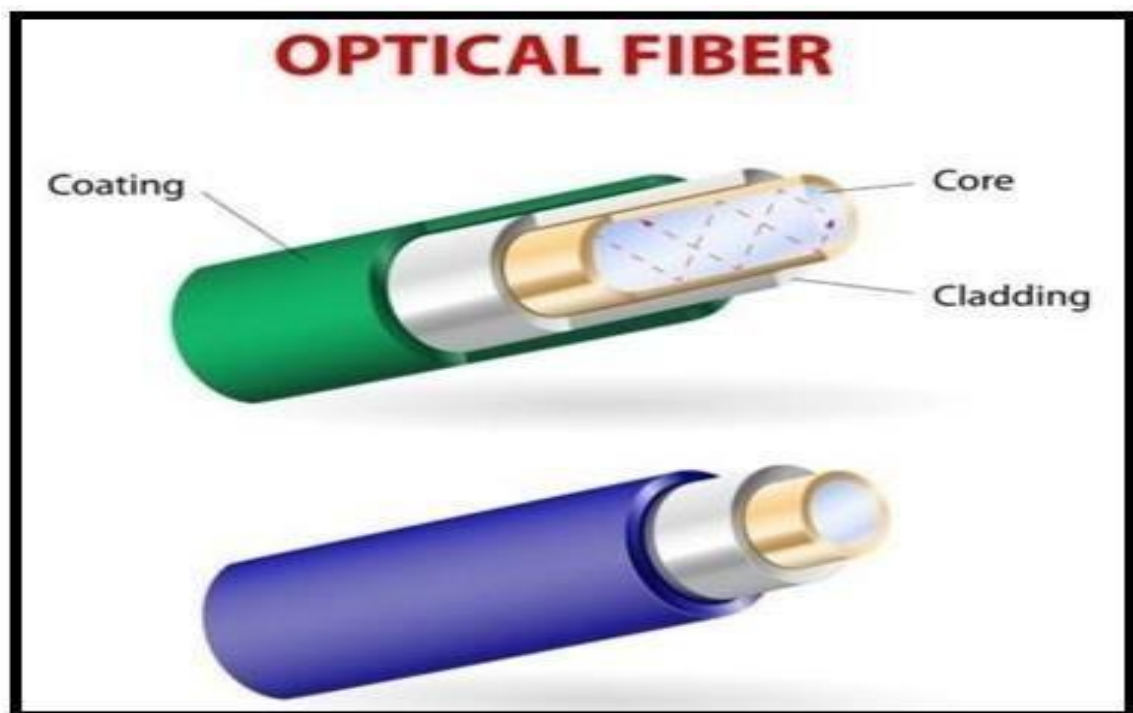
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Core – It is the innermost portion of an optical fiber through which light propagates. It is cylindrical in shape and is made up of a flexible glass of high refractive index. The diameter of the core of a single mode fiber is 8 – 10 μm while multimode fibers are 50 μm in diameter. It is also called the optical waveguide since it is the main channel through which light signals are transmitted.

Cladding – The core is surrounded by a glass cladding. The glass of cladding has a lower refractive index than the core. This enables total internal reflection of light waves in the core, and eventually propagation of light waves within the core.

Outer Coating or Jacket – The outer jacket is a thin plastic sheath or coating that is opaque to light. It prevents light rays from outside to enter the optical fiber. Fibers are typically bundled together, where each bundle is covered by a protective outer sheath that prevents the fibers from physical damage.

The following figure gives a three-dimensional section of a fiber – optic cable.



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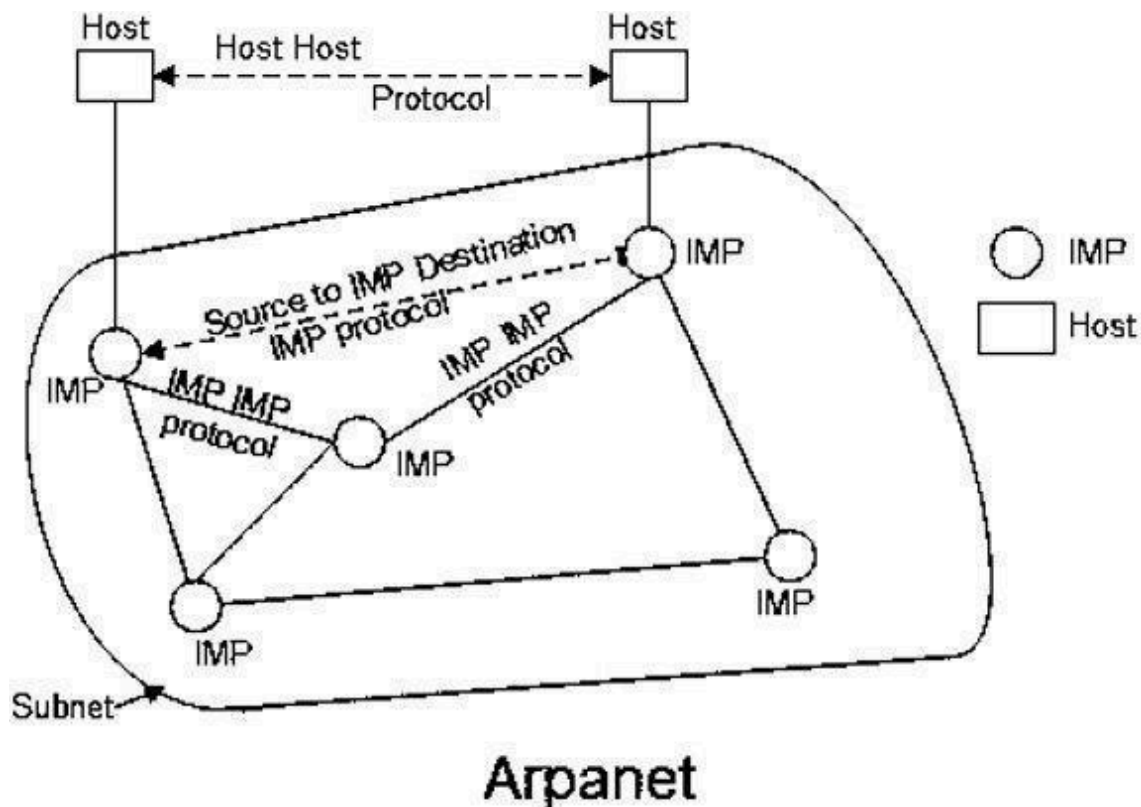
1.6. Examples of Networks: ARPANET:

The full form of ARPANET is **Advanced Research Projects Agency NET** was developed under the direction of the US Advanced Research Projects Agency and was based on a concept first published in 1967. With the interconnection of four university computers in 1969, the concept became a modest reality.

It was the first network to implement the TCP/IP protocol. It was a very early packet switch network. This network became the foundation for the modern day internet.

The protocols used in Arpanet were later developed for joining multiple networks, which gave rise to the modern day internet. It helps in grouping the data in digital communication to packets. It helps by introducing the protocol suit. Now it is the foundation for the internet we are using now.

Given below is the structure of ARPANET –





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Outside the subnet, software was also needed, namely, the host end of the host-IMP connection, the host-host protocol, and the application software. It soon became clear that BBN felt that when it had accepted a message on a host-IMP wire and placed it on the host-IMP wire at the destination, its job was done.

The subnet would consist of minicomputers called **IMPs (Interface Message Processors)** connected by 56-kbps transmission lines.

For high reliability, each IMP would be connected to at least two other IMPs. The subnet was to be a datagram subnet, so if some lines and IMPs were destroyed, messages could be automatically re-routed along alternative paths.

Each node of the network was to consist of an IMP and a host, in the same room, connected by a short wire.

A host could send messages of up to 8063 bits to its IMP, which would then break these up into packets of at most 1008 bits and forward them independently toward the destination. Each packet was received in its entirety before being forwarded, so the subnet was the first electronic store-and-forward packet-switching network.

The software was split into two parts: subnet and host. The subnet software consisted of the IMP end of the host-IMP connection, the IMP-IMP protocol, and a source IMP to destination IMP protocol designed to improve reliability. The original ARPANET design is shown in Fig.

Outside the subnet, software was also needed, namely, the host end of the host-IMP connection, the host-host protocol, and the application software.

It soon became clear that BBN was of the opinion that when it had accepted a message on a host-IMP wire and placed it on the host-IMP wire at the destination, its job was done.

In addition to helping the fledgling ARPANET grow, ARPA also funded research on the use of satellite networks and mobile packet radio networks.

Internet:

The Internet is not really a network at all, but a vast collection of different networks that use certain common protocols and provide certain common services. It is an unusual system in that it was not planned by anyone and is not controlled by anyone. To better understand it, let us start from the beginning and see how it has developed and why.



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Course Code:AS22- 05PC05

Academic Year: 2023-24

Section: A, B & C

LECTURE NOTES

Unit 2
DATA LINK
LAYER

2.1. Data link layer design issues:

- The data link layer uses the services of the physical layer to send and receive bits over communication channels.
- **It has a number of functions, including:**
 - **1. Providing a well-defined service interface to the network layer.**
 - **2. Dealing with transmission errors.**
 - **3. Regulating the flow of data so that slow receivers are not swamped by fast senders.**
- To accomplish these goals, the data link layer takes the packets it gets from the network layer and encapsulates them into frames for transmission.
- Each frame contains a frame header, a payload field for holding the packet, and a frame trailer, as illustrated in Fig.
- Frame management forms the heart of what the data link layer does.

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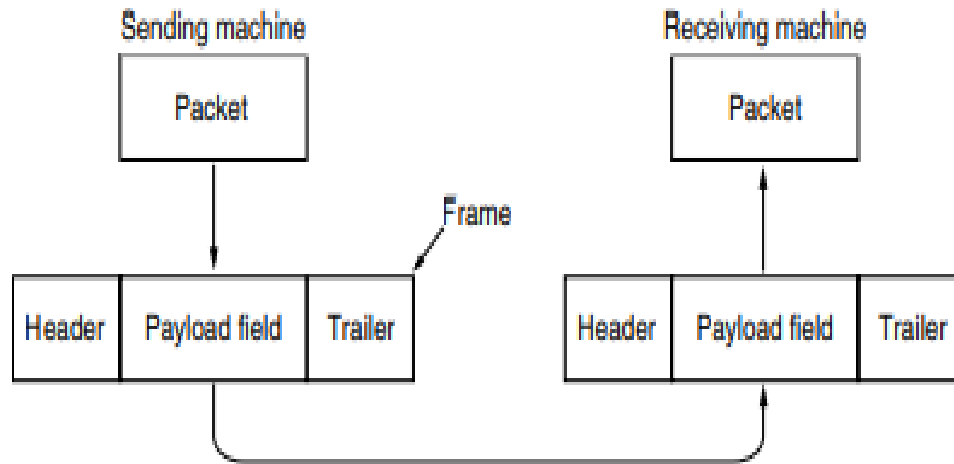


Figure 3-1. Relationship between packets and frames.

2.1.1. Services provided to the network layer:

- The function of the data link layer is to provide services to the network layer.
- The principal service is transferring data from the network layer on the source machine to the network layer on the destination machine.
- On the source machine is an entity, call it a process, in the network layer that hands some bits to the data link layer for transmission to the destination.
- The job of the data link layer is to transmit the bits to the destination machine so they can be handed over to the network layer there, as shown in Fig.

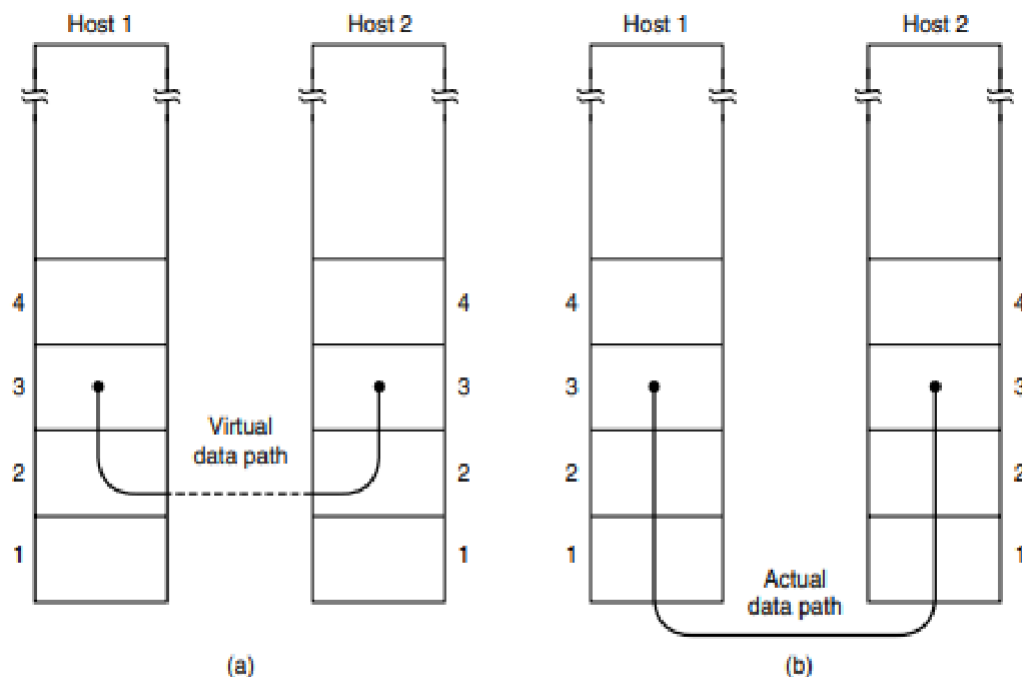


Figure 3-2. (a) Virtual communication. (b) Actual communication.

-
- The data link layer can be designed to offer various services. The actual services that are offered vary from protocol to protocol.
- Three reasonable possibilities that we will consider in turn are:
 - 1. Unacknowledged connectionless service.
 - 2. Acknowledged connectionless service.
 - 3. Acknowledged connection-oriented service.
- **Unacknowledged connectionless service** consists of having the source machine send independent frames to the destination machine without having the destination machine acknowledge them.
- If a frame is lost due to noise on the line no attempt is made to detect the loss or recover from it in the data link layer. This class of service is appropriate when the error rate is very low, so recovery is left to higher layers.
- **Acknowledged connectionless service** : When this service is offered, there are still no logical connections used, but each frame sent is individually acknowledged.

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- In this way, the sender knows whether a frame has arrived correctly or been lost. If it has not arrived within a specified time interval, it can be sent again.
- This service is useful over unreliable channels, such as wireless systems. 802.11 (Wi-Fi) is a good example of this class of service.
- The most sophisticated service the data link layer can provide to the network layer is **connection-oriented service**.
- With this service, the source and destination machines establish a connection before any data are transferred.
- Each frame sent over the connection is numbered, and the data link layer guarantees that each frame sent is indeed received.
- Furthermore, it guarantees that each frame is received exactly once and that all frames are received in the right order.
- Connection-oriented service thus provides the network layer processes with the equivalent of a reliable bit stream.
- When connection-oriented service is used, transfers go through three distinct phases. In the first phase, the connection is established by having both sides initialize variables and counters needed to keep track of which frames have been received and which ones have not. In the second phase, one or more frames are actually transmitted. In the third and final phase, the connection is released, freeing up the variables, buffers, and other resources used to maintain the connection.

2.1.2. Framing:

- To provide service to the network layer, the data link layer must use the service provided to it by the physical layer.
- What the physical layer does is accept a raw bit stream and attempt to deliver it to the destination.
- If the channel is noisy, as it is for most wireless and some wired links, the physical layer will add some redundancy to its signals to reduce the bit error rate to a tolerable level.

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- However, the bit stream received by the data link layer is not guaranteed to be error free.
- Some bits may have different values and the number of bits received may be less than, equal to, or more than the number of bits transmitted.
- It is up to the data link layer to detect and, if necessary, correct errors. **The usual approach is for the data link layer to break up the bit stream into discrete frames, compute a short token called a checksum for each frame, and include the checksum in the frame when it is transmitted.**
- **When a frame arrives at the destination, the checksum is recomputed. If the newly computed checksum is different from the one contained in the frame, the data link layer knows that an error has occurred and takes steps to deal with it (e.g., discarding the bad frame and possibly also sending back an error report).**
- Breaking up the bit stream into frames is more difficult than it at first appears.
- A good design must make it easy for a receiver to find the start of new frames while using little of the channel bandwidth.
- **We will look at four methods:**
 - **1. Byte count.**
 - **2. Flag bytes with byte stuffing.**
 - **3. Flag bits with bit stuffing.**
 - **4. Physical layer coding violations.**
- **1. Byte count :** The first framing method uses a field in the header to specify the number of bytes in the frame.
- When the data link layer at the destination sees the byte count, it knows how many bytes follow and hence where the end of the frame is.
- This technique is shown in Fig3-3(a) for four small example frames of sizes 5, 5, 8, and 8 bytes, respectively.
- The trouble with this algorithm is that the count can be garbled by a transmission error.
- For example, if the byte count of 5 in the second frame of Fig. 3-3(b) becomes a 7 due to a single bit flip, the destination will get out of synchronization.

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- It will then be unable to locate the correct start of the next frame.
- Even if the checksum is incorrect so the destination knows that the frame is bad, it still has no way of telling where the next frame starts.

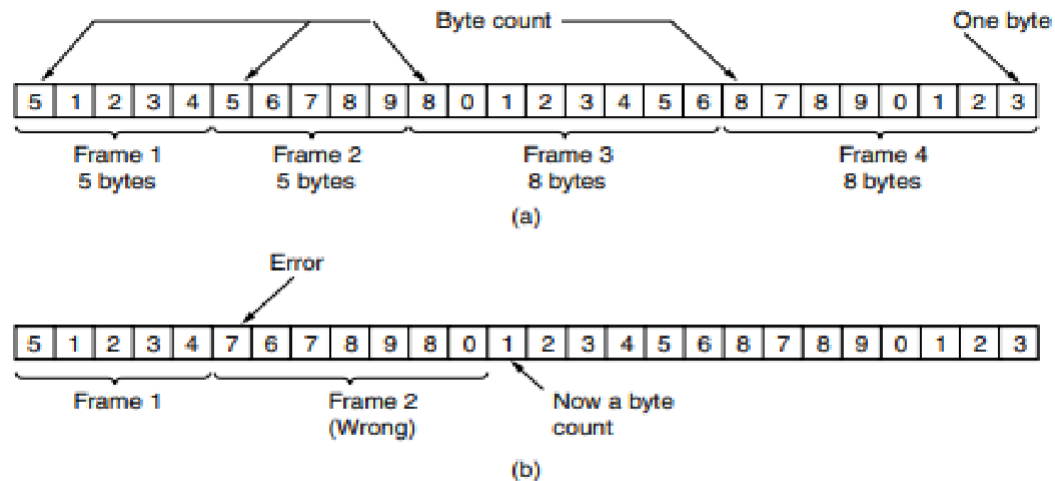


Figure 3-3. A byte stream. (a) Without errors. (b) With one error.



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- The second framing method gets around the problem of resynchronization after an error by having each frame start and end with special bytes.
- Often the same byte, called a flag byte, is used as both the starting and ending delimiter. This byte is shown in Fig. 3-4(a) as FLAG.
- Two consecutive flag bytes indicate the end of one frame and the start of the next.
- Thus, if the receiver ever loses synchronization it can just search for two flag bytes to find the end of the current frame and the start of the next frame.
- It may happen that the flag byte occurs in the data, especially when binary data such as photographs or songs are being transmitted.
- This situation would interfere with the framing. One way to solve this problem is to have the sender's data link layer insert a special escape byte (ESC) just before each "accidental" flag byte in the data.
- Thus, a framing flag byte can be distinguished from one in the data by the absence or presence of an escape byte before it.

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- The data link layer on the receiving end removes the escape bytes before giving the data to the network layer. This technique is called byte stuffing.
- What happens if an escape byte occurs in the middle of the data? The answer is that it, too, is stuffed with an escape byte.
- At the receiver, the first escape byte is removed, leaving the data byte that follows it (which might be another escape byte or the flag byte).

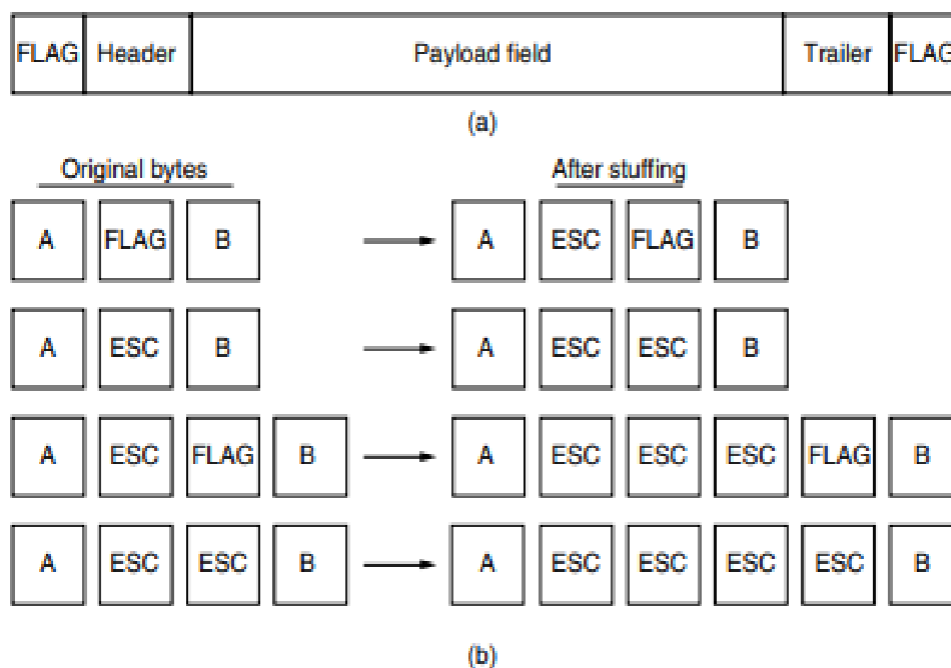


Figure 3-4. (a) A frame delimited by flag bytes. (b) Four examples of byte sequences before and after byte stuffing.



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- **3.Flag bits with bit stuffing** : Each frame begins and ends with a special bit pattern, 01111110 or 0x7E in hexadecimal.
- This pattern is a flag byte. Whenever the sender's data link layer encounters five consecutive 1s in the data, it automatically stuffs a 0 bit into the outgoing bit stream.
- This bit stuffing is analogous to byte stuffing, in which an escape byte is stuffed into the outgoing character stream before a flag byte in the data.

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- It also ensures a minimum density of transitions that help the physical layer maintain synchronization.
- When the receiver sees five consecutive incoming 1 bits, followed by a 0 bit, it automatically destuffs (i.e., deletes) the 0 bit.
- Just as byte stuffing is completely transparent to the network layer in both com- puters, so is bit stuffing.
- If the user data contain the flag pattern, 01111110, this flag is transmitted as 011111010 but stored in the receiver's memory as 01111110.
- Figure 3-5 gives an example of bit.
- With bit stuffing, the boundary between two frames can be unambiguously rec- ognized by the flag pattern.
- Thus, if the receiver loses track of where it is, all it has to do is scan the input for flag sequences, since they can only occur at frame boundaries and never within the data.

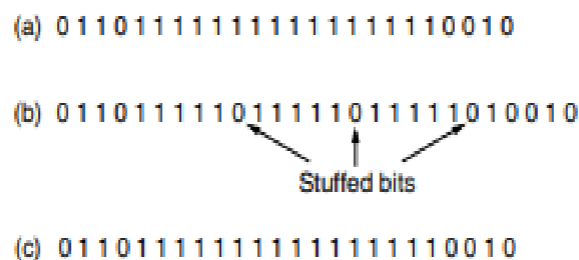


Figure 3-5. Bit stuffing. (a) The original data. (b) The data as they appear on the line. (c) The data as they are stored in the receiver's memory after destuffing.



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- **4.Physical layer coding violations:** The last method of framing is to use a shortcut from the physical layer. This redundancy means that some signals will not occur in regular data. For example, in the 4B/5B line code 4 data bits are mapped to 5 signal bits to ensure sufficient bit transitions.
- This means that 16 out of the 32 signal possibilities are not used. We can use some reserved signals to indicate the start and end of frames.
- In effect, we are using “coding violations” to delimit frames.



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- The beauty of this scheme is that, because they are reserved signals, it is easy to find the start and end of frames and there is no need to stuff the data.

2.1.3. Error Control:

- How to make sure all frames are eventually delivered to the network layer at the destination and in the proper order.
- For unacknowledged connectionless service it might be fine if the sender just kept outputting frames without regard to whether they were arriving properly.
- But for reliable, connection-oriented service it would not be fine at all. The usual way to ensure reliable delivery is to provide the sender with some feedback about what is happening at the other end of the line.
- If the sender receives a positive acknowledgement about a frame, it knows the frame has arrived safely.
- On the other hand, a negative acknowledgement means that something has gone wrong and the frame must be transmitted again.
- An additional complication comes from the possibility that hardware troubles may cause a frame to vanish completely (e.g., in a noise burst).
- Similarly, if the acknowledgement frame is lost, the sender will not know how to proceed.
- It should be clear that a protocol in which the sender transmits a frame and then waits for an acknowledgement, positive or negative, will hang forever if a frame is ever lost due to, for example, malfunctioning hardware or a faulty communication channel.
- This possibility is dealt with by introducing timers into the data link layer.
- When the sender transmits a frame, it generally also starts a timer. The timer is set to expire after an interval long enough for the frame to reach the destination, be processed there, and have the acknowledgement propagate back to the sender. Normally, the frame will be correctly received and the acknowledgement will get back before the timer runs out, in which case the timer will be canceled.

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- However, if either the frame or the acknowledgement is lost, the timer will go off, alerting the sender to a potential problem.
- The obvious solution is to just transmit the frame again.
- However, when frames may be transmitted multiple times there is a danger that the receiver will accept the same frame two or more times and pass it to the network layer more than once.
- To prevent this from happening, it is generally necessary to assign sequence numbers to outgoing frames, so that the receiver can distinguish retransmissions from originals.
- The whole issue of managing the timers and sequence numbers so as to ensure that each frame is ultimately passed to the network layer at the destination exactly once, no more and no less, is an important part of the duties of the data link layer (and higher layers).

2.1.4. Flow Control:

- Another important design issue that occurs in the data link layer (and higher layers as well) is what to do with a sender that systematically wants to transmit frames faster than the receiver can accept them. This situation can occur when the sender is running on a fast, powerful computer and the receiver is running on a slow, low-end machine.
- Two approaches are commonly used. In the first one, **feedback-based flow control**, the receiver sends back information to the sender giving it permission to send more data, or at least telling the sender how the receiver is doing.
- In the second one, **rate-based flow control**, the protocol has a built-in mechanism that limits the rate at which senders may transmit data, without using feedback from the receiver.
- **For example, hardware implementations of the link layer as NICs (Network Interface Cards) are sometimes said to run at "wire speed," meaning that they can handle frames as fast as they can arrive on the link.**

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- A common situation is when a smart phone requests a Web page from a far more powerful server, which then turns on the fire hose and blasts the data at the poor helpless phone until it is completely swamped.

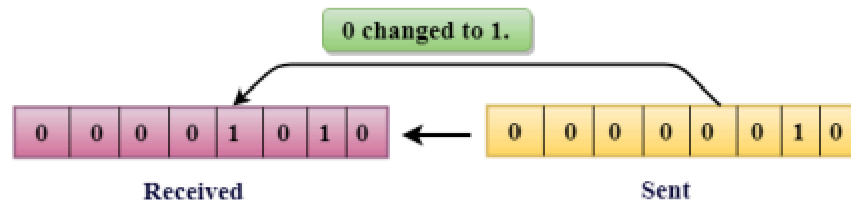
2.2 .Types Of Errors:

Errors can be classified into two categories:

- Single-Bit Error
- Burst Error

Single-Bit Error:

The only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.



In the above figure, the message which is sent is corrupted as single-bit, i.e., 0 bit is changed to 1.

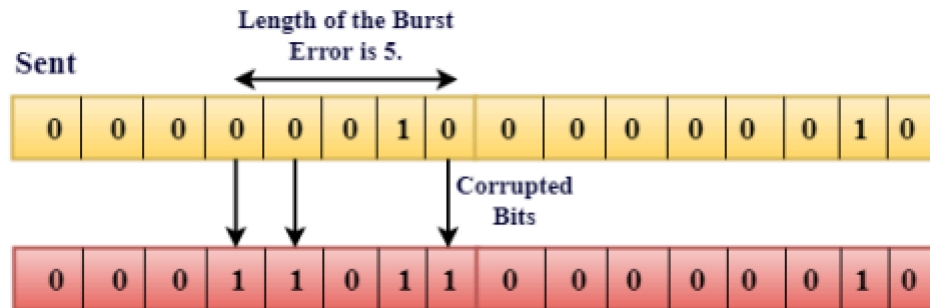
Single-Bit Error does not appear more likely in Serial Data Transmission. For example, Sender sends the

Burst Error:

The two or more bits are changed from 0 to 1 or from 1 to 0 is known as Burst Error.

The Burst Error is determined from the first corrupted bit to the last corrupted bit.

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Received

The duration of noise in Burst Error is more than the duration of noise in Single-Bit.

Burst Errors are most likely to occur in Serial Data Transmission.

The number of affected bits depends on the duration of the noise and data rate.

2.3. Elementary Data Link Protocols:

- To introduce the subject of protocols, we will begin by looking at three proto- cols of increasing complexity.
- Before we look at the protocols, it is useful to make explicit some of the as- sumptions underlying the model of communication.
- To start with, we assume that the physical layer, data link layer, and network layer are independent processes that communicate by passing messages back and forth.
- A common implementation is shown in Fig. 3-10. The physical layer process and some of the data link layer process run on dedicate hardware called a NIC (Network Interface Card).
- The rest of the link layer process and the network layer process run on the main CPU as part of the operating system, with the software for the link layer process often taking the form of a device driver. However, other implementations are also possible (e.g., three processes offloaded to dedicated hardware called a



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network accelerator, or three processes running on the main CPU on a software-defined ratio).

- Actually, the preferred implementation changes from decade to decade with technology trade-offs.
- In any event, treating the three layers as separate processes makes the discussion conceptually cleaner and also serves to emphasize the independence of the layers.
- Another key assumption is that machine A wants to send a long stream of data to machine B, using a reliable, connection-oriented service. Later, we will consider the case where B also wants to send data to A simultaneously.
- A is assumed to have an infinite supply of data ready to send and never has to wait for data to be produced. Instead, when A's data link layer asks for data, the network layer is always able to comply immediately. (This restriction, too, will be dropped later.)
- We also assume that machines do not crash. That is, these protocols deal with communication errors, but not the problems caused by computers crashing and rebooting.
- As far as the data link layer is concerned, the packet passed across the interface to it from the network layer is pure data, whose every bit is to be delivered to the destination's network layer.
- The fact that the destination's network layer may interpret part of the packet

as a header is of no concern to the data link layer.

- When the data link layer accepts a packet, it encapsulates the packet in a frame by adding a data link header and trailer to it (see Fig. 3-1).
- Thus, a frame consists of an embedded packet, some control information (in the header), and a checksum (in the trailer). The frame is then transmitted to the data link layer on the other machine.
- We will assume that there exist suitable library procedures to physical layer to send a frame and from physical layer to receive a frame.
- These procedures compute and append or check the checksum (which is usually done in hardware).

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- It just sits around waiting for something to happen. In the example protocols throughout this chapter we will indicate that the data link layer is waiting for something to happen by

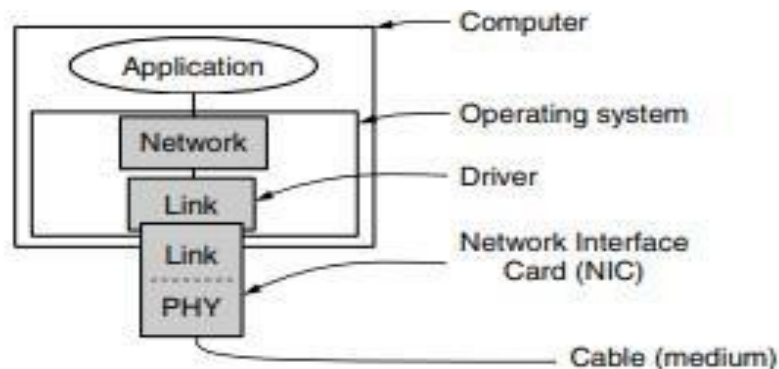


Figure 3-10. Implementation of the physical, data link, and network layers.

the procedure call

wait for event(&event).

- This procedure only returns when something has happened (e.g., a frame has arrived).
- Upon return, the variable event tells what happened. The set of possible events differs for the various protocols to be described and will be de- fined separately for each protocol.
- When a frame arrives at the receiver, the checksum is recomputed. If the checksum in the frame is incorrect (i.e., there was a transmission error), the data link layer is so informed (event = cksum err).
- If the inbound frame arrived undamaged, the data link layer is also in- formed (event = frame arrival) so that it can acquire the frame for inspec- tion using from physical layer.
- As soon as the receiving data link layer has acquired an undamaged frame, it checks the control information in the header, and, if everything is all right, passes the packet portion to the network layer.

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- Under no circumstances is a frame header ever given to a network layer.
- There is a good reason why the network layer must never be given any part of the frame header: to keep the network and data link protocols completely separate.
- As long as the network layer knows nothing at all about the data link protocol or the frame format, these things can be changed without re-quiring changes to the network layer's software.
- This happens whenever a new NIC is installed in a computer. Providing a rigid interface between the network and data link layers greatly simplifies the design task because communication protocols in different layers can evolve independently.
- Figure 3-11 shows some declarations (in C) common to many of the pro- tocols to be discussed later.
- Five data structures are defined there: boolean, seq nr, packet, frame kind, and frame.
- A boolean is an enumerated type and can take on the values true and false.
- A seq nr is a small integer used to number the frames so that we can tell them apart.
- These sequence numbers run from 0 up to and including MAX SEQ, which is defined in each protocol needing it. A packet is the unit of information exchanged between the network layer and the data link layer on the same machine, or between network layer peers.
- In our model it always contains MAX PKT bytes, but more realistically it would be of variable length.
- A frame is composed of four fields: kind, seq, ack, and info, the first three of which contain control information and the last of which may contain actual data to be transferred.
- These control fields are collectively called the frame header. The kind field tells whether there are any data in the frame, because some of the

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protocols distinguish frames containing only control information from those containing data as well.

- The seq and ack fields are used for sequence numbers and acknowledge- ments, respectively; their use will be described in more detail later. The info field of a data frame contains a single packet; the info field of a con- trol frame is not used.
- A more realistic implementation would use a variablelength info field, omitting it altogether for control frames.
- Again, it is important to understand the relationship between a packet and a frame.
- The network layer builds a packet by taking a message from the transport layer and adding the network layer header to it.
- This packet is passed to the data link layer for inclusion in the info field of an outgoing frame.
- When the frame arrives at the destination, the data link layer extracts the packet from the frame and passes the packet to the network layer.
- In this manner, the network layer can act as though machines can ex- change packets directly. A number of procedures are also listed in Fig. 3- 11.
- These are library routines whose details are implementation dependent and whose inner workings will not concern us further in the following discussions.
- The procedure wait for event sits in a tight loop waiting for something to happen, as mentioned earlier.
- The procedures to network layer and from network layer are used by the data link layer to pass packets to the network layer and accept packets from the network layer, respectively.
- Note that from physical layer and to physical layer pass frames between the data link layer and the physical layer.



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- In other words, to network layer and from network layer deal with the interface between layers 2 and 3, whereas from physical layer and to physical layer deal with the interface between layers 1 and 2.
- In most of the protocols, we assume that the channel is unreliable and loses entire frames upon occasion.
- To be able to recover from such calamities, the sending data link layer must start an internal timer or clock whenever it sends a frame. If no re- ply has been received within a certain predetermined time interval, the clock times out and the data link layer receives an interrupt signal. In our protocols this is handled by allowing the procedure wait for event to re- turn event = timeout.
- The procedures start timer and stop timer turn the timer on and off, re- spectively. Timeout events are possible only when the timer is running and before stop timer is called.
- It is explicitly permitted to call start timer while the timer is running; such a call simply resets the clock to cause the next timeout after a full timer interval has elapsed (unless it is reset or turned off).
- The procedures start ack timer and stop ack timer control an auxiliary timer used to generate acknowledgements under certain conditions.
- The procedures enable network layer and disable network layer are used in the more sophisticated protocols, where we no longer assume that the network layer always has packets to send.
- When the data link layer enables the network layer, the network layer is then permitted to interrupt when it has a packet to be sent. We indicate this with event = network layer ready.
- When the network layer is disabled, it may not cause such events. By being careful about when it enables and disables its network layer, the data link layer can prevent the network layer from swamping it with pack- ets for which it has no buffer space.
- Frame sequence numbers are always in the range 0 to MAX SEQ (inclu- sive), where MAX SEQ is different for the different protocols.

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```
#define MAX_PKT 1024                                /* determines packet size in bytes */

typedef enum {false, true} boolean;                  /* boolean type */
typedef unsigned int seq_nr;                          /* sequence or ack numbers */
typedef struct {unsigned char data[MAX_PKT];} packet; /* packet definition */
typedef enum {data, ack, nak} frame_kind;             /* frame_kind definition */

typedef struct {
    frame_kind kind;
    seq_nr seq;
    seq_nr ack;
    packet info;
} frame;                                              /* frames are transported in this layer */
                                                    /* what kind of frame is it? */
                                                    /* sequence number */
                                                    /* acknowledgement number */
                                                    /* the network layer packet */

/* Wait for an event to happen; return its type in event. */
void wait_for_event(event_type *event);

/* Fetch a packet from the network layer for transmission on the channel. */
void from_network_layer(packet *p);

/* Deliver information from an inbound frame to the network layer. */
void to_network_layer(packet *p);

/* Go get an inbound frame from the physical layer and copy it to r. */
void from_physical_layer(frame *r);

/* Pass the frame to the physical layer for transmission. */
void to_physical_layer(frame *s);

/* Start the clock running and enable the timeout event. */
void start_timer(seq_nr k);

/* Stop the clock and disable the timeout event. */
void stop_timer(seq_nr k);

/* Start an auxiliary timer and enable the ack_timeout event. */
void start_ack_timer(void);

/* Stop the auxiliary timer and disable the ack_timeout event. */
void stop_ack_timer(void);

/* Allow the network layer to cause a network_layer_ready event. */
void enable_network_layer(void);

/* Forbid the network layer from causing a network_layer_ready event. */
void disable_network_layer(void);

/* Macro inc is expanded in-line: increment k circularly. */
#define inc(k) if (k < MAX_SEQ) k = k + 1; else k = 0
```

Figure 3-11. Some definitions needed in the protocols to follow. These definitions are located in the file *protocol.h*.

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2.3.1. Utopian Simplex Protocol:

- Data are transmitted in one direction only. Both the transmitting and receiving network layers are always ready.
- Processing time can be ignored. Infinite buffer space is available.
- And best of all, the communication channel between the data link layers never damages or loses frames.
- This thoroughly unrealistic protocol, which we will nickname "Utopia," is simply to show the basic structure on which we will build. Its implementation is shown in Fig. 3-12.
- The protocol consists of two distinct procedures, a sender and a receiver. The sender runs in the data link layer of the source machine, and the receiver runs in the data link layer of the destination machine.
- No sequence numbers or acknowledgements are used here, so MAX SEQ is not needed. The only event type possible is frame arrival (i.e., the arrival of an un- damaged frame).
- The sender is in an infinite while loop just pumping data out onto the line as fast as it can.
- The body of the loop consists of three actions: go fetch a packet from the (al- ways obliging) network layer, construct an outbound frame using the variable s, and send the frame on its way.
- Only the info field of the frame is used by this protocol, because the other fields have to do with error and flow control and there are no errors or flow control restrictions here.
- The receiver is equally simple. Initially, it waits for something to happen, the only possibility being the arrival of an undamaged frame. Eventually, the frame arrives and the procedure wait for event returns, with event set to frame arrival (which is ignored anyway).
- The call to from physical layer removes the newly arrived frame from the hard- ware buffer and puts it in the variable r, where the receiver code can get at it.

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- Finally, the data portion is passed on to the network layer, and the data link layer settles back to wait for the next frame, effectively suspending itself until the frame arrives.

/* Protocol 1 (Utopia) provides for data transmission in one direction only, from sender to receiver. The communication channel is assumed to be error free and the receiver is assumed to be able to process all the input infinitely quickly. Consequently, the sender just sits in a loop pumping data out onto the line as fast as it can. */

```
typedef enum {frame_arrival} event_type;
#include "protocol.h"

void sender1(void)
{
    frame s;                /* buffer for an outbound frame */
    packet buffer;          /* buffer for an outbound packet */

    while (true) {
        from_network_layer(&buffer); /* go get something to send */
        s.info = buffer;           /* copy it into s for transmission */
        to_physical_layer(&s);     /* send it on its way */
    }
    /* Tomorrow, and tomorrow, and tomorrow,
       Creeps in this petty pace from day to day
       To the last syllable of recorded time.
       – Macbeth, V, v */
}

void receiver1(void)
{
    frame r;
    event_type event;        /* filled in by wait, but not used here */

    while (true) {
        wait_for_event(&event);    /* only possibility is frame_arrival */
        from_physical_layer(&r);   /* go get the inbound frame */
        to_network_layer(&r.info); /* pass the data to the network layer */
    }
}
```

Activate Win

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Figure 3-12. A utopian simplex protocol.

- The utopia protocol is unrealistic because it does not handle either flow control or error correction.
- Its processing is close to that of an unacknowledged connectionless service that relies on higher layers to solve these problems, though even an unacknowledged connectionless service would do some error detection.

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2.3.2. Simplex Stop – and –Wait Protocol for an Error – Free Channel:

- Now we will tackle the problem of preventing the sender from flooding the receiver with frames faster than the latter is able to process them.
- This situation can easily happen in practice so being able to prevent it is of great importance.
- The communication channel is still assumed to be error free, however, and the data traffic is still simplex.
- One solution is to build the receiver to be powerful enough to process a continuous stream of back-to-back frames (or, equivalently, define the link layer to be slow enough that the receiver can keep up).
- It must have sufficient buffering and processing abilities to run at the line rate and must be able to pass the frames that are received to the network layer quickly enough. However, this is a worst-case solution.
- It requires dedicated hardware and can be wasteful of resources if the utilization of the link is mostly low. Moreover, it just shifts the problem of dealing with a sender that is too fast elsewhere; in this case to the network layer.
- A more general solution to this problem is to have the receiver provide feedback to the sender.
- After having passed a packet to its network layer, the receiver sends a little dummy frame back to the sender which, in effect, gives the sender permission to transmit the next frame.
- After having sent a frame, the sender is required by the protocol to bide its time until the little dummy (i.e., acknowledgement) frame arrives.
- This delay is a simple example of a flow control protocol. **Protocols in which the sender sends one frame and then waits for an acknowledgement before proceeding are called stop-and-wait.**
- Figure 3-13 gives an example of a simplex stop-and-wait protocol. Although data traffic in this example is simplex, going only from the sender to the receiver, frames do travel in both directions.
- Consequently, the communication channel between the two data link layers needs to be capable of bidirectional information transfer.



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- However, this protocol entails a strict alternation of flow: first the sender sends a frame, then the receiver sends a frame, then the sender sends another frame, then the receiver sends another one, and so on. A halfduplex physical channel would suffice here.
- As in protocol 1, the sender starts out by fetching a packet from the network layer, using it to construct a frame, and sending it on its way. But now, unlike in protocol 1, the sender must wait until an acknowledgement frame arrives before looping back and fetching the next packet from the network layer.
- The sending data link layer need not even inspect the incoming frame as there is only one possibility. The incoming frame is always an acknowledgement.
- The only difference between receiver1 and receiver2 is that after delivering a packet to the network layer, receiver2 sends an acknowledgement frame back to the sender before entering the wait loop again.
- Because only the arrival of the frame back at the sender is important, not its contents, the receiver need not put any particular information in it.

```
typedef enum {frame_arrival} event_type;
#include "protocol.h"

void sender2(void)
{
    frame s;                /* buffer for an outbound frame */
    packet buffer;          /* buffer for an outbound packet */
    event_type event;       /* frame_arrival is the only possibility */

    while (true) {
        from_network_layer(&buffer); /* go get something to send */
        s.info = buffer;             /* copy it into s for transmission */
        to_physical_layer(&s);       /* bye-bye little frame */
        wait_for_event(&event);     /* do not proceed until given the go ahead */
    }
}

void receiver2(void)
{
    frame r, s;              /* buffers for frames */
    event_type event;       /* frame_arrival is the only possibility */
    while (true) {
        wait_for_event(&event); /* only possibility is frame_arrival */
        from_physical_layer(&r); /* go get the inbound frame */
        to_network_layer(&r.info); /* pass the data to the network layer */
        to_physical_layer(&s);     /* send a dummy frame to awaken sender */
    }
}
```

Figure 3-13. A simplex stop-and-wait protocol.

2.3.3. A Simplex Stop-and-Wait Protocol for a Noisy Channel:

- Now let us consider the normal situation of a communication channel that makes errors. Frames may be either damaged or lost completely.
- However, we assume that if a frame is damaged in transit, the receiver hardware will detect this when it computes the checksum.
- If the frame is damaged in such a way that the checksum is nevertheless correct—an unlikely occurrence—this protocol (and all other protocols) can fail (i.e., deliver an incorrect packet to the network layer).
- At first glance it might seem that a variation of protocol 2 would work: adding a timer.
- The sender could send a frame, but the receiver would only send an acknowledgement frame if the data were correctly received.

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- If a damaged frame arrived at the receiver, it would be discarded. After a while the sender would time out and send the frame again.
- This process would be repeated until the frame finally arrived intact. This scheme has a fatal flaw in it though.
- Think about the problem and try to discover what might go wrong before reading further.
- To see what might go wrong, remember that the goal of the data link layer is to provide error-free, transparent communication between network layer processes.
- The network layer on machine A gives a series of packets to its data link layer, which must ensure that an identical series of packets is delivered to the network layer on machine B by its data link layer.
- In particular, the network layer on B has no way of knowing that a packet has been lost or duplicated, so the data link layer must guarantee that no combination of transmission errors, however unlikely, can cause a duplicate packet to be delivered to a network layer.

Consider the following scenario:

1. The network layer on A gives packet 1 to its data link layer. The packet is correctly received at B and passed to the network layer on B. B sends an acknowledgement frame back to A.
2. The acknowledgement frame gets lost completely. It just never arrives at all. Life would be a great deal simpler if the channel mangled and lost only data frames and not control frames, but sad to say, the channel is not very discriminating.
3. The data link layer on A eventually times out. Not having received an acknowledgement, it (incorrectly) assumes that its data frame was lost or damaged and sends the frame containing packet 1 again.
4. The duplicate frame also arrives intact at the data link layer on B and is unwittingly passed to the network layer there. If A is sending a file to B, part of the file will be



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duplicated (i.e., the copy of the file made by B will be incorrect and the error will not have been detected). In other words, the protocol will fail.

- Clearly, what is needed is some way for the receiver to be able to distinguish a frame that it is seeing for the first time from a retransmission.
- The obvious way to achieve this is to have the sender put a sequence number in the header of each frame it sends.
- Then the receiver can check the sequence number of each arriving frame to see if it is a new frame or a duplicate to be discarded.
- Since the protocol must be correct and the sequence number field in the header is likely to be small to use the link efficiently, the question arises: what is the minimum number of bits needed for the sequence number? The header might provide 1 bit, a few bits, 1 byte, or multiple bytes for a sequence number depending on the protocol.
- The important point is that it must carry sequence numbers that are large enough for the protocol to work correctly, or it is not much of a protocol. The only ambiguity in this protocol is between a frame, m , and its direct successor, $m + 1$.
- If frame m is lost or damaged, the receiver will not acknowledge it, so the sender will keep trying to send it.
- Once it has been correctly received, the receiver will send an acknowledgement to the sender.
- It is here that the potential trouble crops up. Depending upon whether the acknowledgement frame gets back to the sender correctly or not, the sender may try to send m or $m + 1$.
- At the sender, the event that triggers the transmission of frame $m + 1$ is the arrival of an acknowledgement for frame m .
- But this situation implies that $m - 1$ has been correctly received, and further- more that its acknowledgement has also been correctly received by the sender.



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Otherwise, the sender would not have begun with m , let alone have been considering $m + 1$.

- As a consequence, the only ambiguity is between a frame and its immediate predecessor or successor, not between the predecessor and successor themselves.
- A 1-bit sequence number (0 or 1) is therefore sufficient. At each instant of time, the receiver expects a particular sequence number next. When a frame containing the correct sequence number arrives, it is accepted and passed to the network layer, then acknowledged.
- Then the expected sequence number is incremented modulo 2 (i.e., 0 becomes 1 and 1 becomes 0).
- Any arriving frame containing the wrong sequence number is rejected as a duplicate. However, the last valid acknowledgement is repeated so that the sender can eventually discover that the frame has been received.
- An example of this kind of protocol is shown in Fig. 3-14. **Protocols in which the sender waits for a positive acknowledgement before advancing to the next data item are often called ARQ (Automatic Repeat reQuest) or PAR (Positive Acknowledgement with Retransmission).**
- Like protocol 2, this one also transmits data only in one direction.
- Protocol 3 differs from its predecessors in that both sender and receiver have a variable whose value is remembered while the data link layer is in the wait state.
- The sender remembers the sequence number of the next frame to send in next frame to send; the receiver remembers the sequence number of the next frame expected in frame expected. Each protocol has a short initialization phase before entering the infinite loop.
- After transmitting a frame, the sender starts the timer running.
- If it was already running, it will be reset to allow another full timer interval. The interval should be chosen to allow enough time for the frame to get to the receiver, for the receiver to process it in the worst case, and for the acknowledgement frame to propagate back to the sender.

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- Only when that interval has elapsed is it safe to assume that either the transmitted frame or its acknowledgement has been lost, and to send a duplicate.
- If the timeout interval is set too short, the sender will transmit unnecessary frames.
- While these extra frames will not affect the correctness of the protocol, they will hurt performance.
- After transmitting a frame and starting the timer, the sender waits for something exciting to happen.
- Only three possibilities exist: an acknowledgement frame arrives undamaged, a damaged acknowledgement frame staggers in, or the timer expires. If a valid acknowledgement comes in, the sender fetches the next packet from its network layer and puts it in the buffer, overwriting the previous packet.
- It also advances the sequence number. If a damaged frame arrives or the timer expires, neither the buffer nor the sequence number is changed so that a duplicate can be sent.
- In all cases, the contents of the buffer (either the next packet or a duplicate) are then sent.
- When a valid frame arrives at the receiver, its sequence number is checked to see if it is a duplicate.
- If not, it is accepted, passed to the network layer, and an acknowledgement is generated.
- Duplicates and damaged frames are not passed to the network layer, but they do cause the last correctly received frame to be acknowledged to signal the sender to advance to the next frame or retransmit a damaged frame.

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```

/* Protocol 3 (PAR) allows unidirectional data flow over an unreliable channel. */
#define MAX_SEQ 1 /* must be 1 for protocol 3 */
typedef enum {frame_arrival, cksum_err, timeout} event_type;
#include "protocol.h"

void sender3(void)
{
    seq_nr next_frame_to_send; /* seq number of next outgoing frame */
    frame s; /* scratch variable */
    packet buffer; /* buffer for an outbound packet */
    event_type event;

    next_frame_to_send = 0; /* initialize outbound sequence numbers */
    from_network_layer(&buffer); /* fetch first packet */
    while (true) {
        s.info = buffer; /* construct a frame for transmission */
        s.seq = next_frame_to_send; /* insert sequence number in frame */
        to_physical_layer(&s); /* send it on its way */
        start_timer(s.seq); /* if answer takes too long, time out */
        wait_for_event(&event); /* frame_arrival, cksum_err, timeout */
        if (event == frame_arrival) {
            from_physical_layer(&s); /* get the acknowledgement */
            if (s.ack == next_frame_to_send) {
                stop_timer(s.ack); /* turn the timer off */
                from_network_layer(&buffer); /* get the next one to send */
                inc(next_frame_to_send); /* invert next_frame_to_send */
            }
        }
    }
}

```

```

void receiver3(void)
{
    seq_nr frame_expected;
    frame r, s;
    event_type event;

    frame_expected = 0;
    while (true) {
        wait_for_event(&event); /* possibilities: frame_arrival, cksum_err */
        if (event == frame_arrival) { /* a valid frame has arrived */
            from_physical_layer(&r); /* go get the newly arrived frame */
            if (r.seq == frame_expected) { /* this is what we have been waiting for */
                to_network_layer(&r.info); /* pass the data to the network layer */
                inc(frame_expected); /* next time expect the other sequence nr */
            }
            s.ack = 1 - frame_expected; /* tell which frame is being acked */
            to_physical_layer(&s); /* send acknowledgement */
        }
    }
}

```

Figure 3-14. A positive acknowledgement with retransmission protocol.



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2.4. Error Correction in Data link Layer:

- Data-link layer uses error control techniques to ensure that frames, i.e. bit streams of data, are transmitted from the source to the destination with a certain extent of accuracy.

Errors:

- **When bits are transmitted over the computer network, they are subject to get corrupted due to interference and network problems. The corrupted bits leads to spurious data being received by the destination and are called errors.**

Error Control:

Error control can be done in two ways

- Two basic strategies to deal with errors:
- **1. Error correcting codes:** Include enough redundant information to enable the receiver to deduce what the transmitted data must have been.
- **2. Error detecting codes:** Include only enough redundancy to allow the receiver to deduce that an error has occurred (but not which error).
- **For both error detection and error correction, the sender needs to send some additional bits along with the data bits. The receiver performs necessary checks based upon the additional redundant bits. If it finds that the data is free from errors, it removes the redundant bits before passing the message to the upper layers.**

A frame consists of – m data bits (message) and – r redundant bits (check).

- **Block code - the r check bits are computed solely as function of the m data bits with which they are associated.**
- **Systemic code – the m data bits are send directly along with the check bits.**



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• Linear code – **the r check bits are computed as a linear function of the m data bits.**

- n – total length of a block (i.e., $n = m + r$)
- (n, m) – code
- n – bit codeword containing n bits.
- m/n – code rate (range $\frac{1}{2}$ for noisy channel and close to 1 for high-quality channel).
- Example : Transmitted: 10001001
- Received: 10110001
- XOR operation gives number of bits that are different.
- XOR: 00111000
- Number of bit positions in which two code words differ is called Hamming Distance. It shows that two codes are d distance apart, and it will require d errors to convert one into the other.

Error Correcting Codes:

We will examine four different error-correcting codes:

1. Hamming codes.
2. Binary convolutional codes.
3. Reed-Solomon codes.
4. Low-Density Parity Check codes.

Hamming Code:

- In Computer Networks, Hamming code is used for the set of error-correction codes which may occur when the data is moved from the sender to the receiver. The hamming method corrects the error by finding the state at which the error has occurred.
- Redundant bits are extra binary bits that are generated and added to the information-carrying bits of data transfer to ensure that no bits were lost during the

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data transfer. The redundancy bits are placed at certain calculated positions to eliminate the errors and the distance between the two redundancy bits is called "Hamming Distance".

Steps to find the Hamming Code –

The hamming method uses the extra parity bits to allow the identification of a single-bit error.

- **Step 1** – First write the bit positions starting from 1 in a binary form (1, 10, 11, 100, etc.)
- **Step 2** – Mark all the bit positions that are powers of two as parity bits (1, 2, 4, 8, 16, 32, 64, etc.)
- **Step 3** – All other bit positions are for the data to be encoded using (3, 5, 6, 7, 9, 10 and 11, etc.)

Each parity bit calculates the parity for some of the bits in the code word. The position of the parity determines the sequence of bits that it alternatively checks and skips.

- **Position 1** – Check 1 bit, then skip 1 bit, check 1 bit and then skip 1 bit and so on (Ex – 1,3,5,7,11, etc.)
- **Position 2** – Check 2 bit, then skip 2 bit, check 2 bit, then skip 2 bit (Ex – 2,3,6,7,10,11,14,15, etc.)
- **Position 4** – Check 4 bit, then skip 4 bit, check 4 bit, then skip 4 bit (Ex – 4, 5, 6, 7, 12, 13, 14, 15, etc.)
- **Position 8** – Check 8 bit, then skip 8 bit, check 8 bit, then skip 8 bit (Ex – 8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31).

Note – Set the parity bit 1 if the total number of 1s in the positions it checks odd or set the parity bit 0 if the total number of 1s in the positions it checks even.

Example:

Construct the even parity Hamming code word for a data byte

1001101. The number (1001101) of bits is 7.



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The value of r is calculated as –

$$2^r \geq M + R + 1$$

$$\Rightarrow 2^4 \geq 7 + 4 + 1$$

Therefore, the number of redundancy bits = 4

Now, let's calculate the required number of parity bits.

We take $P = 2$, then $2^P = 2^2 = 4$ and $n + P + 1 = 4 + 2 + 1$

= 7 The 2 parity bits are not sufficient for the 4-bit data.

Now, we will take $P = 3$, then $2^P = 2^3 = 8$ and $n + P + 1 = 4 + 3 + 1 = 8$

Therefore, 3 parity bits are sufficient for 4-bit data.

The total bits in the codeword are – $4 + 3 = 7$

Position 1: checks the bits 1,3,5,7,9 and 11.

? _1_0 0 1_1 0 1 0. In position 1 even parity so set position 1 to a 0: 0_1_0 0 1_1 0 1 0.

0	1	0	1		1	0	0		1		0
---	---	---	---	--	---	---	---	--	---	--	---

Position 2: checks bits 2,3,6,7,10,11.

0 ? 1_0 0 1_1 0 1 0. In position 2 odd parity so set position 2 to a 1: 0 1 1_0 0 1_1 0 1 0

0	1	0	1		1	0	0		1	1	0
---	---	---	---	--	---	---	---	--	---	---	---

Position 4 checks bits 4,5,6,7,12.

0 1 1 ? 0 0 1_1 0 1 0. In position 4 odd parity so set position 4 to a 1: 0 1 1 1 0 0 1_1 0 1 0

0	1	0	1		1	0	0	1	1	1	0
---	---	---	---	--	---	---	---	---	---	---	---

Position 8 checks bits 8,9,10,11,12.

0 1 1 1 0 0 1 ? 1 0 1 0. In position 8 even parity so set position 8 to a 1: 0 1 1 1 0 0 1 0 1 0 1 0

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0	1	0	1	0	1	0	0	1	1	1	0
---	---	---	---	---	---	---	---	---	---	---	---

Code Word = 011100101010

0	1	1	1	0	0	1	0	1	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---

Binary convolutional codes:

- In block codes, the data comprises of a block of data of a definite length.
- However, in convolutional codes, the input data bits are not divided into block but are instead fed as streams of data bits, which convolve to output bits based upon the logic function of the encoder.
- This code is the only one we will cover that is not a block code.
- In a convolutional code, an encoder processes a sequence of input bits and generates a sequence of output bits.
- There is no natural message size or encoding boundary as in a block code. The output depends on the current and previous input bits.
- That is, the encoder has memory. The number of previous bits on which the output depends is called the constraint length of the code.
- Convolutional codes are specified in terms of their rate and constraint length.
- Convolutional codes are widely used in deployed networks, for example, as part of the GSM mobile phone system, in satellite communications, and in 802.11.
- As an example, a popular convolutional code is shown in Fig. 3-7. This code is known as the NASA convolutional code of $r = 1/2$ and $k = 7$.

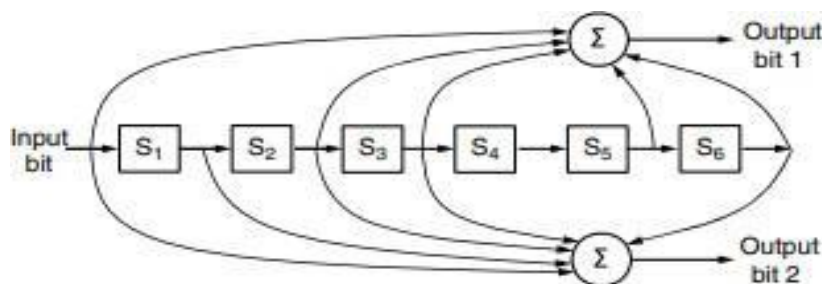


Figure 3-7. The NASA binary convolutional code used in 802.11.

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- In Fig. 3-7, each input bit on the left-hand side produces two output bits on the right-hand side that are XOR sums of the input and internal state.
- Since it deals with bits and performs linear operations, this is a binary, linear convolutional code.
- Since 1 input bit produces 2 output bits, the code rate is $1/2$. It is not systematic since none of the output bits is simply the input bit.
- The internal state is kept in six memory registers. Each time another bit is input the values in the registers are shifted to the right.
- For example, if 111 is input and the initial state is all zeros, the internal state, written left to right, will become 100000, 110000, and 111000 after the first, second, and third bits have been input.
- The output bits will be 11, followed by 10, and then 01. It takes seven shifts to flush an input completely so that it does not affect the output.
- The constraint length of this code is thus $k = 7$.

Reed – Solomon Code:

- The third kind of error-correcting code we will describe is the Reed-Solomon code.
- Like Hamming codes, Reed-Solomon codes are linear block codes, and they are often systematic too.
- Unlike Hamming codes, which operate on individual bits, Reed-Solomon codes operate on m bit symbols.
- Naturally, the mathematics are more involved, so we will describe their operation by analogy. Reed-Solomon codes are based on the fact that every n degree polynomial is uniquely determined by $n + 1$ points.
- For example, a line having the form $ax + b$ is determined by two points. Extra points on the same line are redundant, which is helpful for error correction.
- Imagine that we have two data points that represent a line and we send those two data points plus two check points chosen to lie on the same line. If one of the points is received in error, we can still recover the data points by fitting a line to the received points.

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- Three of the points will lie on the line, and one point, the one in error, will not. By finding the line we have corrected the error. Reed-Solomon codes are actually defined as polynomials that operate over finite fields, but they work in a similar manner.
- For m bit symbols, the codewords are $2^m - 1$ symbols long. A popular choice is to make $m = 8$ so that symbols are bytes. A codeword is then 255 bytes long. The (255, 233) code is widely used; it adds 32 redundant symbols to 233 data symbols.
- Reed-Solomon codes are widely used in practice because of their strong error-correction properties, particularly for burst errors.

Low density Parity check:

- LDPC codes are linear block codes that were invented by Robert Gallager in his doctoral thesis (Gallagher, 1962).

Like most theses, they were promptly forgotten, only to be reinvented in 1995 when advances in computing power had made them practical. In an LDPC code, each output bit is formed from only a fraction of the input bits.

- This leads to a matrix representation of the code that has a low density of 1s, hence the name for the code.
- The received code words are decoded with an approximation algorithm that iteratively improves on a best fit of the received data to a legal codeword.
- This corrects errors. LDPC codes are practical for large block sizes and have excellent error-correction abilities that outperform many other codes (including the ones we have looked at) in practice.
- They are part of the standard for digital video broadcasting, 10 Gbps Ethernet, power-line networks, and the latest version of 802.11.

2.5.Error Detection and Correction in Data link Layer:

- Data-link layer uses error control techniques to ensure that frames, i.e. bit streams of data, are transmitted from the source to the destination with a certain extent of accuracy.



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Errors:

- When bits are transmitted over the computer network, they are subject to get corrupted due to interference and network problems. The corrupted bits leads to spurious data being received by the destination and are called errors.

Error Control:

Error control can be done in two ways

- Two basic strategies to deal with errors:
- **1. Error correcting codes**:. Include enough redundant information to enable the receiver to deduce what the transmitted data must have been.
- **2. Error detecting codes**. Include only enough redundancy to allow the receiver to deduce that an error has occurred (but not which error).
- **For both error detection and error correction, the sender needs to send some additional bits along with the data bits. The receiver performs necessary checks based upon the additional redundant bits.** If it finds that the data is free from errors, it removes the redundant bits before passing the message to the upper layers.

A frame consists of – m data bits (message) and – r redundant bits (check).

- Block code - **the r check bits are computed solely as function of the m data bits with which they are associated.**
- Systemic code – **the m data bits are send directly along with the check bits.**
- Linear code – **the r check bits are computed as a linear function of the m data bits.**
 - n – total length of a block (i.e., $n = m + r$)
 - (n, m) – code
 - n – bit codeword containing n bits.
 - m/n – code rate (range $\frac{1}{2}$ for noisy channel and close to 1 for high-quality channel).



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- Example : Transmitted: 10001001
- Received: 10110001
- XOR operation gives number of bits that are different.
- XOR: 00111000
- Number of bit positions in which two code words differ is called Hamming Distance. It shows that two codes are d distance apart, and it will require d errors to convert one into the other.

Error Correcting Codes:

We will examine four different error-correcting codes:

1. Hamming codes.
2. Binary convolutional codes.
3. Reed-Solomon codes.
4. Low-Density Parity Check codes.

Hamming Code:

- In Computer Networks, Hamming code is used for the set of error-correction codes which may occur when the data is moved from the sender to the receiver. The hamming method corrects the error by finding the state at which the error has occurred.
- Redundant bits are extra binary bits that are generated and added to the information-carrying bits of data transfer to ensure that no bits were lost during the data transfer. The redundancy bits are placed at certain calculated positions to eliminate the errors and the distance between the two redundancy bits is called "Hamming Distance".

Steps to find the Hamming Code –

The hamming method uses the extra parity bits to allow the identification of a single-bit error.

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- **Step 1** – First write the bit positions starting from 1 in a binary form (1, 10, 11, 100, etc.)
- **Step 2** – Mark all the bit positions that are powers of two as parity bits (1, 2, 4, 8, 16, 32, 64, etc.)
- **Step 3** – All other bit positions are for the data to be encoded using (3, 5, 6, 7, 9, 10 and 11, etc.)

Each parity bit calculates the parity for some of the bits in the code word. The position of the parity determines the sequence of bits that it alternatively checks and skips.

- **Position 1** – Check 1 bit, then skip 1 bit, check 1 bit and then skip 1 bit and so on (Ex – 1, 3, 5, 7, 11, etc.)
- **Position 2** – Check 2 bit, then skip 2 bit, check 2 bit, then skip 2 bit (Ex – 2, 3, 6, 7, 10, 11, 14, 15, etc.)
- **Position 4** – Check 4 bit, then skip 4 bit, check 4 bit, then skip 4 bit (Ex – 4, 5, 6, 7, 12, 13, 14, 15, etc.)
- **Position 8** – Check 8 bit, then skip 8 bit, check 8 bit, then skip 8 bit (Ex – 8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31).

Note – Set the parity bit 1 if the total number of 1s in the positions it checks odd or set the parity bit 0 if the total number of 1s in the positions it checks even.

Example:

Construct the even parity Hamming code word for a data byte

1001101. The number (1001101) of bits is 7.

The value of r is calculated as –

$$2^r \geq M + R + 1$$

$$\Rightarrow 2^4 \geq 7 + 4 + 1$$

Therefore, the number of redundancy bits = 4



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Now, let's calculate the required number of parity bits.

We take $P = 2$, then $2^P = 2^2 = 4$ and $n + P + 1 = 4 + 2 + 1$
 $= 7$ The 2 parity bits are not sufficient for the 4-bit data.

Now, we will take $P = 3$, then $2^P = 2^3 = 8$ and $n + P + 1 = 4 + 3 + 1 = 8$

Therefore, 3 parity bits are sufficient for 4-bit data.

The total bits in the codeword are $- 4 + 3 = 7$

Position 1: checks the bits 1,3,5,7,9 and 11.

? _1_0 0 1_1 0 1 0. In position 1 even parity so set position 1 to a 0: 0_1_0 0 1_1 0 1 0.

0	1	0	1		1	0	0		1		0
---	---	---	---	--	---	---	---	--	---	--	---

Position 2: checks bits 2,3,6,7,10,11.

0 ? 1_0 0 1_1 0 1 0. In position 2 odd parity so set position 2 to a 1: 0 1 1_0 0 1_1 0 1 0

0	1	0	1		1	0	0		1	1	0
---	---	---	---	--	---	---	---	--	---	---	---

Position 4 checks bits 4,5,6,7,12.

0 1 1 ? 0 0 1_1 0 1 0. In position 4 odd parity so set position 4 to a 1: 0 1 1 1 0 0 1_1 0 1 0

0	1	0	1		1	0	0	1	1	1	0
---	---	---	---	--	---	---	---	---	---	---	---

Position 8 checks bits 8,9,10,11,12.

0 1 1 1 0 0 1 ? 1 0 1 0. In position 8 even parity so set position 8 to a 1: 0 1 1 1 0 0 1 0 1 0 1 0

0	1	0	1	0	1	0	0	1	1	1	0
---	---	---	---	---	---	---	---	---	---	---	---

Code Word = 011100101010

0	1	1	1	0	0	1	0	1	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---

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Binary convolutional codes:

- In block codes, the data comprises of a block of data of a definite length.
- However, in convolutional codes, the input data bits are not divided into block but are instead fed as streams of data bits, which convolve to output bits based upon the logic function of the encoder.
- This code is the only one we will cover that is not a block code.
- In a convolutional code, an encoder processes a sequence of input bits and generates a sequence of output bits.
- There is no natural message size or encoding boundary as in a block code. The output depends on the current and previous input bits.
- That is, the encoder has memory. The number of previous bits on which the output depends is called the constraint length of the code.
- Convolutional codes are specified in terms of their rate and constraint length.
- Convolutional codes are widely used in deployed networks, for example, as part of the GSM mobile phone system, in satellite communications, and in 802.11.
- As an example, a popular convolutional code is shown in Fig. 3-7. This code is known as the NASA convolutional code of $r = 1/2$ and $k = 7$.

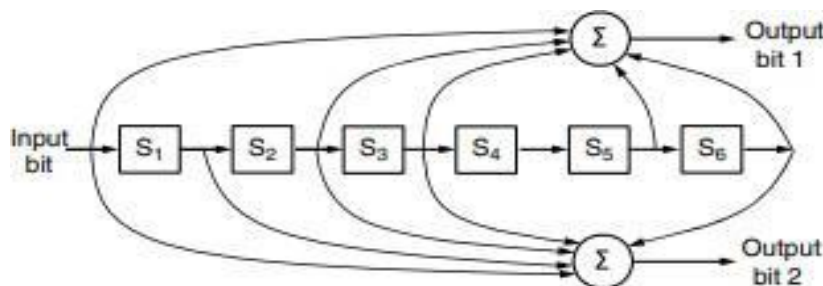


Figure 3-7. The NASA binary convolutional code used in 802.11.

-
- In Fig. 3-7, each input bit on the left-hand side produces two output bits on the right-hand side that are XOR sums of the input and internal state.
- Since it deals with bits and performs linear operations, this is a binary, linear convolutional code.
- Since 1 input bit produces 2 output bits, the code rate is $1/2$. It is not systematic since none of the output bits is simply the input bit.



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- The internal state is kept in six memory registers. Each time another bit is input the values in the registers are shifted to the right.
- For example, if 111 is input and the initial state is all zeros, the internal state, written left to right, will become 100000, 110000, and 111000 after the first, second, and third bits have been input.
- The output bits will be 11, followed by 10, and then 01. It takes seven shifts to flush an input completely so that it does not affect the output.
- The constraint length of this code is thus $k = 7$.

Reed – Solomon Code:

- The third kind of error-correcting code we will describe is the Reed-Solomon code.
- Like Hamming codes, Reed-Solomon codes are linear block codes, and they are often systematic too.
- Unlike Hamming codes, which operate on individual bits, Reed-Solomon codes operate on m bit symbols.
- Naturally, the mathematics are more involved, so we will describe their operation by analogy. Reed-Solomon codes are based on the fact that every n degree polynomial is uniquely determined by $n + 1$ points.
- For example, a line having the form $ax + b$ is determined by two points. Extra points on the same line are redundant, which is helpful for error correction.
- Imagine that we have two data points that represent a line and we send those two data points plus two check points chosen to lie on the same line. If one of the points is received in error, we can still recover the data points by fitting a line to the received points.
- Three of the points will lie on the line, and one point, the one in error, will not. By finding the line we have corrected the error. Reed-Solomon codes are actually defined as polynomials that operate over finite fields, but they work in a similar manner.
- For m bit symbols, the codewords are $2m-1$ symbols long. A popular choice is

to make $m = 8$ so that symbols are bytes. A codeword is then 255 bytes long.



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The (255, 233) code is widely used; it adds 32 redundant symbols to 233 data symbols.

- Reed-Solomon codes are widely used in practice because of their strong error- correction properties, particularly for burst errors.

Low density Parity check:

- LDPC codes are linear block codes that were invented by Robert Gallager in his doctoral thesis (Gallagher, 1962).

Like most theses, they were promptly forgotten, only to be reinvented in 1995 when advances in computing power had made them practical. In an LDPC code, each output bit is formed from only a fraction of the input bits.

- This leads to a matrix representation of the code that has a low density of 1s, hence the name for the code.
- The received codewords are decoded with an approximation algorithm that iteratively improves on a best fit of the received data to a legal codeword.
- This corrects errors. LDPC codes are practical for large block sizes and have excellent error-correction abilities that outperform many other codes (including the ones we have looked at) in practice.
- They are part of the standard for digital video broadcasting, 10 Gbps Ethernet, power-line networks, and the latest version of 802.11.

Steps to be followed in Checksum Generator:

- In the sender, the checksum generator subdivides the data unit into equal segments of n bits.
- These segments are added with each other by using one's complement arithmetic in such a way that the total is also n bits long.
- That total is then complemented and appended to the end of the

data unit. Error Detection :



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An error occurs when the output information does not match the input information. Digital signals suffer from noise during transmission, which can create errors in binary bits travelling from one system to another. That is, a 0 bit may become a 1 bit, or a 1 bit may become a 0.

When a message is sent, it may be jumbled by noise or the data may be damaged. To avoid this, we employ error-detecting codes, which are bits of extra data appended to a digital message to assist us detect whether an error occurred during transmission.

2.6.Error Detection Techniques:

- We will examine three different error-detecting codes. They are all linear, systematic block codes:
- 1. Parity bit.
- 2. Checksums.
- 3. Cyclic Redundancy Checks (CRCs).

1. Parity bit :

- To see how they can be more efficient than error-correcting codes, consider the first error-detecting code, in which a single parity bit is appended to the data.
- The parity bit is chosen so that the number of 1 bits in the codeword is even (or odd).
- Doing this is equivalent to computing the (even) parity bit as the modulo 2 sum or XOR of the data bits.
- For example, when 1011010 is sent in even parity, a bit is added to the end to make it 10110100. With odd parity 1011010 becomes 10110101.
- A code with a single parity bit has a distance of 2, since any single-bit error produces a codeword with the wrong parity. This means that it can detect single-bit errors.
- One difficulty with this scheme is that a single parity bit can only reliably detect a single-bit error in the block.
- If the block is badly garbled by a long burst error, the probability that the error will be detected is only 0.5, which is hardly acceptable.

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- However, there is something else we can do that provides better protection against burst errors: we can compute the parity bits over the data in a different order than the order in which the data bits are transmitted. **Doing so is called interleaving.**
- In this case, we will compute a parity bit for each of the n columns and send all the data bits as k rows, sending the rows from top to bottom and the bits in each row from left to right in the usual manner.
- At the last row, we send the n parity bits. This transmission order is shown in Fig. 3-8 for $n = 7$ and $k = 7$.

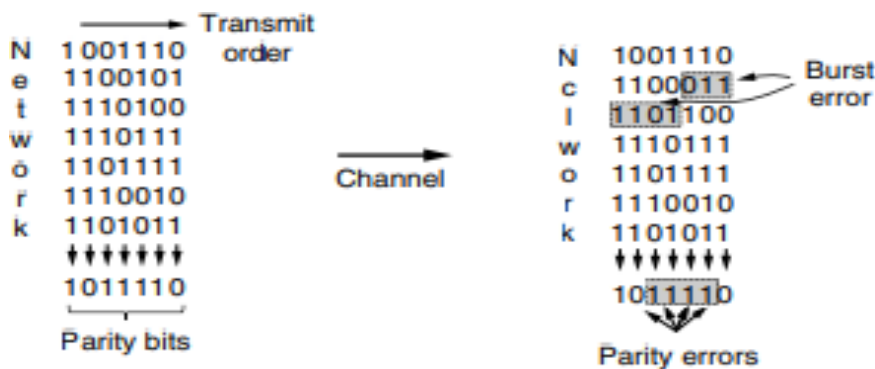


Figure 3-8. Interleaving of parity bits to detect a burst error.

- Interleaving is a general technique to convert a code that detects (or corrects) isolated errors into a code that detects (or corrects) burst errors. In Fig. 3-8, when a burst error of length $n = 7$ occurs, the bits that are in error are spread across different columns.
- (A burst error does not imply that all the bits are wrong; it just implies that at least the first and last are wrong. In Fig. 3-8, 4 bits were flipped over a range of 7 bits.)
- At most 1 bit in each of the n columns will be affected, so the parity bits on those columns will detect the error. This method uses n parity bits on blocks of kn data bits to detect a single burst error of length n or less.



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Steps Involved-

Error detection using single parity check involves the following steps-

Step-01:

At sender side,

- Total number of 1's in the data unit to be transmitted is counted.
- The total number of 1's in the data unit is made even in case of even parity.
- The total number of 1's in the data unit is made odd in case of odd parity.
- This is done by adding an extra bit called as **parity bit**.

Step-02:

- The newly formed code word (Original data + parity bit) is transmitted to the receiver.

Step-03:

At receiver side,

- Receiver receives the transmitted code word.
- The total number of 1's in the received code word is counted.

Then, following cases are possible-



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- If total number of 1's is even and even parity is used, then receiver assumes that no error occurred.
- If total number of 1's is even and odd parity is used, then receiver assumes that error occurred.
- If total number of 1's is odd and odd parity is used, then receiver assumes that no error occurred.
- If total number of 1's is odd and even parity is used, then receiver assumes that error occurred.

Parity Check Example-

Consider the data unit to be transmitted is 1001001 and even parity is used. Then,

At Sender Side-

- Total number of 1's in the data unit is counted.
- Total number of 1's in the data unit = 3.
- Clearly, even parity is used and total number of 1's is odd.
- So, parity bit = 1 is added to the data unit to make total number of 1's even.
- Then, the code word 10010011 is transmitted to the receiver.

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1	0	0	1	0	0	1
---	---	---	---	---	---	---

Original data unit



Parity bit



1	0	0	1	0	0	1	1
---	---	---	---	---	---	---	---

Transmitted data unit

At Receiver Side-

- After receiving the code word, total number of 1's in the code word is counted.
- Consider receiver receives the correct code word = 10010011.
- Even parity is used and total number of 1's is even.
- So, receiver assumes that no error occurred in the data during the transmission. Advantage-

- This technique is guaranteed to detect an odd number of bit errors (one, three, five and so on).
- If odd number of bits flip during transmission, then receiver can detect by counting the number of 1's.

Limitation-

- This technique can not detect an even number of bit errors (two, four, six and so on).
- If even number of bits flip during transmission, then receiver can not catch the error.



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2. Checksum-

Checksum is an error detection method.

- The second kind of error-detecting code, the checksum, is closely related to groups of parity bits. The word “checksum” is often used to mean a group of check bits associated with a message, regardless of how are calculated.
- A group of parity bits is one example of a checksum. However, there are other, stronger checksums based on a running sum of the data bits of the message.
- The checksum is usually placed at the end of the message, as the complement of the sum function.
- This way, errors may be detected by summing the entire received codeword, both data bits and checksum. If the result comes out to be zero, no error has been detected.

Error detection using checksum method involves the following steps-

Step-01:

At sender side,

- If m bit checksum is used, the data unit to be transmitted is divided into segments of m bits.
- All the m bit segments are added.
- The result of the sum is then complemented using 1's complement arithmetic.
- The value so obtained is called as **checksum**.

Step-02:



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- The data along with the checksum value is transmitted to the receiver.

Step-03:

At receiver side,

- If m bit checksum is being used, the received data unit is divided into segments of m bits.
- All the m bit segments are added along with the checksum value.
- The value so obtained is complemented and the result is checked.

Checksum Example-

Consider the data unit to be transmitted is- 10011001111000100010010010000100

Consider 8 bit checksum is used.

Step-01:

At sender side,

The given data unit is divided into segments of 8 bits as-

10011001	11100010	00100100	10000100
----------	----------	----------	----------

Now, all the segments are added and the result is obtained as-

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- $10011001 + 11100010 + 00100100 + 10000100 = 1000100011$
- Since the result consists of 10 bits, so extra 2 bits are wrapped around.
- $00100011 + 10 = 00100101$ (8 bits)
- Now, 1's complement is taken which is 11011010.
- Thus, checksum value = 11011010

Step-02:

- The data along with the checksum value is transmitted to the receiver.

Step-03:

At receiver side,

- The received data unit is divided into segments of 8 bits.
- All the segments along with the checksum value are added.
- Sum of all segments + Checksum value = $00100101 + 11011010 = 11111111$
- Complemented value = 00000000
- Since the result is 0, receiver assumes no error occurred in the data and therefore accepts it.

3. Cyclic Redundancy Checks (CRCs):

- A third and stronger kind of error-detecting code is in widespread use at the link layer: the CRC (Cyclic Redundancy Check), also known as a polynomial code.
- Polynomial codes are based upon treating bit strings as representations of poly- nomials with coefficients of 0 and 1 only.
- When the polynomial code method is employed, the sender and receiver must agree upon a generator polynomial, $G(x)$, in advance.



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- Both the high- and low order bits of the generator must be 1. To compute the CRC for some frame with m bits corresponding to the polynomial $M(x)$, the frame must be longer than the generator polynomial.
- The idea is to append a CRC to the end of the frame in such a way that the polynomial represented by the checksummed frame is divisible by $G(x)$. When the receiver gets the checksummed frame, it tries dividing it by $G(x)$. If there is a remainder, there has been a transmission error.
- The algorithm for computing the CRC is as follows:
 1. Let r be the degree of $G(x)$. Append r zero bits to the low-order end of the frame so it now contains $m + r$ bits and corresponds to the polynomial $x^r M(x)$.
 2. Divide the bit string corresponding to $G(x)$ into the bit string corresponding to $x^r M(x)$, using modulo 2 division.
 3. Subtract the remainder (which is always r or fewer bits) from the bit string corresponding to $x^r M(x)$ using modulo 2 subtraction. The result is the check- summed frame to be transmitted. Call its polynomial $T(x)$. Figure 3-9 illustrates the calculation for a frame

1101011111 using the generator $G(x) = x^4 + x + 1$

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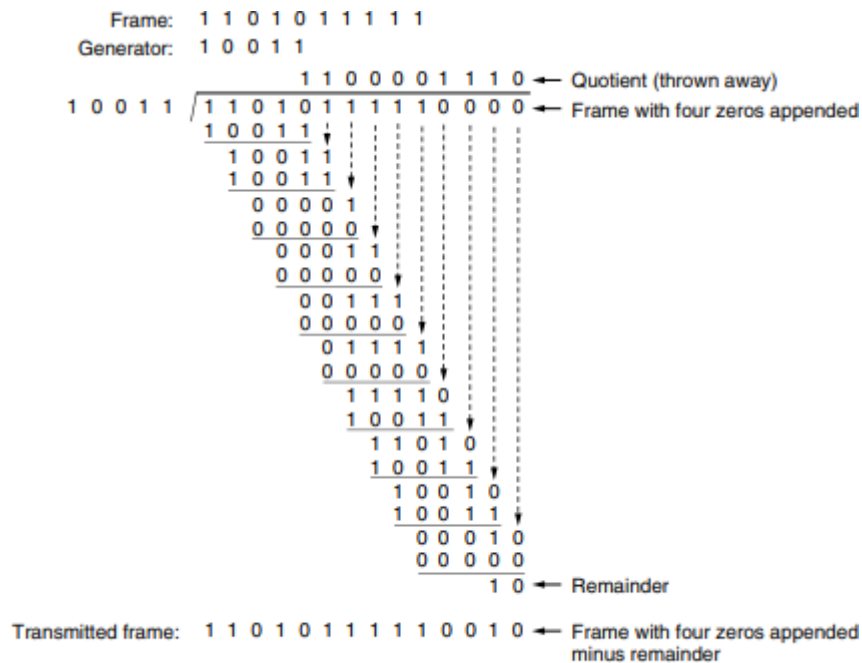


Figure 3-9. Example calculation of the CRC.

1. A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is x^4+x+1 . What is the actual bit string transmitted?

Solution-

- The generator polynomial $G(x) = x^4 + x + 1$ is encoded as 10011.
- Clearly, the generator polynomial consists of 5 bits.
- So, a string of 4 zeroes is appended to the bit stream to be transmitted.
- The resulting bit stream is 1101011011**0000**.

Now, the binary division is performed as-



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$$\begin{array}{r}
 \begin{array}{cccccccccccc}
 & & & & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
 1 & 0 & 0 & 1 & 1 & | & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\
 & & & & & & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 & & & & & & 1 & 0 & 0 & 1 & 1 & & & & & & & \\
 & & & & & & \hline
 & & & & & & 1 & 0 & 0 & 1 & 1 & & & & & & & \\
 & & & & & & \hline
 & & & & & & 0 & 0 & 0 & 0 & 1 & & & & & & & \\
 & & & & & & \hline
 & & & & & & 0 & 0 & 0 & 1 & 0 & & & & & & & \\
 & & & & & & \hline
 & & & & & & 0 & 0 & 1 & 0 & 1 & & & & & & & \\
 & & & & & & \hline
 & & & & & & 0 & 1 & 0 & 1 & 1 & & & & & & & \\
 & & & & & & \hline
 & & & & & & 1 & 0 & 1 & 1 & 0 & & & & & & & \\
 & & & & & & \hline
 & & & & & & 0 & 1 & 0 & 1 & 0 & & & & & & & \\
 & & & & & & \hline
 & & & & & & 1 & 0 & 1 & 0 & 0 & & & & & & & \\
 & & & & & & \hline
 & & & & & & 0 & 1 & 1 & 1 & 0 & & & & & & & \\
 & & & & & & \hline
 & & & & & & 1 & 1 & 1 & 0 & & & & & & & & \leftarrow \text{Remainder}
 \end{array}
 \end{array}$$

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From here, CRC = 1110.

Now,

- The code word to be transmitted is obtained by replacing the last 4 zeroes of 11010110110000 with the CRC.
- Thus, the code word transmitted to the receiver = 11010110111110.

2. Message D = 1010001101 (10 bits)

Predetermined P = 110101 (6 bits)

FCS R = to be calculated 5 bits

Hence, $n = 15$ $K = 10$ and $(n - k) =$

5

The message is generated through 2^5 :accommodating 1010001101000

The product is divided by P.

$$\begin{array}{r}
 \begin{array}{c} 1101010110 \leftarrow Q \\ \hline 110101 \overline{) 101000110100000} \\ \underline{110101} \\ 111011 \\ \underline{110101} \\ 111010 \\ \underline{110101} \\ 111110 \\ \underline{110101} \\ 101100 \\ \underline{110101} \\ 110010 \\ \underline{110101} \\ 01110 \leftarrow R \end{array}
 \end{array}$$



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The remainder is inserted to 2^sD to provide T = 101000110101110 that is sent.

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LECTURE NOTES

UNIT 3

A One-Bit Sliding Window Protocol:

Sliding window protocols are data link layer protocols for reliable and sequential delivery of data frames. The sliding window is also used in Transmission Control Protocol. In these protocols, the sender has a buffer called the sending window and the receiver has buffer called the receiving window.

In one – bit sliding window protocol, the size of the window is 1. So the sender transmits a frame, waits for its acknowledgment, then transmits the next frame. Thus it uses the concept of stop and waits for the protocol. This protocol provides for full – duplex communications. Hence, the acknowledgment is attached along with the next data frame to be sent by piggybacking.

Working Principle

The data frames to be transmitted additionally have an acknowledgment field, *ack* field that is of a few bits length. The *ack* field contains the sequence number of the last frame received without error. If this sequence number matches with the sequence number of the frame to be sent, then it is inferred that there is no error and the frame is transmitted. Otherwise, it is inferred that there is an error in the frame and the previous frame is retransmitted.

Since this is a bi-directional protocol, the same algorithm applies to both the communicating parties.



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The algorithm of One – bit Sliding Window Protocol

```
begin
    frame s, r; // s and r denotes frames to be sent and
    received SeqNo=0; // Initialise sequence number of
    outbound frame RSeqNo=0; // Initialise sequence
    number of expected frame while(true) // check
    repeatedly
do
    Wait_For_Event(); // wait for availability of packet
    if(Event(Request_For_Transfer) AND
        canSend) then
        Get_Data_From_Network_Layer();
        s
        =Make_Frame(SeqNo);
        Store_Copy_Frame(s);
        Start_Timer(s);
        SeqNo=SeqNo+1;
    endif;
    Wait_For_Event(); // wait for arrival of frame
    if(Event(Frame_Arrival) then
        r=Receive_Frame_From_Physical_Layer();
        if(
r.SeqNo=RSeqNo) then
        Extract_Data(r);
        Deliver_Data_To_Network_Layer(r);
        Stop_Timer(r);
        RSeqNo=RSeqNo+1;
    en
dif
endi
f
s.ack = r.SeqNo;
```



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Send_Frame_To_Physical_Layer(s)

; Start_Timer(s);

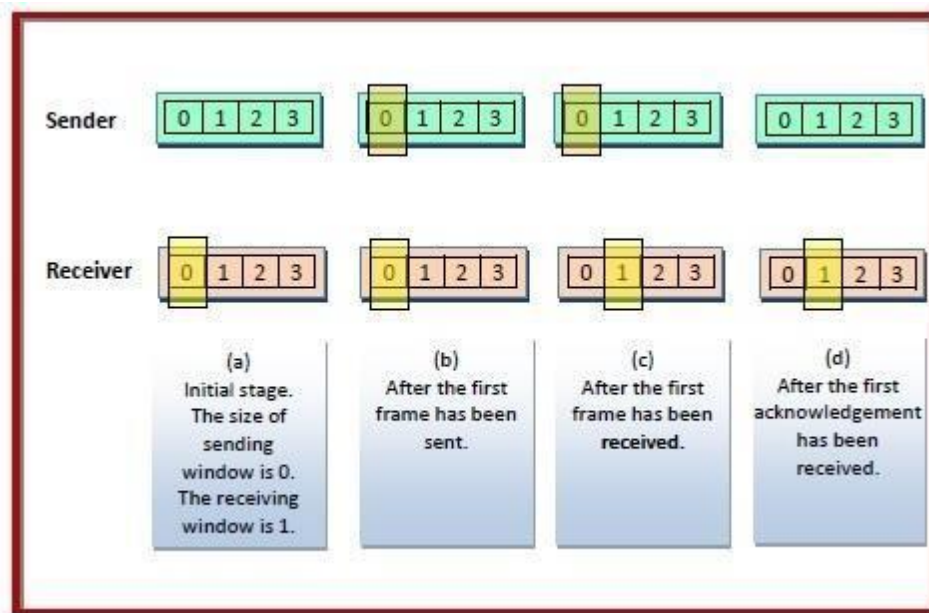
SeqNo=SeqNo+1;

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```
endwhile  
end
```

Illustrative Example

The following diagram depicts a scenario with sequence numbers 0, 1, 2, 3, 0, 1, 2 and so on. It depicts the sliding windows in the sending and the receiving stations during frame transmission.



A Protocol Using Go-Back-N:

Go-Back-N protocol, also called Go-Back-N Automatic Repeat reQuest, is a data link layer protocol that uses a sliding window method for reliable and sequential delivery of data frames. It is a case of sliding window protocol having to send window size of N and receiving window size of 1.

Working Principle

Go - Back - N ARQ provides for sending multiple frames before receiving the acknowledgment for the first frame. The frames are sequentially numbered and a finite number of frames. The maximum number of frames that can be sent depends upon the size of the sending window. If the

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acknowledgment of a frame is not received within an agreed upon time period, all frames starting from that frame are retransmitted.

The size of the sending window determines the sequence number of the outbound frames. If the sequence number of the frames is an n -bit field, then the range of sequence numbers that can be assigned is 0 to $2^n - 1$. Consequently, the size of the sending window is $2^n - 1$. Thus in order to accommodate a sending window size of $2^n - 1$, a n -bit sequence number is chosen.

The sequence numbers are numbered as modulo- n . For example, if the sending window size is 4, then the sequence numbers will be 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, and so on. The number of bits in the sequence number is 2 to generate the binary sequence 00, 01, 10, 11.

The size of the receiving window is 1.

Sender Site Algorithm of Go-Back-N Protocol

```
begin
  frame s;           //s denotes frame to be sent
  frame t;           //t is temporary frame
  S_window = power(2,m)-1;    //Assign maximum
  window size SeqFirst=0;      // Sequence number of first
  frame in window SeqN=0;     // Sequence number of Nth frame
  window while(true)         //check repeatedly
  do
    Wait_For_Event();        //wait for availability of packet
  if(Event(Request_For_Transfer))then
    //check if window is full
    if(SeqN-SeqFirst>= S_window)then
      doNothing();
    endif;
    Get_Data_From_Network_Layer();
    s=Make_Frame();
    s.seq=SeqN;
    Store_Copy_Frame(s);
```

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```
Send_Fram
e(s);
Start_Timer
(s);
SeqN=Seq
N+1;
endif;
if(Event(Frame_Arrival)then
  r=Receive_Acknowledgement();
  if(AckNo>SeqFirst&&AckNo<SeqN)then
    while(SeqFirst<=AckNo)
      Remove_copy_frame(s.seq(SeqFirst));
      SeqFirst=SeqFirst+1;
    endwhile
    Stop_Timer(s);
  en
dif
endi
f
// Resend all frames if acknowledgement havn't been received
if(Event(Time_Out))the
  n TempSeq=SeqFirst;
  while(TempSeq<SeqN)
    t
    =Retrieve_Copy_Frame(s.seq(SeqFirst
    )); Send_Frame(t);
    Start_Timer(t);
    TempSeq=TempSeq
    +1;
  endwh
ile
endif
end
```



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Receiver Site Algorithm of Go-Back-N Protocol

Begin

frame f;

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```
RSeqNo=0;// Initialise sequence number of expected frame
while(true)//check repeatedly
do
    Wait_For_Event();//wait for arrival of frame
    if(Event(Frame_Arrival)then
        Receive_Frame_From_Physical_Layer();
        if(Corrupted( f.SeqNo)
            doNothing();
        elseif( f.SeqNo=RSeqNo)then
            Extract_Data();
            Deliver_Data_To_Network_Layer();
            RSeqNo=RSeqNo+1;
            Send_ACK(RSeqNo);
        endif
    endif
endwhile
end
```

A Protocol Using Selective Repeat:

Selective repeat protocol, also called Selective Repeat ARQ (Automatic Repeat reQuest), is a data link layer protocol that uses sliding window method for reliable delivery of data frames. Here, only the erroneous or lost frames are retransmitted, while the good frames are received and buffered.

It uses two windows of equal size: a sending window that stores the frames to be sent and a receiving window that stores the frames received by the receiver. The size is half the maximum sequence number of the frame. For example, if the sequence number is from 0 – 15, the window size will be 8.



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Working Principle

Selective Repeat protocol provides for sending multiple frames depending upon the availability of frames in the sending window, even if it does not receive acknowledgement for any frame in the interim. The maximum number of frames that can be sent depends upon the size of the sending window.

The receiver records the sequence number of the earliest incorrect or un-received frame. It then fills the receiving window with the subsequent frames that it has received. It sends the sequence number of the missing frame along with every acknowledgement frame.

The sender continues to send frames that are in its sending window. Once, it has sent all the frames in the window, it retransmits the frame whose sequence number is given by the acknowledgements. It then continues sending the other frames.

Sender Site Algorithm of Selective Repeat Protocol

```
begin
  frame s;//s denotes frame to be sent
  frame t;//t is temporary frame
  S_window = power(2,m-1);//Assign maximum window size
  SeqFirst=0;// Sequence number of first frame in window
  SeqN=0;// Sequence number of Nth frame window
  while(true)//check repeatedly
  do
    Wait_For_Event();//wait for availability of packet
    if(Event(Request_For_Transfer))then
      //check if window is full
      if(SeqN-SeqFirst>= S_window)then
        doNothing();
      endif;
      Get_Data_From_Network_Layer();
      s =Make_Frame();
      s.seq =SeqN;
      Store_Copy_Frame(s);
```

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Send_Fram

e(s);

Start_Timer(s);

SeqN=SeqN+1;

endif;

if(Event(Frame_Arrival)then

r=Receive_Acknowledgement();

//Resend frame whose sequence number is with ACK

if(r.type = NAK)then

if(NAK_No >SeqFirst&& NAK_No

<SeqN)then Retransmit(s.seq(NAK_No));

Start_Timer(s);

endif

//Remove frames from sending window with positive ACK

elseif(r.type = ACK)then

Remove_Frame(s.seq(SeqFirst))

; Stop_Timer(s);

SeqFirst=SeqFirst+1;

en

dif

endif

// Resend frame if acknowledgement haven't been received

if(Event(Time_Out))th

en Start_Timer(s);

Retransmit_Frame(s);

en

di

f

en

d

Receiver Site Algorithm of Selective Repeat Protocol

Begin

frame f;

RSeqNo=0; // Initialise sequence number of expected frame



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```
NAKsent=false;
se; ACK
=false;
For each slot in receive_window
Mark(slot)=false;
while(true)//check repeatedly
do
    Wait_For_Event();//wait for arrival of frame
    if(Event(Frame_Arrival)then
        Receive_Frame_From_Physical_Layer();
        if(Corrupted( f.SeqNo) AND
        NAKsent=false)then
            SendNAK(f.Seq
            No);
            NAKsent=true;
        endif
        if( f.SeqNo!=RSeqNo AND
        NAKsent=false)then SendNAK(f.SeqNo);
            NAKsent=true;
            if( f.SeqNoisin receive_window
            )then
                if(Mark(RSeqNo)=false)then
                    Store_frame(f.SeqN
                    o);
                    Mark(RSeqNo)=tr
                    ue;
                en
            dif
        endi
        f
    else
        while(Mark(RSeqNo))
```



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```
Extract_Data(RSeqNo);  
Deliver_Data_To_Network_Layer(  
); RSeqNo=RSeqNo+1;  
Send_ACK(RSeqNo);  
endwhile
```

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```
endif
endif
endwhile
end
```

EXAMPLE DATA LINK PROTOCOLS :

- Within a single building, LANs are widely used for interconnection, but most wide-area network infrastructure is built up from point-to-point lines.
- Here we will examine the data link protocols found on point-to-point lines in the Internet in two common situations.
- The first situation is when packets are sent over SONET optical fiber links in wide-area networks.
- These links are widely used, for example, to connect routers in the different locations of an ISP's network.
- The second situation is for ADSL links running on the local loop of the telephone network at the edge of the Internet.
- These links connect millions of individuals and businesses to the Internet. The Internet needs point-to-point links for these uses, as well as dial-up modems, leased lines, and cable modems, and so on.
- A standard protocol called PPP (Point-to-Point Protocol) is used to send packets over these links.
- SONET and ADSL links both apply PPP, but in different ways.
- It provides a bit stream that runs at a

well-defined rate. 1. Packet Over SONET:

- To carry packets across these links, some framing mechanism is needed to distinguish occasional packets from the continuous bitstream in which they are transported.

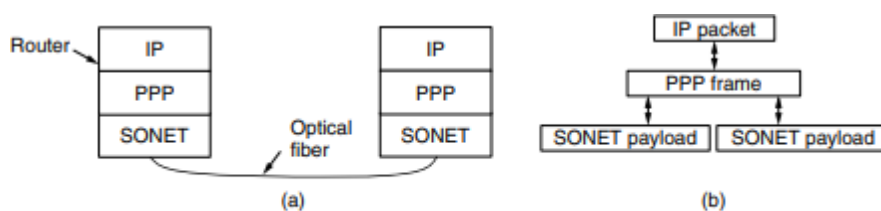


Figure 3-23. Packet over SONET. (a) A protocol stack. (b) Frame relationships.

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- PPP improves on an earlier, simpler protocol called SLIP (Serial Line Internet Protocol) and is used to handle error detection link configuration, support multiple protocols, permit authentication, and more.
- With a wide set of options, PPP provides three main features:
 1. A framing method that unambiguously delineates the end of one frame and the start of the next one. The frame format also handles error detection.
 2. A link control protocol for bringing lines up, testing them, negotiating options, and bringing them down again gracefully when they are no longer needed. This protocol is called LCP (Link Control Protocol).
 3. A way to negotiate network-layer options in a way that is independent of the network layer protocol to be used. The method chosen is to have a different NCP (Network Control Protocol) for each network layer supported.
- The PPP frame format was chosen to closely resemble the frame format of HDLC (High-level Data Link Control), a widely used instance of an earlier family of protocols, since there was no need to reinvent the wheel.
- **The primary difference between PPP and HDLC is that PPP is byte oriented rather than bit oriented. In particular, PPP uses byte stuffing and all frames are an integral number of bytes. HDLC uses bit stuffing and allows frames of, say, 30.25 bytes.**
- There is a second major difference in practice, however. HDLC provides reliable transmission with a sliding window, acknowledgements, and timeouts in the manner we have studied. PPP can also provide reliable transmission in noisy environments, such as wireless networks;

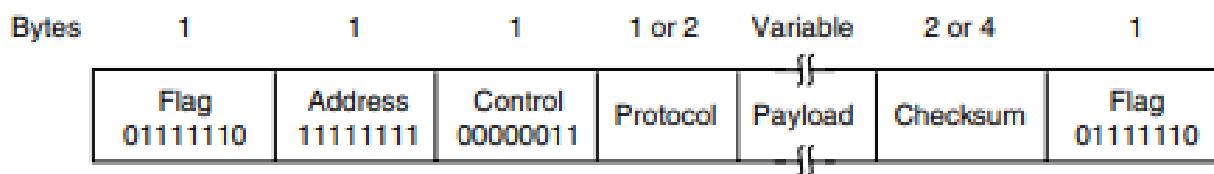


Figure 3-24. The PPP full frame format for unnumbered mode operation.

- All PPP frames begin with the standard HDLC flag byte of 0x7E (01111110).
- The PPP full frame format for unnumbered mode operation. **The Address field is followed by the Control field, the default value of which is 00000011.**

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- This value indicates an unnumbered frame. Since the Address and Control fields are always constant in the default configuration, LCP provides the necessary mechanism for the two parties to negotiate an option to omit them altogether and save 2 bytes per frame.
- The fourth PPP field is the Protocol field. Its job is to tell what kind of packet is in the Payload field. Codes starting with a 0 bit are defined for IP version 4, IP version 6, and other network layer protocols that might be used.
- Codes starting with a 1 bit are used for PPP configuration protocols, including LCP and a different NCP for each network layer protocol supported.
- The default size of the Protocol field is 2 bytes, but it can be negotiated down to 1 byte using LCP.
- The Payload field is variable length, up to some negotiated maximum. If the length is not negotiated using LCP during line setup, a default length of 1500 bytes is used. Padding may follow the payload if it is needed.
- After the Payload field comes the Checksum field, which is normally 2 bytes, but a 4-byte checksum can be negotiated.
- PPP is a framing mechanism that can carry the packets of multiple protocols over many types of physical layers.
- Before PPP frames can be carried over SONET lines, the PPP link must be established and configured.
- The phases that the link goes through when it is brought up, used, and taken down again are shown in Fig. 3-25.

THE DATA LINK LAYER

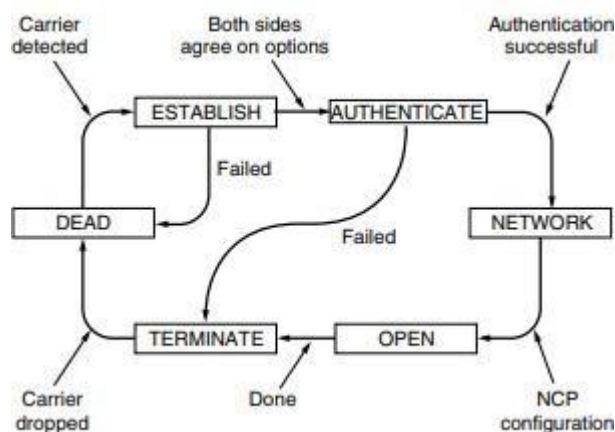


Figure 3-25. State diagram for bringing a PPP link up and down.

- Before PPP frames can be carried over SONET lines, the PPP link must be established and configured.

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- The phases that the link goes through when it is brought up, used, and taken down again are shown in Fig. 3-25.
- The link starts in the DEAD state, which means that there is no connection at the physical layer. When a physical layer connection is established, the link moves to ESTABLISH.
- At this point, the PPP peers exchange a series of LCP packets, each carried in the Payload field of a PPP frame, to select the PPP options for the link from the possibilities mentioned above.
- The initiating peer proposes options, and the responding peer either accepts or rejects them, in whole or part. The responder can also make alternative proposals.
- If LCP option negotiation is successful, the link reaches the AUTHENTICATE state. Now the two parties can check each other's identities, if desired.
- If authentication is successful, the NETWORK state is entered and a series of NCP packets are sent to configure the network layer.
- It is difficult to generalize about the NCP protocols because each one is specific to some network layer protocol and allows configuration requests to be made that are specific to that protocol.
- For IP, for example, the assignment of IP addresses to both ends of the link is the most important possibility.
- Once OPEN is reached, data transport can take place. It is in this state that IP packets are carried in PPP frames across the SONET line.
- When data transport is finished, the link moves into the TERMINATE state, and from there it moves back to the DEAD state when the physical layer connection is dropped.

ADSL (Asymmetric Digital Subscriber Loop):

- ADSL connects millions of home subscribers to the Internet at megabit/sec rates over the same telephone local loop that is used for plain old telephone service.
- The overall picture for the protocols and devices used with ADSL is shown in Fig. 3-26.
- Different protocols are deployed in different networks, so we have chosen to show the most popular scenario.
- Inside the home, a computer such as a PC sends IP packets to the DSL modem using a link layer like Ethernet.
- The DSL modem then sends the IP packets over the local loop to the DSLAM using the protocols that we are about to study.
- At the DSLAM (or a router connected to it depending on the implementation) the IP packets are extracted and enter an ISP network so that they may reach any destination on the Internet.

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The protocols shown over the ADSL link in Fig. 3-26 start at the bottom with the ADSL physical layer. They are based on a digital modulation scheme called

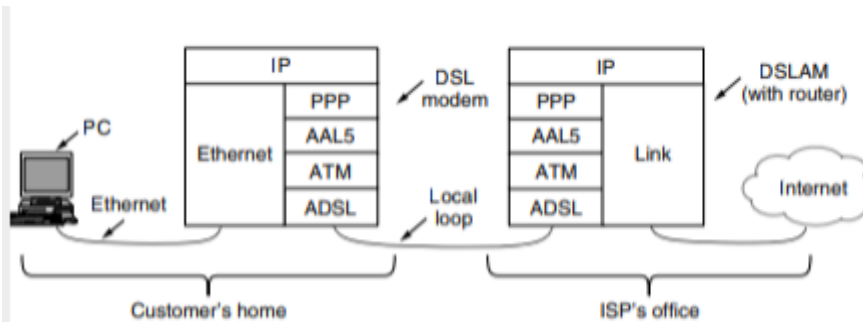


Figure 3-26. ADSL protocol stacks.

orthogonal frequency division multiplexing (also known as discrete multitone).

- Near the top of the stack, just below the IP network layer, is PPP.
- This protocol is the same PPP that we have just studied for packet over SONET transports.
- It works in the same way to establish and configure the link and carry IP packets.
- In between ADSL and PPP are ATM and AAL5. These are new protocols that we have not seen before.
- ATM (Asynchronous Transfer Mode) was designed in the early 1990s and launched with incredible hype.
- It promised a network technology that would solve the world's telecommunications problems by merging voice, data, cable television, telegraph, carrier pigeon, tin cans connected by strings, tom toms, and everything else into an integrated system that could do everything for everyone. This did not happen.
- In large part, the problems of ATM were similar to those we described concerning the OSI protocols, that is, bad timing, technology, implementation, and politics.
- Nevertheless, ATM was much more successful than OSI. While it has not taken over the world, it remains widely used in niches including broadband access lines such as DSL, and WAN links inside telephone networks. ATM is a link layer that is based on the transmission of fixed-length cells of information. The "Asynchronous" in its name means that the cells do not always need to be sent in the way that bits are continuously sent over synchronous lines, as in SONET. Cells only need to be sent when there is information to carry.
- ATM is a connection-oriented technology. Each cell carries a virtual circuit identifier in its header and devices use this identifier to forward cells along the paths of established connections.
- The cells are each 53 bytes long, consisting of a 48-byte payload plus a 5-byte header. By using small cells, ATM can flexibly divide the bandwidth of a physical layer link among different users in fine slices.

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- This ability is useful when, for example, sending both voice and data over one link without having long data packets that would cause large variations in the delay of the voice samples.
- The unusual choice for the cell length (e.g., compared to the more natural choice of a 250 THE DATA LINK LAYER CHAP. 3 power of 2) is an indication of just how political the design of ATM was. The 48-byte size for the payload was a compromise to resolve a deadlock between Europe, which wanted 32-byte cells, and the U.S., which wanted 64-byte cells.
- To send data over an ATM network, it needs to be mapped into a sequence of cells. This mapping is done with an ATM adaptation layer in a process called segmentation and reassem- bly.
- Several adaptation layers have been defined for different services, ranging from periodic voice samples to packet data.
- The main one used for packet data is AAL5 (ATM Adaptation Layer 5). An AAL5 frame is shown in Fig. 3-27.
- Instead of a header, it has a trailer that gives the length and has a 4-byte CRC for error detec- tion.
- Naturally, the CRC is the same one used for PPP and IEEE 802 LANs like Ethernet. As well as a payload, the AAL5 frame has padding.
- This rounds out the overall length to be a multiple of 48 bytes so that the frame can be evenly divided into cells.
- No addresses are needed on the frame as the virtual circuit identifier carried in each cell will get it to the right destination.

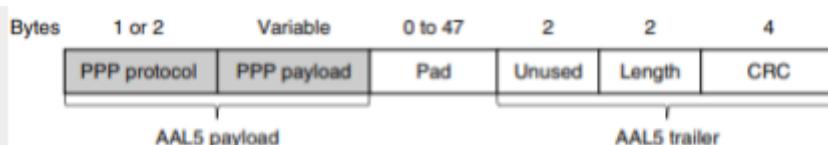


Figure 3-27. AAL5 frame carrying PPP data.

- Now that we have described ATM, we have only to describe how PPP makes use of ATM in the case of ADSL.
- It is done with yet another standard called PPPoA (PPP over ATM). This standard is not really a protocol (so it does not appear in Fig. 3-26) but more a specification of how to work with both PPP and AAL5 frames.
- It is described in RFC 2364 (Gross et al., 1998). Only the PPP protocol and payload fields are placed in the AAL5 payload, as shown in Fig. 3-27.
- The protocol field indicates to the DSLAM at the far end whether the payload is an IP packet or a packet from another protocol such as LCP.



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- The far end knows that the cells contain PPP information because an ATM virtual circuit is set up for this purpose.
- Within the AAL5 frame, PPP framing is not needed as it would serve no purpose; ATM and AAL5 already provide the framing.
- More framing would be worthless. The PPP CRC is also not needed because AAL5 already includes the very same CRC. This error detection mechanism supplements the ADSL physical layer coding of a Reed-Solomon code for error correction and a 1-byte CRC for the detection of any remaining errors not otherwise caught.
- This scheme has a much more sophisticated error-recovery mechanism than when packets are sent over a SONET line because ADSL is a much noisier channel.

Introduction

Network links can be divided into two categories: those using point-to-point connections and those using broadcast channels.

In any broadcast network, the key issue is how to determine who gets to use the channel when there is competition for it. To make this point, consider a conference call in which six people, on six different telephones, are all connected so that each one can hear and talk to all the others. Broadcast channels are sometimes referred to as **multiaccess channels** or **random access channels**.

The protocols used to determine who goes next on a multiaccess channel belong to a sublayer of the data link layer called the **MAC (Medium Access Control)** sublayer. The MAC sublayer is especially important in LANs, particularly wireless ones because wireless is naturally a broadcast channel. WANs, in contrast, use point-to-point links, except for satellite networks.

1. THE CHANNEL ALLOCATION PROBLEM

The central theme of this chapter is how to allocate a single broadcast channel among competing users. The channel might be a portion of the wireless spectrum in a geographic region, or a single wire or optical fiber to which multiple nodes are connected. It does not matter. In both cases, the channel connects each user to all other users and any user who makes full use of the channel interferes with other users who also wish to use the channel.

- Static Channel Allocation

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The traditional way of allocating a single channel, such as a telephone trunk, among multiple competing users is to chop up its capacity by using one of the multiplexing schemes we described in Sec. 2.5, such as FDM (Frequency Division Multiplexing). If there are N users, the bandwidth is divided into N equal-sized portions, with each user being assigned one portion. Since each user has a private frequency band, there is now no interference among users. When there is only a small and constant number of users, each of which has a steady stream or a heavy load of traffic, this division is a simple and efficient allocation mechanism.

When the number of senders is large and varying or the traffic is bursty, FDM presents some problems. If the spectrum is cut up into N regions and fewer than N users are currently interested in communicating, a large piece of valuable spectrum will be wasted. And if more than N users want to communicate, some of them will be denied permission for lack of bandwidth.

A static allocation is a poor fit to most computer systems, in which data traffic is extremely bursty, often with peak traffic to mean traffic ratios of 1000:1. Consequently, most of the channels will be idle most of the time.

The poor performance of static FDM can easily be seen with a simple queueing theory calculation. Let us start by finding the mean time delay, T , to send a frame onto a channel of capacity C bps. We assume that the frames arrive randomly with an average arrival rate of λ frames/sec, and that the frames vary in length with an average length of $1/\mu$ bits. With these parameters, the service rate of the channel is μC frames/sec. A standard queueing theory result is

$$T = \frac{1}{\mu C - \lambda}$$

(For the curious, this result is for an "M/M/1" queue. It requires that the randomness of the times between frame arrivals and the frame lengths follow an exponential distribution, or equivalently be the result of a Poisson process.)

In our example, if C is 100 Mbps, the mean frame length, $1/\mu$, is 10,000 bits, and the frame arrival rate, λ , is 5000 frames/sec, then $T = 200 \mu\text{sec}$. Note that if we ignored the queueing delay and just asked how long it takes to send a 10,000-bit frame on a 100-Mbps network, we would get the (incorrect) answer of 100 μsec . That result only holds when there is no contention for the channel.

Now let us divide the single channel into N independent subchannels, each with capacity C/N bps. The mean input rate on each of the subchannels will now be λ/N . Recomputing T ,

we
get

$$T_N = \frac{1}{\mu(C/N) - (\lambda/N)} =$$



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$$\frac{N}{\mu C - \lambda} = NT$$

Assumptions for Dynamic Channel Allocation

Underlying all the work done in this area are the following five key assumptions:

1. **Independent Traffic.** The model consists of N independent **stations** (e.g., com- puters, telephones), each with a program or user that gener- ates frames for transmission. The expected number of frames gener- ated in an interval of length Δt is $\lambda \Delta t$, where λ is a constant (the arri- val rate of new frames). Once a frame has been generated, the sta- tion is blocked and does nothing until the frame has been suc- cessfully transmitted.
2. **Single Channel.** A single channel is available for all communica- tion. All sta- tions can transmit on it and all can receive from it. The stations are assumed to be equally capable, though protocols may assign them different roles (e.g., pri- orities).
3. **Observable Collisions.** If two frames are transmitted simultan- eously, they overlap in time and the resulting signal is garbled. This event is called a **colli- sion**. All stations can detect that a collision has occurred. A collided frame must be transmitted again later. No er- rors other than those generated by collisions occur.
4. **Continuous or Slotted Time.** Time may be assumed continuous, in which case frame transmission can begin at any instant. Alterna- tively, time may be slotted or divided into discrete intervals (called slots). Frame transmissions must then begin at the start of a slot. A slot may contain 0, 1, or more frames, correspond- ing to an idle slot, a successful transmission, or a collision, respectively.
5. **Carrier Sense or No Carrier Sense.** With the carrier sense as- sumption, sta- tions can tell if the channel is in use before trying to use it. No station will attempt to use the channel while it is sensed as busy. If there is no carrier sense, stations cannot sense the channel before trying to use it. They just go ahead and transmit. Only later can they determine whether the transmission was success- ful.

MULTIPLE ACCESS PROTOCOLS

Many algorithms for allocating a multiple access channel are known.

ALOHA



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The story of our first MAC starts out in pristine Hawaii in the early 1970s. In this case, “pristine” can be interpreted as “not having a working telephone system.” This did not make life more pleasant for researcher Norman Abramson and his colleagues at the University of Hawaii who were trying to connect users on remote islands to the main computer in Honolulu. Stringing their own cables under the Pacific Ocean was not in the cards, so they looked for a different solution.

The one they found used short-range radios, with each user terminal sharing the same upstream frequency to send frames to the central computer. It included a simple and elegant method to solve the channel allocation problem. Their work has been extended by many researchers since then (Schwartz and Abramson, 2009). Although Abramson’s work, called the ALOHA system, used ground-based radio broadcasting, the basic idea is applicable to any system in which uncoordinated users are competing for the use of a single shared channel.

We will discuss two versions of ALOHA here: pure and slotted. They differ with respect to whether time is continuous, as in the pure version, or divided into discrete slots into which all frames must fit.

Pure ALOHA

The basic idea of an ALOHA system is simple: let users transmit whenever they have data to be sent. There will be collisions, of course, and the colliding frames will be damaged. Senders need some way to find out if this is the case. In the ALOHA system, after each station has sent its frame to the central computer, this computer rebroadcasts the frame to all of the stations. A sending station can thus listen for the broadcast from the hub to see if its frame has gotten through. In other systems, such as wired LANs, the sender might be able to listen for collisions while transmitting. If the frame was destroyed, the sender just waits a random amount of time and sends it again. The waiting time must be random or the same frames will collide over and over, in lockstep. Systems in which multiple users share a common channel in a way that can lead to conflicts are known as **contention** systems.

A sketch of frame generation in an ALOHA system is given in Fig. 4-1. We have made the frames all the same length because the throughput of ALOHA systems is maximized by having a uniform frame size rather than by allowing variable-length frames.

Whenever two frames try to occupy the channel at the same time, there will be a collision (as seen in Fig. 4-1) and both will be garbled. If the first bit of a new frame overlaps with just the last bit of a frame that has almost finished, both frames will be totally destroyed (i.e., have incorrect checksums) and both will have to be retransmitted later. The checksum does not (and should not) distinguish between a total loss and a near miss. Bad is bad.



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In pure ALOHA, frames are transmitted at completely arbitrary times.

An interesting question is: what is the efficiency of an ALOHA channel? In other words, what fraction of all transmitted frames escape collisions under these chaotic circumstances? Let us first consider an infinite collection of users typing at their terminals (stations). A user is always in one of two states: typing or waiting. Initially, all users are in the typing state. When a line is finished, the user stops typing, waiting for a response. The station then transmits a frame containing the line over the shared channel to the central computer and checks the channel to see if it was successful. If so, the user sees the reply and goes back to typing. If not, the user continues to wait while the station retransmits the frame over and over until it has been successfully sent.

Let the “frame time” denote the amount of time needed to transmit the standard, fixed-length frame (i.e., the frame length divided by the bit rate). If $N > 1$, the user community is generating frames at a higher rate than the channel can handle, and nearly every frame will suffer a collision. For reasonable throughput, we would expect $0 < N < 1$. In addition to the new frames, the stations also generate retransmissions of frames that previously suffered collisions. Let us further assume that the old and new frames combined are well modeled by a Poisson distribution, with mean of G frames per frame time. Clearly, $G \geq N$. At low load (i.e., $N \sim 0$), there will be few collisions, hence few retransmissions, so $G \sim N$. At high load, there will be many collisions, so $G > N$. Under all loads, the throughput, S , is just the offered load, G , times the probability, P_0 , of a transmission succeeding—that is, $S = GP_0$, where P_0 is the probability that a frame does not suffer a collision.

A frame will not suffer a collision if no other frames are sent within one frame time of its start, as shown in Fig.

Let t be the time required to send one frame. If any other user has generated a frame between time t_0 and $t_0 + t$, the end of that frame will collide with the beginning of the shaded one. In fact, the shaded frame's fate was already sealed even before the first bit was sent, but since in pure ALOHA a station does not listen to the channel before transmitting, it has no way of knowing that another frame was already



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underway.

Vulnerable



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Vulnerable period for the shaded frame.

The probability that k frames are generated during a given frame time, in which G frames are expected, is given by the Poisson distribution

$$\Pr[k] =$$

$$\frac{G^k e^{-G}}{k!} \quad (4-2)$$

so the probability of zero frames is just e^{-G} . In an interval two frame times long, the mean number of frames generated is $2G$. The probability of no frames being initiated during the entire vulnerable period is thus given by $P_0 = e^{-2G}$. Using $S = GP_0$, we get

$$S = Ge^{-2G}$$

The relation between the offered traffic and the throughput is shown in Fig. The maximum throughput occurs at $G = 0.5$, with $S = 1/2e$, which is about 0.184

Slotted ALOHA

Roberts (1972) published a method for doubling the capacity of an ALOHA system. His proposal was to divide time into discrete intervals called **slots**, each interval corresponding to one frame. This

In Roberts' method, which has come to be known as **slotted ALOHA**—in contrast to Abramson's **pure ALOHA**—a station is not permitted to send whenever the user types a line. Instead, it is required to wait for the beginning of the next slot. The probability of no other traffic during the same slot as our test frame is then e^{-G} , which leads to

$$S = Ge^{-G}$$

As you can see from Fig. 4-3, slotted ALOHA peaks at $G = 1$, with a throughput of $S = 1/e$ or about 0.368, twice that of pure ALOHA. If the system is operating at $G = 1$, the probability of an empty slot is 0.368 (from Eq. 4-2). The best we can hope for using slotted ALOHA is 37% of the slots empty, 37% successes, and 26% collisions.

$$= \sum_{k=0}^{\infty} k e^{-G} (1 - e^{-G})^{k-1} = e^{-G}$$

Carrier Sense Multiple Access Protocols

- With slotted ALOHA, the best channel utilization that can be achieved is $1/e$.



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- Protocols in which stations listen for a carrier (i.e., a transmission) and act accordingly are called carrier sense protocols.

Persistent and Nonpersistent

CSMA 1-persistent CSMA

- The first carrier sense protocol that we will study here is called 1-persistent
 - CSMA (Carrier Sense Multiple Access). That is a bit of a mouthful for the simplest CSMA scheme. When a station has data to send, it first listens to the channel to see if anyone else is transmitting at that moment.
 - If the channel is idle, the stations sends its data. Otherwise, if the channel is busy, the station just waits until it becomes idle. Then the station transmits a frame.
 - If a collision occurs, the station waits a random amount of time and starts all over again. The protocol is called 1-persistent because the station transmits with a probability of 1 when it finds the channel idle.
 - You might expect that this scheme avoids collisions except for the rare case of simultaneous sends, but it in fact it does not.
 - If two stations become ready in the middle of a third station's transmission, both will wait politely until the transmission ends, and then both will begin transmitting exactly simultaneously, resulting in a collision. If they were not so impatient, there would be fewer collisions.
-
- Even so, this protocol has better performance than pure ALOHA because both stations have the decency to desist from interfering with the third station's frame.
 - Exactly the same holds for slotted ALOHA.

nonpersistent CSMA

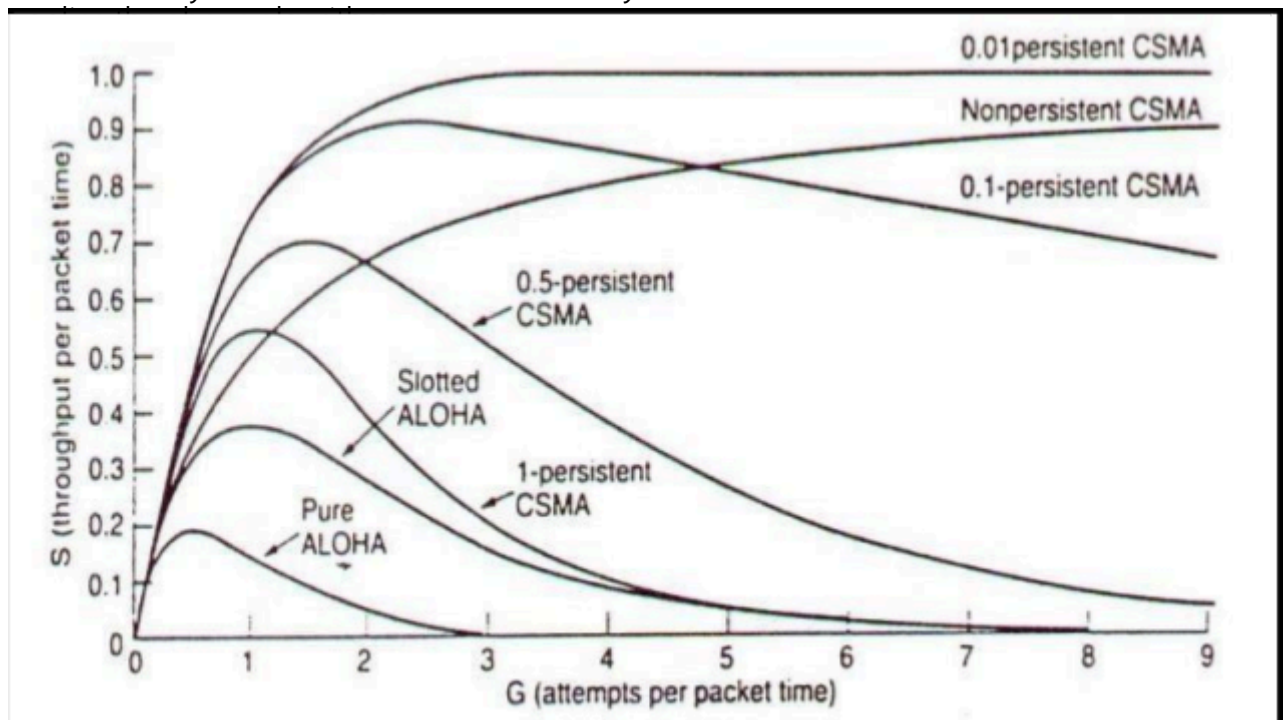
- A second carrier sense protocol is nonpersistent CSMA. In this protocol, a conscious attempt is made to be less greedy than in the previous one. As before, a station senses the channel when it wants to send a frame, and if no one else is sending, the station begins doing so itself.
- However, if the channel is already in use, the station does not continually sense it for the purpose of seizing it immediately upon detecting the end of the previous transmission. Instead, it waits a random period of time and then repeats the algorithm.
- Consequently, this algorithm leads to better channel utilization but longer delays than 1-persistent CSMA.

p-persistent CSMA

- The last protocol is p-persistent CSMA. It applies to slotted channels and works as follows. When a station becomes ready to send, it senses the channel.

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- If it is idle, it transmits with a probability p . With a probability $q = 1 - p$, it defers until the next slot. If that slot is also idle, it either transmits or defers again, with probabilities p and q .
- This process is repeated until either the frame has been transmitted or another station has begun transmitting. In the latter case, the unlucky station acts as if there had been a collision (i.e., it waits a random time and starts again). If the station initially senses that the channel is busy, it waits until the next slot and



discuss in Sec. 4.4. Figure 4-4 shows the computed throughput versus offered traffic for all three protocols, as well as for pure and slotted ALOHA.

CSMA with Collision Detection

- Persistent and nonpersistent CSMA protocols are definitely an improvement over ALOHA because they ensure that no station begins to transmit while the channel is busy.
- However, if two stations sense the channel to be idle and begin transmitting simultaneously, their signals will still collide.

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- Another improvement is for the stations to quickly detect the collision and abruptly stop transmitting, (rather than finishing them) since they are irretrievably garbled anyway.
 - This strategy saves time and bandwidth. This protocol, known as CSMA/CD (CSMA with Collision Detection), is the basis of the classic Ethernet LAN, so it is worth devoting some time to looking at it in detail.
 - It is important to realize that collision detection is an analog process. The station's hardware must listen to the channel while it is transmitting.
 - If the signal it reads back is different from the signal it is putting out, it knows that a collision is occurring.
-
- CSMA/CD, as well as many other LAN protocols, uses the conceptual model of Fig. 4-5. At the point marked t_0 , a station has finished transmitting its frame.
 - Any other station having a frame to send may now attempt to do so. If two or more stations decide to transmit simultaneously, there will be a collision. If a station detects a collision, it aborts its transmission, waits a random period of time, and then tries again (assuming that no other station has started transmitting in the meantime).
 - Therefore, our model for CSMA/CD will consist of alternating contention and transmission periods, with idle periods occurring when all stations are quiet (e.g., for lack of work).

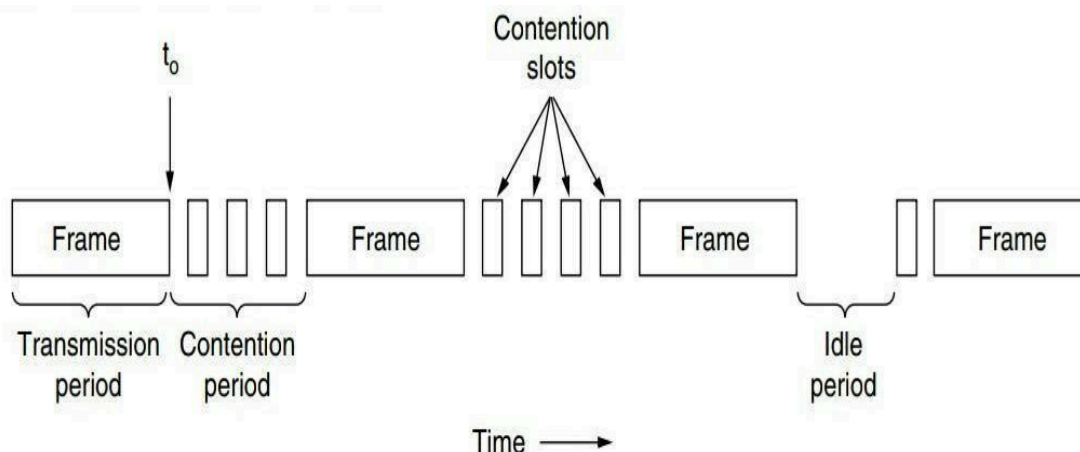


Figure 4-5. CSMA/CD can be in contention, transmission, or idle state.

- Now let us look at the details of the contention algorithm. Suppose that two stations both begin transmitting at exactly time t_0 . How long will it take them to realize that they have



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collided? The answer is vital to determining the length of the contention period and hence what the delay and throughput will be.

- The minimum time to detect the collision is just the time it takes the signal to propagate from one station to the other.
- With this understanding, we can think of CSMA/CD contention as a slotted ALOHA system with a slot width of 2τ . On a 1-km long coaxial cable, $\tau \sim 5 \mu\text{sec}$.

Collision:

If two frames are transmitted simultaneously, they overlap in time and the resulting signal is garbled. This event is called a collision. All stations can detect that a collision has occurred. A collided frame must be transmitted again later. No errors other than those generated by collisions occur.

Adhoc network

Ad hoc networks are mostly wireless local area networks (LANs). The devices communicate with each other directly instead of relying on a base station or access points as in wireless LANs for data transfer co-ordination. Each device participates in routing activity, by determining the route using the routing algorithm and forwarding data to other devices via this route.

Bridge

A bridge is a network device that connects two or more LANs (local area networks) to form a larger LAN. Network bridging refers to the process of aggregating networks. A bridge connects the various components, making them appear to be part of a single network.

Carrier sense protocols

Protocols in which stations listen for a carrier and act accordingly are called carrier sense protocols.

rate adaption



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Rate adaptation is a criteria that determines the performance of IEEE 802.11 wireless networks, or WiFi. Rate adaptation allows transmission to be done at different rates within the wireless network, depending upon the network conditions.

In wireless networks, the signal may be strong or weak. Through rate adaptation technique, the data transfer rate can be changed depending upon the signal strength, i.e. when the signal is strong, high data rates are adopted, while low data rates are adopted during weak signals.

Virtual LAN

Virtual Local Area Networks or Virtual LANs (VLANs) are a logical group of computers that appear to be on the same LAN irrespective of the configuration of the underlying physical network. Network administrators partition the networks to match the functional requirements of the VLANs so that each VLAN comprise of a subset of ports on a single or multiple switches or bridges.

CSMA/CA protocol

The **Carrier Sense Multiple Access/ Collision Detection** protocol is used to detect a collision in the media access control (MAC) layer. Once the collision was detected, the CSMA CD immediately stopped the transmission by sending the signal so that the sender does not waste all the time to send the data packet. Suppose a collision is detected from each station while broadcasting the packets

Bit map protocol

First collision-free protocol, the basic bit-map method, each contention period consists of exactly N slots. If station 0 has a frame to send, it transmits a 1 bit during the zeroth slot. No other station is allowed to transmit during this slot. Regardless of what station 0 does, station 1 gets the opportunity to transmit a 1 during slot 1, but only if it has a frame queued. In general, station j may announce that it has a frame to send by inserting a 1 bit into slot j. After all N slots have passed by, each station has complete knowledge of which stations wish to transmit. At that point, they begin transmitting in numerical order.

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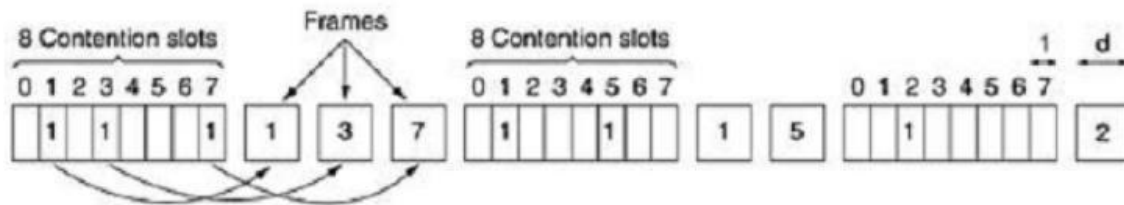


Fig.6 The basic bit-map protocol

Since everyone agrees on who goes next, there will never be any collisions. After the last ready station has transmitted its frame, an event all stations can easily monitor, another N bit contention period is begun. If a station becomes ready just after its bit slot has passed by, it is out of luck and must remain silent until every station has had a chance and the bit map has come around again. Protocols like this in which the desire to transmit is broadcast before the actual transmission are called reservation protocols.

Binary countdown protocol

Binary Countdown Protocol is a collision-free protocol that operates in the Medium Access Control (MAC) layer of the OSI model. Collision-free protocols resolves channel access while the stations are contending for the shared channel, thus eliminating any possibilities of collisions.

Working Principle of Binary Countdown

In a binary countdown protocol, each station is assigned a binary address. The binary addresses are bit strings of equal lengths. When a station wants to transmit, it broadcasts its address to all the stations in the channel, one bit at a time starting with the highest order bit.

In order to decide which station gets the channel access, the addresses of the stations which are broadcasted are ORed. The higher numbered station gets the channel access.

Example

Suppose that six stations contend for channel access which have the addresses: 1011, 0010, 0101, 1100, 1001 and 1101.

The iterative steps are –

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- All stations broadcast their most significant bit, i.e. 1, 0, 0, 1, 1, 1. Stations 0010 and 0101 sees 1 bit in other stations, and so they give up competing for the channel.
- The stations 1011, 1100, 1001 and 1101 continue. They broadcast their next bit,
i.e. 0, 1, 0, 1. Stations 1011 and 1001 sees 1 bit in other stations, and so they give up competing for the channel.
- The stations 1100 and 1101 continue. They broadcast their next bit, i.e. 0, 0. Since both of them have same bit value, both of them broadcast their next bit.
- The stations 1100 and 1101 broadcast their least significant bit, i.e. 0 and 1. Since station 1101 has 1 while the other 0, station 1101 gets the access to the channel.
- After station 1101 has completed frame transmission, or there is a time-out, the next contention cycle starts.

Station Address	Bit Time				Station Status
	0	1	2	3	
1011	1	0	-	-	Gives up after bit time 1
0010	0	-	-	-	Gives up after bit time 0
0101	0	-	-	-	Gives up after bit time 0
1100	1	1	0	0	Gives up after bit time 3
1001	1	0	-	-	Gives up after bit time 0
1101	1	1	0	1	Gets channel access

Uses of bridges

- Bridges connects two or more different LANs that has a similar protocol and provides communication between the devices (nodes) in them.
- By joining multiple LANs, bridges help in multiplying the network capacity of a single LAN.
- Since they operate at data link layer, they transmit data as data frames. On receiving a data frame, the bridge consults a database to decide whether to pass, transmit or discard the frame.

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- If the frame has a destination MAC (media access control) address in the same network, the bridge passes the frame to that node and then discards it.
- If the frame has a destination MAC address in a connected network, it will forward the frame toward it.

By deciding whether to forward or discard a frame, it prevents a single faulty node from bringing down the entire network.

In cases where the destination MAC address is not available, bridges can broadcast data frames to each node. To discover new segments, they maintain the MAC address table.

In order to provide full functional support, bridges ideally need to be transparent. No major hardware, software or architectural changes should be required for their installation.

Bridges can switch any kind of packets, be it IP packets or AppleTalk packets, from the network layer above. This is because bridges do not examine the payload field of the data frame that arrives, but simply looks at the MAC address for switching. Bridges also connect virtual LANs (VLANs) to make a larger VLAN.

802.11 physical layer

All of the transmission methods also define multiple rates. If the wireless signal is weak, a low rate can be used. If the signal is clear, the highest rate can be used. This adjustment is called rate adaptation. The first transmission method we shall look at is 802.11b. It is a spread-spectrum method that supports rates of 1, 2, 5.5, and 11 Mbps, though in practice the operating rate is nearly always 11 Mbps. The spreading sequence used by 802.11b is a Barker sequence. 802.11a, which supports rates up to 54 Mbps in the 5-GHz ISM band. You might have expected that 802.11a to come before 802.11b, but that was not the case. The 802.11a method is based on OFDM (Orthogonal Frequency Division Multiplexing) because OFDM uses the spectrum efficiently and resists wireless signal degradations such as multipath. 802.11g copies the OFDM modulation methods of 802.11a but operates in the narrow 2.4-GHz ISM band along with 802.11b. It offers the same rates as 802.11a (6 to 54 Mbps) plus of course compatibility with any 802.11b devices that happen to be nearby. The goal for 802.11n was throughput of at least 100 Mbps after all the wireless overheads were removed. This goal called for a raw speed increase of at least a factor of four. r 802.11n was throughput of at least 100

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Mbps after all the wireless overheads were removed. This goal called for a raw speed increase of at least a factor of four.

Usage of Bridge, Router in datalink layer and network layer

Router:

It is an interconnecting device that works at the physical, data link and network layer of OSI reference model. It may connect LAN and WAN in the network and can pass or relay the packets among them. It has access to the network or logical address. The following figure displays the router acts as a station in the network.



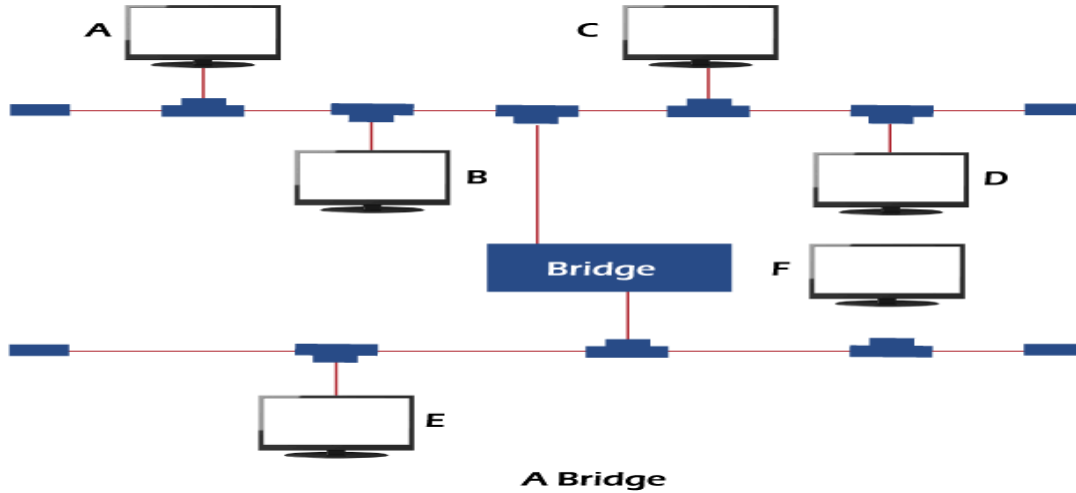
Routers is an internet

They have links with two or more networks and contain the addresses of the stations on all these networks. It receives the packets from one connected network and passes them to a second connected network. If a received packet contains the address of a node that is on some other network, the Router determines which of its connected networks the best next relay point for that packet.

Bridge

A bridge is a networking device that works in both the physical and data link layer in a network. This devices can divide a large network into smaller segments and pass the frames between two originally separated LANs. A bridge maintains a MAC address of various stations attached to it. When a frames enters a bridge, it checks the address contained in the frame and compares it with a table of all the stations on both segments.

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The 802.1Q Ethernet frame format

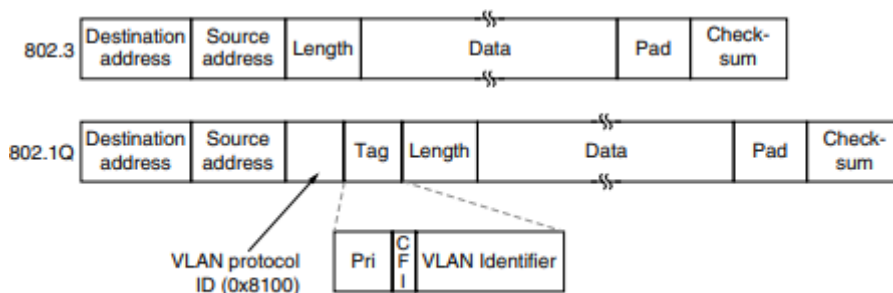


Figure 4-49. The 802.3 (legacy) and 802.1Q Ethernet frame formats.

The first one is the VLAN protocol ID. It always has the value 0x8100. The second 2- byte field contains three subfields. The main one is the VLAN identifier, occupying the low-order 12 bits. The last field, CFI (Canonical format indicator), should have been called the CEI (Corporate ego indicator). It was originally intended to indicate the order of the bits in the MAC addresses . With 802.1Q, frames are colored depending on the port on which they are received. For this method to work, all machines on a port must belong to the same VLAN, which reduces flexibility.

- Preamble – An 8-byte starting field that provides alert and timing pulse for transmission.
- Destination Address – A 6-byte field containing physical address of destination stations.

Source Address – A 6-byte field containing the physical address of the sending station.



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Channel allocation problems

Channel Allocation Schemes

Channel Allocation may be done using two schemes –

- Static Channel Allocation
- Dynamic Channel Allocation

Static Channel Allocation

In static channel allocation scheme, a fixed portion of the frequency channel is allotted to each user. For N competing users, the bandwidth is divided into N channels using frequency division multiplexing (FDM), and each portion is assigned to one user.

This scheme is also referred as fixed channel allocation or fixed channel assignment.

In this allocation scheme, there is no interference between the users since each user is assigned a fixed channel. However, it is not suitable in case of a large number of users with variable bandwidth requirements.

Dynamic Channel Allocation

In dynamic channel allocation scheme, frequency bands are not permanently assigned to the users. Instead channels are allotted to users dynamically as needed, from a central pool. The allocation is done considering a number of parameters so that transmission interference is minimized.

This allocation scheme optimises bandwidth usage and results in faster transmissions.

Dynamic channel allocation is further divided into centralised and distributed allocation.

Assumptions for Dynamic Channel Allocation

The key assumptions of these methods are –



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- **Independent Traffic** – It is assumed that the users are independent of each other,
i.e. each user has to send or receive data which is independent of the data transmission of the other users in the network system. Each user has a program for generating frames for transmission. Once the user generates a frame for transmission, it is blocked until the frame has been successfully transmitted.
- **Single Channel** – The algorithms assume that all contending stations request for transmission via a single channel. All stations are considered to be equally capable of transmission. However, the algorithm may assign different priorities to them for contention.
- **Detectable Collisions** – If two frames from different stations are simultaneously transmitted, the resulting signal is distorted, and a collision is said to occur. If a collision occurs, all stations should be able to detect the collision. The frames lost due to collisions are retransmitted.
- **Continuous Time or Slotted Time** – Depending upon the allocation scheme, time may be considered as continuous or slotted. In continuous time, frame transmission can start at any instant. In slotted time, time is divided into discrete slots. If a slot does not contain any frame, it is called an idle slot; if it contains a single frame, then the transmission is successful; if it contains more than one frames, then a collision is said to occur.
- **Carrier Sense or No Carrier Sense** – The stations may or may not be capable of detecting whether the channel is in use before sending the frames. In algorithms which are based upon carrier sense, a station sends frame only when it senses that the channel is not busy. On the other hand, in algorithms based upon no carrier sense, the stations transmit a frame when it is available and later are informed whether successful transmission had occurred or not.

Limited contention protocols

Limited Contention Protocols are the media access control (MAC) protocols that combines the advantages of collision based protocols and collision free protocols. They behave like slotted ALOHA under light loads and bitmap protocols under heavy loads.

Limited contention protocols divide the contending stations into groups, which may or not be disjoint. At slot 0, only stations in group 0 can compete for channel access. At



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slot 1, only stations in group 1 can compete for channel access and so on. In this process, if a station successfully acquires the channel, then it transmits its data frame. If there is a collision or there are no stations competing for a given slot in a group, the stations of the next group can compete for the slot.

By dynamically changing the number of groups and the number of stations allotted in a group according to the network load, the protocol changes from slotted ALOHA under low loads to bit map protocol under high loads. Under low loads, only one group is there containing all stations, which is the case of slotted ALOHA. As the load increases, more groups are added and the size of each group is reduced. When the load is very high, each group has just one station, i.e. only one station can compete at a slot, which is the case of bit map protocol.

The performance of limited contention protocol is highly dependent upon the algorithm to dynamically adjust the group configurations to the changes in network environment.

802.11 MAC sub layer Protocol

The 802.11 MAC sublayer protocol is quite different from that of Ethernet, due to two factors that are fundamental to wireless communication. 802.11 tries to avoid collisions with a protocol called CSMA/CA (CSMA with Collision Avoidance). This protocol is conceptually similar to Ethernet's CSMA/CD, with channel sensing before sending and exponential back off after collisions. An example timeline is shown in Fig. 4-25. Station A is the first to send a frame. While A is sending, stations B and C become ready to send. They see that the channel is busy and wait for it to become idle. Shortly after A receives an acknowledgement, the channel goes idle. However, rather than sending a frame right away and colliding, B and C both perform a backoff. C picks a short backoff,

and thus sends first. B pauses its countdown while it senses that C is using the channel, and resumes after C has received an acknowledgement. B soon completes its backoff and sends its frame.

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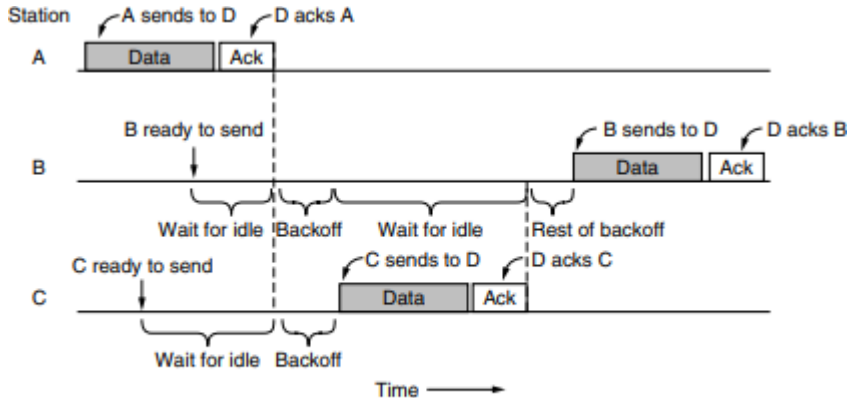


Figure 4-25. Sending a frame with CSMA/CA.

Compared to Ethernet, there are two main differences. First, starting backoffs early helps to avoid collisions. This avoidance is worthwhile because collisions are expensive, as the entire frame is transmitted even if one occurs. Second, acknowledgements are used to infer collisions because collisions cannot be detected.

This mode of operation is called DCF (Distributed Coordination Function) because each station acts independently, without any kind of central control. The standard also includes an optional mode of operation called PCF (Point Coordination Function) in which the access point controls all activity in its cell, just like a cellular base station.

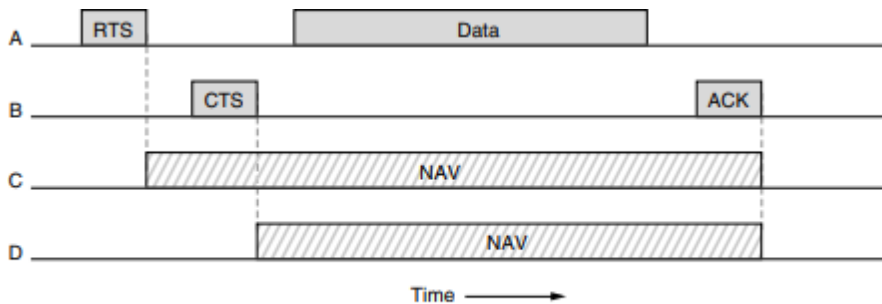


Figure 4-27. Virtual channel sensing using CSMA/CA.

An optional RTS/CTS mechanism uses the NAV to prevent terminals from sending frames at the same time as hidden terminals. It is shown in Fig. 4-27. In this example, A wants to send to B. C is a station within range of A (and possibly within range of B, but that does not matter). D is a station within range of B but not within range of A. The protocol starts when A decides it wants to send data to B. A begins by sending an



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RTS frame to B to request permission to send it a frame. If B receives this request, it answers with a CTS frame to indicate that the channel is clear to send. Upon receipt of the CTS, A sends its frame and starts an ACK timer. Upon correct receipt of the data frame, B responds with an ACK frame, completing the exchange. If A's ACK timer expires before the ACK gets back to it, it is treated as a collision and the whole protocol is run again after a backoff.

Services provided by 802.11

The **association service** is used by mobile stations to connect themselves to APs. Typically, it is used just after a station moves within radio range of the AP. Upon arrival, the station learns the identity and capabilities of the AP, either from beacon frames or by directly asking the AP.

Reassociation lets a station change its preferred AP. This facility is useful for mobile stations moving from one AP to another AP in the same extended 802.11 LAN, like a handover in the cellular network.

Stations must also **authenticate** before they can send frames via the AP, but authentication is handled in different ways depending on the choice of security scheme.

If the 802.11 network is "open," anyone is allowed to use it. Otherwise, credentials are needed to authenticate. The recommended scheme, called WPA2 (WiFi Protected Access 2), implements security as defined in the 802.11i standard.

The scheme that was used before WPA is called WEP (Wired Equivalent Privacy). For this scheme, authentication with a preshared key happens before association.

Once frames reach the AP, the distribution service determines how to route them. If the destination is local to the AP, the frames can be sent out directly over the air. Otherwise, they will have to be forwarded over the wired network.

Integration service handles any translation that is needed for a frame to be sent outside the 802.11 LAN, or to arrive from outside the 802.11 LAN.

Data transmission is what it is all about, so 802.11 naturally provides a **data delivery service**.



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Wireless is a broadcast signal. For information sent over a wireless LAN to be kept confidential, it must be encrypted. This goal is accomplished with a **privacy service** that manages the details of encryption and decryption.

To handle traffic with different priorities, there is a **QOS traffic scheduling service**.

Finally, there are two services that help stations manage their use of the spectrum. The transmit power control service gives stations the information they need to meet regulatory limits on transmit power that vary from region to region. The dynamic frequency selection service give stations the information they need to avoid transmitting on frequencies in the 5-GHz band that are being used for radar in the proximity.

Spanning tree bridges

To increase reliability, redundant links can be used between bridges. the bridges to communicate with each other and overlay the actual topology with a spanning tree that reaches every bridge. In effect, some potential connections between bridges are ignored in the interest of constructing a fictitious loop-free topology that is a subset of the actual topology.

For example, in Fig. 4-44 we see five bridges that are interconnected and also have stations connected to them. Each station connects to only one bridge. There are some redundant connections between the bridges so that frames will be forwarded in loops if all of the links are used. This topology can be thought of as a graph in which the bridges are the nodes and the point-to-point links are the edges. The graph can be reduced to a spanning tree, which has no cycles by definition, by dropping the links shown as dashed lines in Fig. 4-44. Using this spanning tree, there is exactly one path from every station to every other station. Once the bridges have agreed on the spanning tree, all forwarding between stations follows the spanning tree. Since there is a unique path from each source to each destination, loops are impossible.

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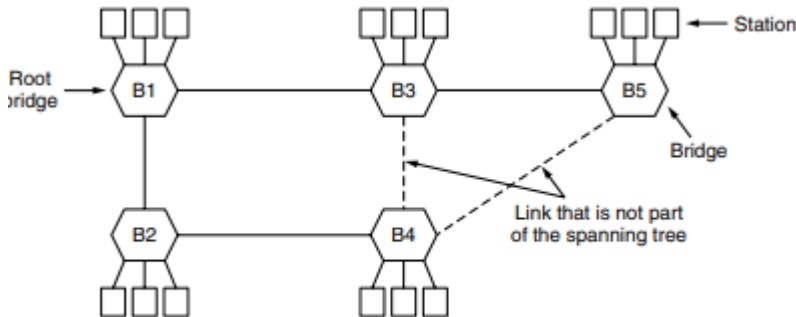


Figure 4-44. A spanning tree connecting five bridges. The dashed lines are links that are not part of the spanning tree.

To build the spanning tree, the bridges run a distributed algorithm. Each bridge periodically broadcasts a configuration message out all of its ports to its neighbors and processes the messages it receives from other bridges, as described next. These messages are not forwarded, since their purpose is to build the tree, which can then be used for forwarding. The bridges must first choose one bridge to be the root of the spanning tree. To make this choice, they each include an identifier based on their MAC address in the configuration message, as well as the identifier of the bridge they believe to be the root. MAC addresses are installed by the manufacturer and guaranteed to be unique worldwide, which makes these identifiers convenient and unique. The bridges choose the bridge with the lowest identifier to be the root. After enough messages have been exchanged to spread the news, all bridges will agree on which bridge is the root. In Fig. 4-44, bridge B1 has the lowest identifier and becomes the root. Next, a tree of shortest paths from the root to every bridge is constructed. In Fig. 4-44, bridges B2 and B3 can each be reached from bridge B1 directly, in one hop that is a shortest path. Bridge B4 can be reached in two hops, via either B2 or B3. To break this tie, the path via the bridge with the lowest identifier is chosen, so B4 is reached via B2. Bridge B5 can be reached in two hops via B3. To find these shortest paths, bridges include the distance from the root in their configuration messages. Each bridge remembers the shortest path it finds to the root. The bridges then turn off ports that are not part of the shortest path.

Learning bridges

The topology of two LANs bridged together is shown in Fig. 4-41 for two cases. On the left-hand side, two multidrop LANs, such as classic Ethernet, are joined by a special station—the bridge—that sits on both LANs. On the right-hand side, LANs with

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point-to-point cables, including one hub, are joined together. The bridges are the devices to which the stations and hub are attached. If the LAN technology is Ethernet, the bridges are better known as Ethernet switches.

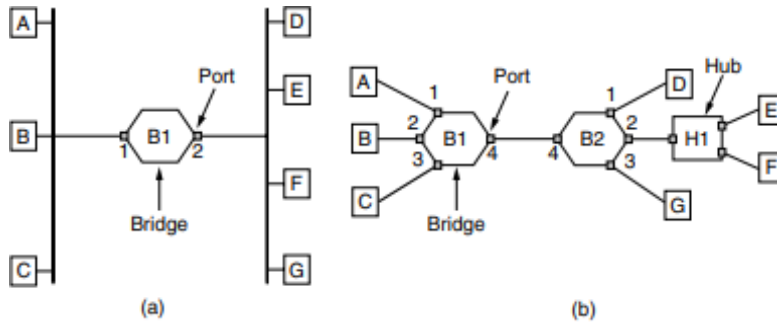


Figure 4-41. (a) Bridge connecting two multidrop LANs. (b) Bridges (and a hub) connecting seven point-to-point stations.

Bridges were developed when classic Ethernets were in use, so they are often shown in topologies with multidrop cables, as in Fig. 4-41(a). However, all the topologies that are encountered today are comprised of point-to-point cables and switches. The bridges work the same way in both settings. To bridge multidrop LANs, a bridge is added as a new station on each of the multidrop LANs, as in Fig. 4-41(a). To bridge point-to-point LANs, the hubs are either connected to a bridge or, preferably, replaced with a bridge to increase performance. In Fig. 4-41(b), bridges have replaced all but one hub. This arrangement is useful for bridging LANs in different buildings. Now let us consider what happens inside the bridges. Each bridge operates in promiscuous mode, that is, it accepts every frame transmitted by the stations attached to each of its ports. The bridge must decide whether to forward or discard each frame, and, if the former, on which port to output the frame. This decision is made by using the destination address. As an example, consider the topology of Fig. 4-41(a). If station A sends a frame to station B, bridge B1 will receive the frame on port 1. This frame can be immediately discarded without further ado because it is already on the correct port. However, in the topology of Fig. 4-41(b) suppose that A sends a frame to D. Bridge B1 will receive the frame on port 1 and output it on port 4. Bridge B2 will then receive the frame on its port 4 and output it on its port 1. A simple way to implement this scheme is to have a big (hash) table inside the bridge. The table can list each possible destination and which output port it belongs on. For example, in Fig. 4-41(b), the table at B1 would list D as belonging to port 4, since all B1 has to know is which port to put frames on to reach D. That, in fact, more forwarding will happen later when the frame hits B2 is not of interest to



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B1 When the bridges are first plugged in, all the hash tables are empty. None of the bridges know where any of the destinations are, so they use a flooding algorithm: every incoming frame for an unknown destination is output on all the ports to which the bridge is connected except the one it arrived on. As time goes on, the bridges learn where destinations are. Once a destination is known, frames destined for it are put only on the proper port; they are not flooded. The algorithm used by the bridges is backward learning. As mentioned above, the bridges operate in promiscuous mode, so they see every frame sent on any of their ports. By looking at the source addresses, they can tell which machines are accessible on which ports. For example, if bridge B1 in Fig. 4- 41(b) sees a frame on port 3 coming from C, it knows that C must be reachable via port 3, so it makes an entry in its hash table. Any subsequent frame addressed to C coming in to B1 on any other port will be forwarded to port 3.

ALOHA

The ALOHA system, used groundbased radio broadcasting, the basic idea is applicable to any system in which uncoordinated users are competing for the use of a single shared channel.

two versions of ALOHA here: **pure and slotted**. They differ with respect to whether time is continuous, as in the pure version, or divided into discrete slots into which all frames must fit.

Pure ALOHA The basic idea of an ALOHA system is simple: let users transmit when- ever they have data to be sent. There will be collisions, of course, and the colliding frames will be damaged. Senders need some way to find out if this is the case. In the ALOHA system, after each station has sent its frame to the central computer, this com- puter rebroadcasts the frame to all of the stations. A sending station can thus listen for the broadcast from the hub to see if its frame has gotten through. In other systems, such as wired LANs, the sender might be able to listen for collisions while transmitting. If the frame was destroyed, the sender just waits a random amount of time and sends it again. The waiting time must be random or the same frames will collide over and over, in lockstep. Systems in which multiple users share a common channel in a way that can lead to conflicts are known as contention systems.

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A sketch of frame generation in an ALOHA system is given in Fig. 4-1. We have made the frames all the same length because the throughput of ALOHA systems is maximized by having a uniform frame size rather than by allowing variable-length frames.

Whenever two frames try to occupy the channel at the same time, there will be a collision (as seen in Fig. 4-1) and both will be garbled. If the first bit of a new frame overlaps with just the last bit of a frame that has almost finished, both frames will be totally destroyed (i.e., have incorrect checksums) and both will have to be retransmitted later. The checksum does not (and should not) distinguish between a total loss and a near miss.

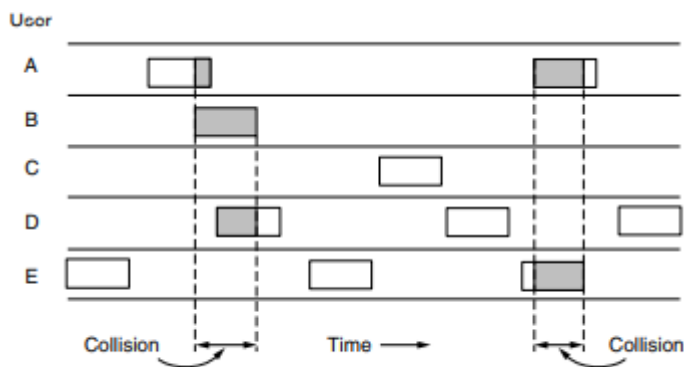


Figure 4-1. In pure ALOHA, frames are transmitted at completely arbitrary times.

Let the “frame time” denote the amount of time needed to transmit the standard, fixed-length frame (i.e., the frame length divided by the bit rate). At this point, we assume that the new frames generated by the stations are well modeled by a Poisson distribution with a mean of N frames per frame time. (The infinite population assumption is needed to ensure that N does not decrease as users become blocked.) If $N > 1$, the user community is generating frames at a higher rate than the channel can handle, and nearly every frame will suffer a collision. For reasonable throughput, we would expect $0 < N < 1$.

1. In addition to the new frames, the stations also generate retransmissions of frames that previously suffered collisions. Let us further assume that the old and new frames combined are well modeled by a Poisson distribution, with mean of G frames per frame time. Clearly, $G \geq N$. At low load (i.e., $N \sim 0$), there will be few collisions, hence few retransmissions, so $G \sim N$. At high load, there will be many collisions, so $G > N$.

N. Under all loads, the throughput, S , is just the offered load, G , times the probability,

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P_0 , of a transmission succeeding—that is, $S = GP_0$, where P_0 is the probability that a frame does not suffer a collision.

The probability that k frames are generated during a given frame time, in which G frames are expected, is given by the Poisson distribution

$$\Pr[k] = \frac{G^k e^{-G}}{k!} \quad (4-2)$$

so the probability of zero frames is just e^{-G} . In an interval two frame times long, the mean number of frames generated is $2G$. The probability of no frames being initiated during the entire vulnerable period is thus given by $P_0 = e^{-2G}$. Using $S = GP_0$, we get

$$S = Ge^{-2G}$$

The relation between the offered traffic and the throughput is shown in Fig. 4-3. The maximum throughput occurs at $G = 0.5$, with $S = 1/2e$, which is about 0.184. In other words, the best we can hope for is a channel utilization of 18%. This result is not very encouraging, but with everyone transmitting at will, we could hardly have expected a 100% success rate.

Slotted ALOHA Soon after ALOHA came onto the scene, Roberts (1972) published a method for doubling the capacity of an ALOHA system. His proposal was to divide time into discrete intervals called slots, each interval corresponding to one frame.

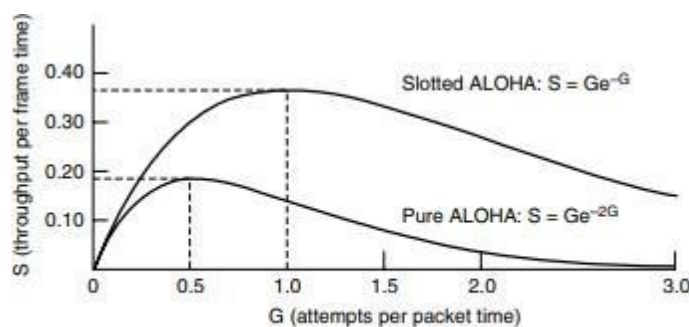


Figure 4-3. Throughput versus offered traffic for ALOHA systems.

In Roberts' method, which has come to be known as slotted ALOHA—in contrast to Abramson's pure ALOHA—a station is not permitted to send whenever the user types a line. Instead, it is required to wait for the beginning of the next slot. Thus, the

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continuous time ALOHA is turned into a discrete time one. This halves the vulnerable period.

To see this, look at Fig. 4-3 and imagine the collisions that are now possible. The probability of no other traffic during the same slot as our test frame is then e^{-G} , which leads to $S = Ge^{-G}$ (4-3). As you can see from Fig. 4-3, slotted ALOHA peaks at $G = 1$, with a throughput of $S = 1/e$ or about 0.368, twice that of pure ALOHA. If the system is operating at $G = 1$, the probability of an empty slot is 0.368.

802.11 Architecture and Protocol Stack

802.11 networks can be used in two modes. The most popular mode is to connect clients, such as laptops and smart phones, to another network, such as a company intranet or the Internet. This mode is shown in Fig. 4-23(a). In infrastructure mode, each client is associated with an AP (Access Point) that is in turn connected to the other network. The client sends and receives its packets via the AP. Several access points may be connected together, typically by a wired network called a distribution system, to form an extended 802.11 network.

In this case, clients can send frames to other clients via their APs. The other mode, shown in Fig. 4-23(b), is an ad hoc network. This mode is a collection of computers that are associated so that they can directly send frames to each other. There is no access point. Since Internet access is the killer application for wireless, ad hoc networks are not very popular. Now we will look at the protocols.

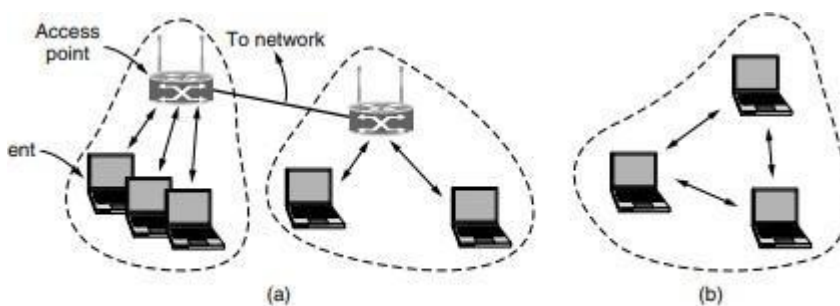


Figure 4-23. 802.11 architecture. (a) Infrastructure mode. (b) Ad-hoc mode.

The physical layer corresponds fairly well to the OSI physical layer, but the data link layer in all the 802 protocols is split into two or more sublayers. In 802.11, the MAC

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(Medium Access Control) sublayer determines how the channel is allocated, that is, who gets to transmit next. Above it is the LLC (Logical Link Control) sublayer, whose job it is to hide the differences between the different 802 variants and make them indistinguishable as far as the network layer is concerned. This could have been a significant responsibility, but these days the LLC is a glue layer that identifies the protocol (e.g., IP) that is carried within an 802.11 frame.

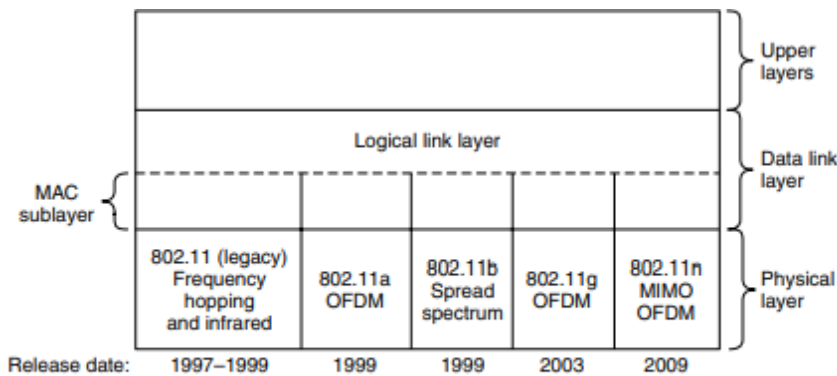


Figure 4-24. Part of the 802.11 protocol stack.

Several transmission techniques have been added to the physical layer as 802.11 has evolved since it first appeared in 1997. Two of the initial techniques, infrared in the manner of television remote controls and frequency hopping in the 2.4-GHz band, are now defunct. The third initial technique, direct sequence spread spectrum at 1 or 2 Mbps in the 2.4-GHz band, was extended to run at rates up to 11 Mbps and quickly became a hit. It is now known as 802.11b. To give wireless junkies a much-wanted speed boost, new transmission techniques based on the OFDM (Orthogonal Frequency Division Multiplexing). The first is called 802.11a and uses a different frequency band, 5 GHz. The second stuck with 2.4 GHz and compatibility. It is called 802.11g. Both give rates up to 54 Mbps. Most recently, transmission techniques that simultaneously use multiple antennas at the transmitter and receiver for a speed boost were finalized as 802.11n in Oct. 2009. With four antennas and wider channels, the 802.11 standard now defines rates up to a startling 600 Mbps.

802.11 Frame Structure

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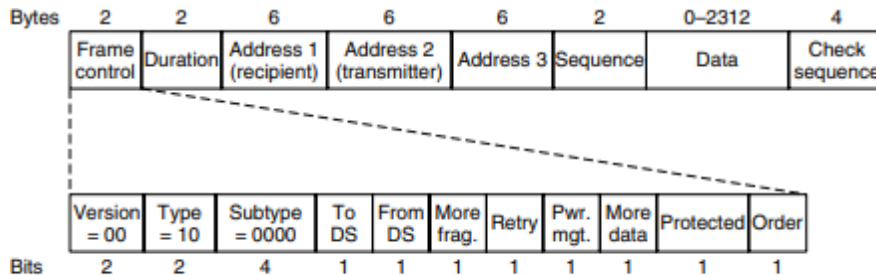


Figure 4-29. Format of the 802.11 data frame.

The 802.11 standard defines three different classes of frames in the air: data, control, and management. Each of these has a header with a variety of fields used within the MAC sublayer.

First comes the Frame control field, which is made up of 11 subfields. The first of these is the Protocol version, set to 00. It is there to allow future versions of 802.11 to operate at the same time in the same cell. Then come the Type (data, control, or management) and Subtype fields (e.g., RTS or CTS). For a regular data frame (without quality of service), they are set to 10 and 0000 in binary. The To DS and From DS bits are set to indicate whether the frame is going to or coming from the network connected to the APs, which is called the distribution system. The More fragments bit means that more fragments will follow. The Retry bit marks a retransmission of a frame sent earlier. The Power management bit indicates that the sender is going into power-save mode. The More data bit indicates that the sender has additional frames for the receiver. The Protected Frame bit indicates that the frame body has been encrypted for security. We will discuss security briefly in the next section. Finally, the Order bit tells the receiver that the higher layer expects the sequence of frames to arrive strictly in order.

The second field of the data frame, the Duration field, tells how long the frame and its acknowledgement will occupy the channel, measured in microseconds.

Next come addresses. Data frames sent to or from an AP have three addresses, all in standard IEEE 802 format. The first address is the receiver, and the second address is the transmitter.

The Sequence field numbers frames so that duplicates can be detected. Of the 16 bits available, 4 identify the fragment and 12 carry a number that is advanced with each new transmission. The Data field contains the payload, up to 2312 bytes. The first bytes of this payload are in a format known as LLC (Logical Link Control). This layer is the



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glue that identifies the higher-layer protocol (e.g., IP) to which the payloads should be passed. Last comes the Frame check sequence, which is the same 32-bit CRC. Management frames have the same format as data frames, plus a format for the data portion that varies with the subtype (e.g., parameters in beacon frames). Control frames are short. Like all frames, they have the Frame control, Duration, and Frame check sequence fields.

Career sense multiple access protocols

Protocols in which stations listen for a carrier (i.e., a transmission) and act accordingly are called carrier sense protocols.

Persistent and Nonpersistent CSMA : The first carrier sense protocol that we will study here is called 1-persistent CSMA (Carrier Sense Multiple Access). That is a bit of a mouthful for the simplest CSMA scheme. When a station has data to send, it first listens to the channel to see if anyone else is transmitting at that moment. If the channel is idle, the station sends its data. Otherwise, if the channel is busy, the station just waits until it becomes idle. Then the station transmits a frame. If a collision occurs, the station waits a random amount of time and starts all over again. The protocol is called 1-persistent because the station transmits with a probability of 1 when it finds the channel idle.

You might expect that this scheme avoids collisions except for the rare case of simultaneous sends, but in fact it does not. If two stations become ready in the middle of a third station's transmission, both will wait politely until the transmission ends, and then both will begin transmitting exactly simultaneously, resulting in a collision. **A second carrier sense protocol is nonpersistent CSMA.** In this protocol, a conscious attempt is made to be less greedy than in the previous one. As before, a station senses the channel when it wants to send a frame, and if no one else is sending, the station begins doing so itself. However, if the channel is already in use, the station does not continually sense it for the purpose of seizing it immediately upon detecting the end of the previous transmission. Instead, it waits a random period of time and then repeats the algorithm. Consequently, this algorithm leads to better channel utilization but longer delays than 1-persistent CSMA.

The last protocol is p-persistent CSMA. It applies to slotted channels and works as follows. When a station becomes ready to send, it senses the channel. If it is idle, it transmits with a probability p . With a probability $q = 1 - p$, it defers until the next slot. If that slot is also idle, it either transmits or defers again, with probabilities p

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and q. This process is repeated until either the frame has been transmitted or another station has begun transmitting.

CSMA with Collision Detection Persistent and nonpersistent CSMA protocols are definitely an improvement over ALOHA because they ensure that no station begins to transmit while the channel is busy. However, if two stations sense the channel to be idle and begin transmitting simultaneously, their signals will still collide. Another improvement is for the stations to quickly detect the collision and abruptly stop transmitting, (rather than finishing them) since they are irretrievably garbled anyway. This strategy saves time and bandwidth. This protocol, known as CSMA/CD (CSMA with Collision Detection), is the basis of the classic Ethernet LAN, so it is worth devoting some time to looking at it in detail. It is important to realize that collision detection is an analog process.

3.b) Explain about collision free protocols in detail.

Almost collisions can be avoided in CSMA/CD. They can still occur during the contention period. The collision during contention period adversely affects the system performance, this happens when the cable is long and length of packet are short. This problem becomes serious as fiber optics network come into use. Here we shall discuss some protocols that resolve the collision during the contention period.

Bit-map Protocol

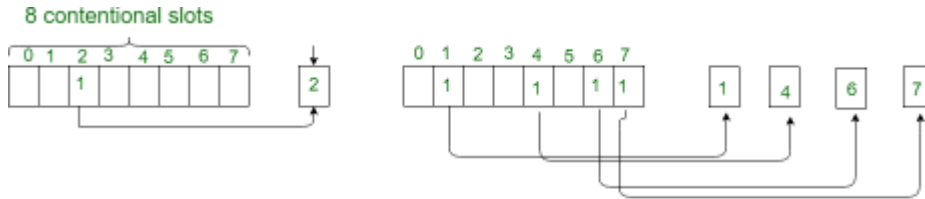
Binary Countdown

Limited Contention Protocols

The Adaptive Tree Walk Protocol

- 1. Bit-map Protocol:** Bit map protocol is collision free Protocol. In bitmap protocol method, each contention period consists of exactly N slots. If any station has to send frame, then it transmits a 1 bit in the respective slot. For example if station 2 has a frame to send, it transmits a 1 bit during the second slot. In general Station 1 Announce the fact that it has a frame questions by inserting a 1 bit into slot 1. In this way, each station has complete knowledge of which station wishes to transmit. There will never be any collisions because everyone agrees on who goes next. Protocols like this in which the desire to transmit is broadcasting for the actual transmission are called *Reservation Protocols*.

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A Bit-map Protocol.

Generally, high numbered stations have to wait for half a scan before starting to transmit low numbered stations have to wait for half a scan($N/2$ bit slots) before starting to transmit, low numbered stations have to wait on an average $1.5 N$ slots.

Binary Countdown:

Binary countdown protocol is used to overcome the overhead 1 bit per binary station. In binary countdown, binary station addresses are used. A station wanting to use the channel broadcast its address as binary bit string starting with the high order bit. All addresses are assumed of the same length. Here, we will see the example to illustrate the working of the binary countdown.

In this method, different station addresses are ORed together who decide the priority of transmitting. If these stations 0001, 1001, 1100, 1011 all are trying to seize the channel for transmission. All the station at first broadcast their most significant address bit that is 0, 1, 1, 1 respectively. The most significant bits are ORed together. Station 0001 see the 1MSB in another station addresses and knows that a higher numbered station is competing for the channel, so it gives up for the current round.

Other three stations 1001, 1100, 1011 continue. The next bit is 1 at station 1100, station 1011 and 1001 give up. Then station 110 starts transmitting a frame, after which another bidding cycle starts.



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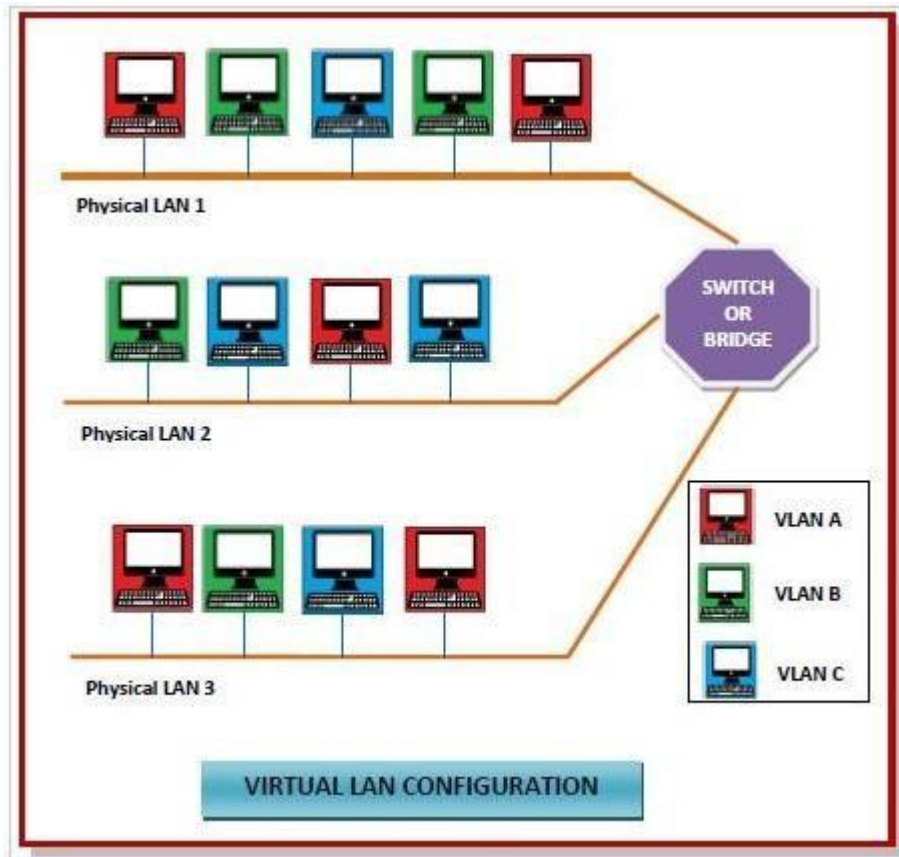
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Network administrators partition the networks to

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match the functional requirements of the VLANs so that each VLAN comprise of a subset of ports on a single or multiple switches or bridges. This allows computers and devices in a VLAN to communicate in the simulated environment as if it is a separate LAN.



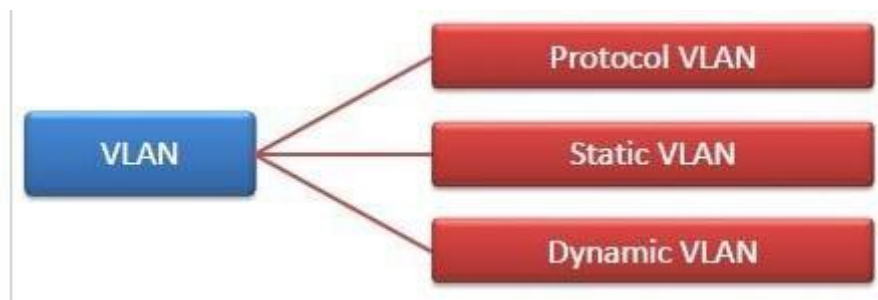
Features of VLANs

- A VLAN forms sub-network grouping together devices on separate physical LANs.
- VLAN's help the network manager to segment LANs logically into different broadcast domains.
- VLANs function at layer 2, i.e. Data Link Layer of the OSI model.
- There may be one or more network bridges or switches to form multiple, independent VLANs.

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- Using VLANs, network administrators can easily partition a single switched network into multiple networks depending upon the functional and security requirements of their systems.
- VLANs eliminate the requirement to run new cables or reconfiguring physical connections in the present network infrastructure.
- VLANs help large organizations to re-partition devices aiming improved traffic management.
- VLANs also provide better security management allowing partitioning of devices according to their security criteria and also by ensuring a higher degree of control connected devices.
- VLANs are more flexible than physical LANs since they are formed by logical connections. This aids in quicker and cheaper reconfiguration of devices when the logical partitioning needs to be changed.

Types of VLANs



4.b) Explain about IEEE 802.1Q standard in detail? Refer Question no 6 in 3 marks questions.

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Academic Year: 2023-24

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Section: C



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LECTURE NOTES

UNIT 4

Services provided to the transport layer:

The services provided to the transport layer are as follows –

- **Logical Addressing** – Network layer adds header to incoming packet which includes logical address to identify sender and receiver.
- **Routing** – It is the mechanism provided by Network Layer for routing the packets to the final destination in the fastest possible and efficient way.
- **Flow control** – This layer routes the packet to another way, If too many packets are present at the same time preventing bottlenecks and congestion.
- **Breaks Large Packets** – Breaks larger packets into small packets.
- **Connection Oriented service** – It is a network communication mode, where a communication session is established before any useful data can be transferred and where a stream of data is delivered in the same order as it was sent.
- **Connectionless Service** – It is a data transmission method used in packet switching networks by which each data unit is individually addressed and routed based on information carried in each unit, rather than in the setup information of a prearranged, fixed data channel as in connection-oriented communication.
- **DataGram** – A datagram is a basic transfer unit associated with a packet-switched network. The delivery, arrival time and order of arrival need not be guaranteed by the network.
- **A virtual circuit** – It is a means of transporting data over a packet switched computer network in such a way that it appears as though there is a dedicated physical layer link between the source and destination end system of this data.



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Virtual Circuit network and data gram network:

Virtual – circuit network is a category of packet switching network, where a virtual path is established between the source and the destination systems for data communication to occur. This path appears to the user as if it is a dedicated physical path, but actually is a logical circuit allocated from a managed pool of circuit resources as per traffic requirements.

In a connectionless communication systems, datagram refers to the smallest unit via which data is transmitted. Datagrams are data packets which contain adequate header information so that they can be individually routed by all intermediate network switching devices to the destination. These networks are called datagram networks since communication occurs via datagrams. They exist in packet switching networks.

Routing algorithm

A routing algorithm is a procedure that lays down the route or path to transfer data packets from source to the destination. They help in directing Internet traffic efficiently. After a data packet leaves its source, it can choose among the many different paths to reach its destination. Routing algorithm mathematically computes the best path, i.e. “least – cost path” that the packet can be routed through.

Sink tree

Sink trees of a given node in a computer network is a tree formed by connecting the node with all the other nodes of in the network with the optimal path, or more precisely the shortest path. Or, in other words, a sink tree is a minimal cost tree formed with the given node as the destination and all other nodes as sources. A popular measure for the path or the cost is the number of hops.

Forwarding in network layer

Forwarding is simply defined as **the action applied by each router when a packet arrives at one of its interfaces**. When a router receives a packet from one of its attached networks, it needs to forward the packet to another attached network (unicast routing) or to some attached networks(in case of multicast routing).



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Compare flow control versus congestion control.

Both Flow control and Congestion control are traffic controlling methods in computer networks. Flow control is a technique that allows two stations working at different speeds to communicate with each other. It is a set of measures taken to regulate the amount of data that a sender sends so that a fast sender does not overwhelm a slow receiver.

Congestion occurs in a network when it receives too many packets than its designed capacity. Network congestion occurs in case of traffic overloading. There are mechanisms to control such congestion scenarios in a network.

Multiprotocol router.

A router that supports two or more communications protocols, such as IP and IPX. It is used to switch network traffic between different LANs located throughout the enterprise as well as to switch LAN traffic to WANs.

Adaptive algorithms and non adaptive algorithms.

Adaptive routing algorithm is also called a dynamic routing algorithm. In this algorithm, the routing decisions are made based on network traffic and topology. The parameters which are used in adaptive routing algorithms are distance, hop, estimated transit time and count.

Non-adaptive routing algorithm is also called a static routing algorithm. In a non-adaptive routing algorithm, the routing decisions are not made based on network traffic and topology. This algorithm is used by static routing. Non-adaptive routing algorithms are simple as compared to Adaptive routing algorithms in terms of complexity.

Internetwork routing.

Internet routing is **the process of transmitting and routing IP packets over the Internet between two or more nodes.** It is the same as standard routing procedures but incorporates packet routing techniques and processes on external networks or those that are hosted or Internet enabled. Internet routers route packets from internal networks to

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external Internet-based routers. Internet routing enables a user to access web pages and other data stored on a remote website. Internet routing involves broadcasting or sending a message from an internal network to an external network using Internet-based networks. Such routing generally involves sending a message that travels between several Internet service providers (ISP) or autonomous systems (AS) before reaching the destination.

Differences between IPv4 and IPv6

	Ipv4	Ipv6
Address length	IPv4 is a 32-bit address.	IPv6 is a 128-bit address.
Fields	IPv4 is a numeric address that consists of 4 fields which are separated by dot (.).	IPv6 is an alphanumeric address that consists of 8 fields, which are separated by colon.
Classes	IPv4 has 5 different classes of IP address that includes Class A, Class B, Class C, Class D, and Class E.	IPv6 does not contain classes of IP addresses.
Number of IP address	IPv4 has a limited number of IP addresses.	IPv6 has a large number of IP addresses.
VLSM	It supports VLSM (Virtual Length Subnet Mask). Here, VLSM means that Ipv4 converts	It does not support VLSM.

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	IP addresses into a subnet of different sizes.	
Address configuration	It supports manual and DHCP configuration.	It supports manual, DHCP, auto-configuration, and renumbering.
Address space	It generates 4 billion unique addresses	It generates 340 undecillion unique addresses.
End-to-end connection integrity	In IPv4, end-to-end connection integrity is unachievable.	In the case of IPv6, end-to-end connection integrity is achievable.
Security features	In IPv4, security depends on the application. This IP address is not developed in keeping the security feature in mind.	In IPv6, IPSEC is developed for security purposes.
Address representation	In IPv4, the IP address is represented in decimal.	In IPv6, the representation of the IP address in hexadecimal.
Fragmentation	Fragmentation is done by the senders and the forwarding routers.	Fragmentation is done by the senders only.
Packet flow identification	It does not provide any mechanism for packet flow identification.	It uses flow label field in the header for the packet flow identification.

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Checksum field	The checksum field is available in IPv4.	The checksum field is not available in IPv6.
Transmission scheme	IPv4 is broadcasting.	On the other hand, IPv6 is multicasting, which provides efficient network operations.
Encryption and Authentication	It does not provide encryption and authentication.	It provides encryption and authentication.
Number of octets	It consists of 4 octets.	It consists of 8 fields, and each field contains 2 octets. Therefore, the total number of octets in IPv6 is 16.

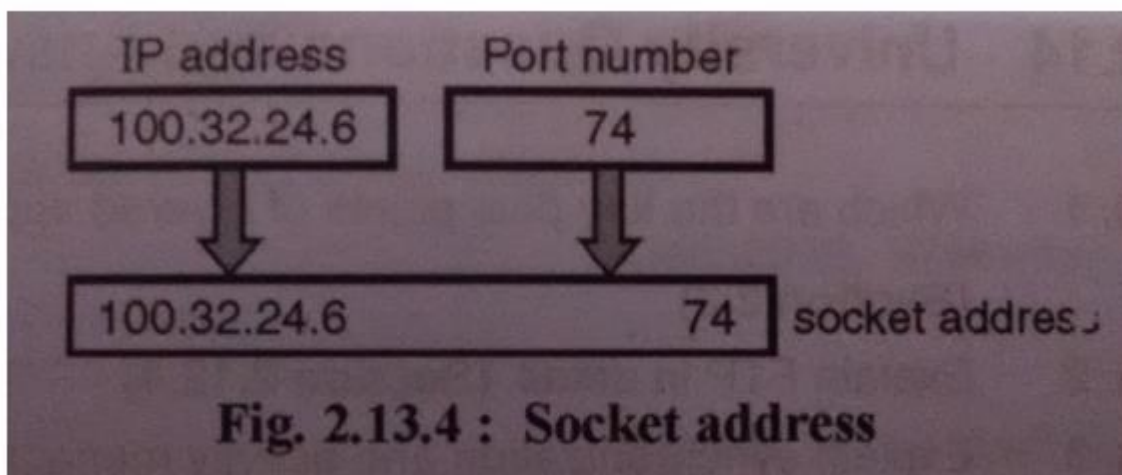
11. What is Multicast routing.

Multicast routing is **a networking method for efficient distribution of one-to-many traffic**. A multicast source, such as a live video conference, sends traffic in one stream to a multicast group. The multicast group contains receivers such as computers, devices, and IP phones.

12. What is Socket address.

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1. Process to process delivery (transport layer communication) needs two identifiers, one is IP address and the other is port number at each end to make a connection.
2. Socket address is the combinations of IP address and port number as shown in the figure.



The client socket address defines the client process uniquely whereas the server socket address defines the server process uniquely.

A transport layer protocol requires the client socket address as well as the server socket address. These two addresses contain four pieces.

These four pieces go into the IP header and the transport layer protocol header.

The IP header contains the IP addresses while the UDP and TCP headers contain the port numbers.

1. Explain store and forward packet switching in network layer.

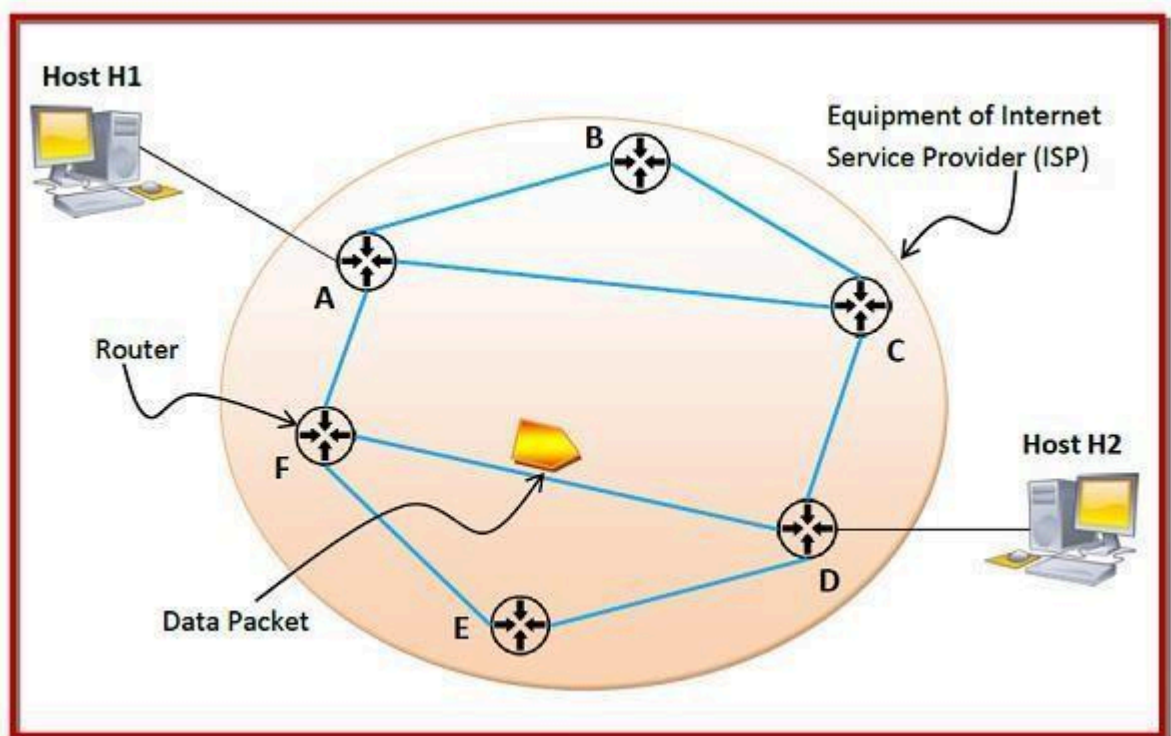
In telecommunications, store – and – forward packet switching is a technique where the data packets are stored in each intermediate node, before they are forwarded to the next node. The intermediate node checks whether the packet is error-free before transmitting, thus ensuring integrity of the data packets. In general, the network layer operates in an environment that uses store and forward packet switching.

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Working Principle

The node which has a packet to send, delivers it to the nearest node, i.e. router. The packet is stored in the router until it has fully arrived and its checksum is verified for error detection. Once, this is done, the packet is transmitted to the next router. The same process is continued in each router until the packet reaches its destination.

The following scenario exemplifies the mechanism –



In the above diagram, we can see that the Internet Service Provider (ISP) has six routers (A to F) connected by transmission lines shown in blue lines. There are two hosts, host H1 is connected to router A, while host H2 is connected to router D. Suppose that H1 wants to send a data packet to H2. H1 sends the packet to router A. The packet is stored in router A until it has arrived fully. Router A verifies the checksum using CRC (cyclic redundancy check) code. If there is a CRC error, the packet is discarded, otherwise it is transmitted to the next hop, here router F. The same process is followed by router F which then transmits the packet to router D. Finally router D delivers the packet to host H2.

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Advantages and Disadvantages

Store – and forward packet switching ensures high quality data packet transmission. Since erroneous packets are discarded at each router, bad packets or invalid packets in the network are mostly eliminated.

However, error – free packet transmission is achieved by compromising on the overall speed of transmission. Switch latency is introduced due to waiting for entire packet to arrive as well as computation of CRC. Though the latency at each router may seem small enough, the cumulative latency at all routers make it inappropriate for time – critical online applications.

2. What is Optimality Principle in network Layer.

The optimality principle in computer networks is stated as follows –

Given a network of routers for data transmission, if a router 'J' lies on the optimal path from router 'I' to router 'K', then the optimal path from router 'J' to router 'K' also falls along the same route.

Explanation

The purpose of a routing algorithm at a router is to decide which output line an incoming packet should go. The optimal path from a particular router to another may be the least cost path, the least distance path, the least time path, the least hops path or a combination of any of the above.

The optimality principle can be logically proved as follows –

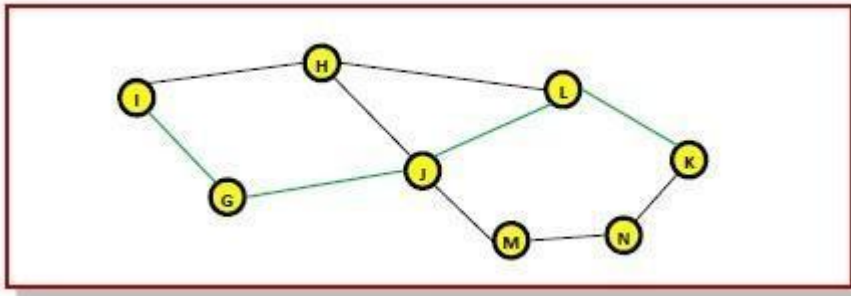
- If a better route could be found between router J and router K, the path from router I to router K via J would be updated via this route. Thus, the optimal path from J to K will again lie on the optimal path from I to K.

Example

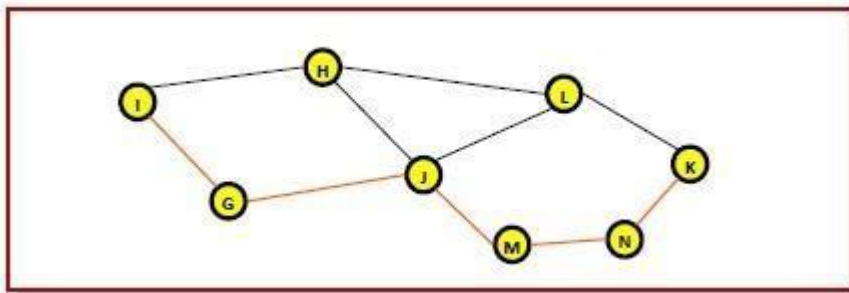
- Consider a network of routers, {G, H, I, J, K, L, M, N} as shown in the figure. Let the optimal route from I to K be as shown via the green path, i.e. via the

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route I-G-J-L-K. According to the optimality principle, the optimal path from J to K will be along the same route, i.e. J-L-K.



- Now, suppose we find a better route from J to K is found, say along J-M-N-K. Consequently, we will also need to update the optimal route from I to K as I-G-J-M-N-K, since the previous route ceases to be optimal in this situation. This new optimal path is shown in orange lines in the following figure –

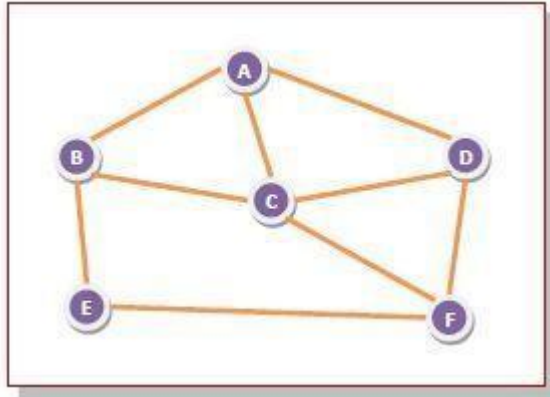


3. What is flooding.

Flooding is a non-adaptive routing technique following this simple method: when a data packet arrives at a router, it is sent to all the outgoing links except the one it has arrived on.

For example, let us consider the network in the figure, having six routers that are connected through transmission lines.

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Using flooding technique –

- An incoming packet to A, will be sent to B, C and D.
- B will send the packet to C and E.
- C will send the packet to B, D and F.
- D will send the packet to C and F.
- E will send the packet to F.
- F will send the packet to C

and E. Types of Flooding

Flooding may be of three types –

- **Uncontrolled flooding** – Here, each router unconditionally transmits the incoming data packets to all its neighbours.
- **Controlled flooding** – They use some methods to control the transmission of packets to the neighbouring nodes. The two popular algorithms for controlled flooding are Sequence Number Controlled Flooding (SNCF) and Reverse Path Forwarding (RPF).
- **Selective flooding** – Here, the routers don't transmit the incoming packets only along those paths which are heading towards approximately in the right direction, instead of every available paths.

Advantages of Flooding

- It is very simple to setup and implement, since a router may know only its neighbours.

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- It is extremely robust. Even in case of malfunctioning of a large number routers, the packets find a way to reach the destination.
- All nodes which are directly or indirectly connected are visited. So, there are no chances for any node to be left out. This is a main criteria in case of broadcast messages.
- The shortest path is always chosen by

flooding. Limitations of Flooding

- Flooding tends to create an infinite number of duplicate data packets, unless some measures are adopted to damp packet generation.
- It is wasteful if a single destination needs the packet, since it delivers the data packet to all nodes irrespective of the destination.
- The network may be clogged with unwanted and duplicate data packets. This may hamper delivery of other data packets.

4. Define broadcasting routing and multicasting routing.

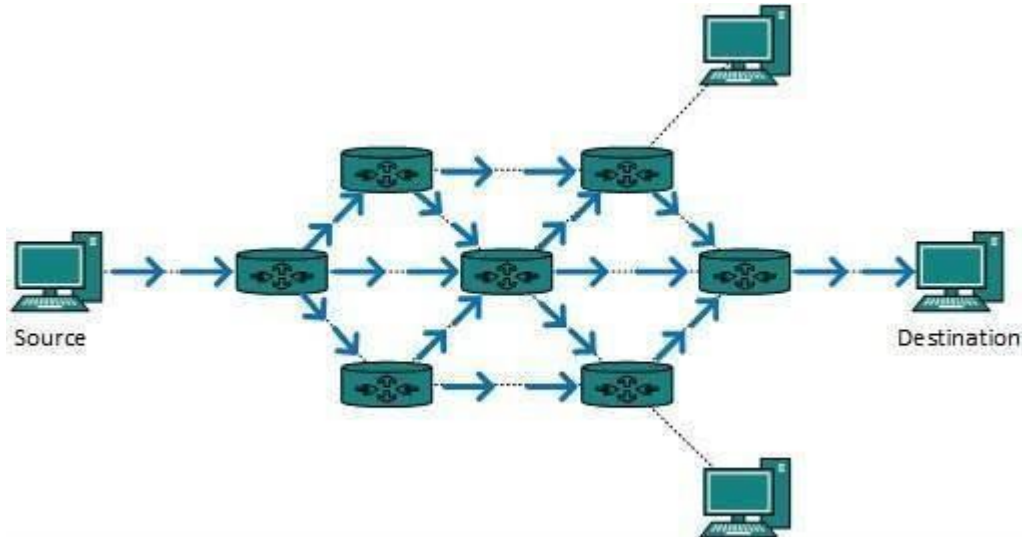
Broadcast routing

By default, the broadcast packets are not routed and forwarded by the routers on any network. Routers create broadcast domains. But it can be configured to forward broadcast messages in some special cases. A broadcast message is destined to all network devices.

Broadcast routing can be done in two ways (algorithm):

- A router creates a data packet and then sends it to each host one by one. In this case, the router creates multiple copies of single data packet with different destination addresses. All packets are sent as unicast but because they are sent to all, it simulates as if router is broadcasting. This method consumes lots of bandwidth and router must destination address of each node.
- Secondly, when router receives a packet that is to be broadcasted, it simply floods those packets out of all interfaces. All routers are configured in the same way.

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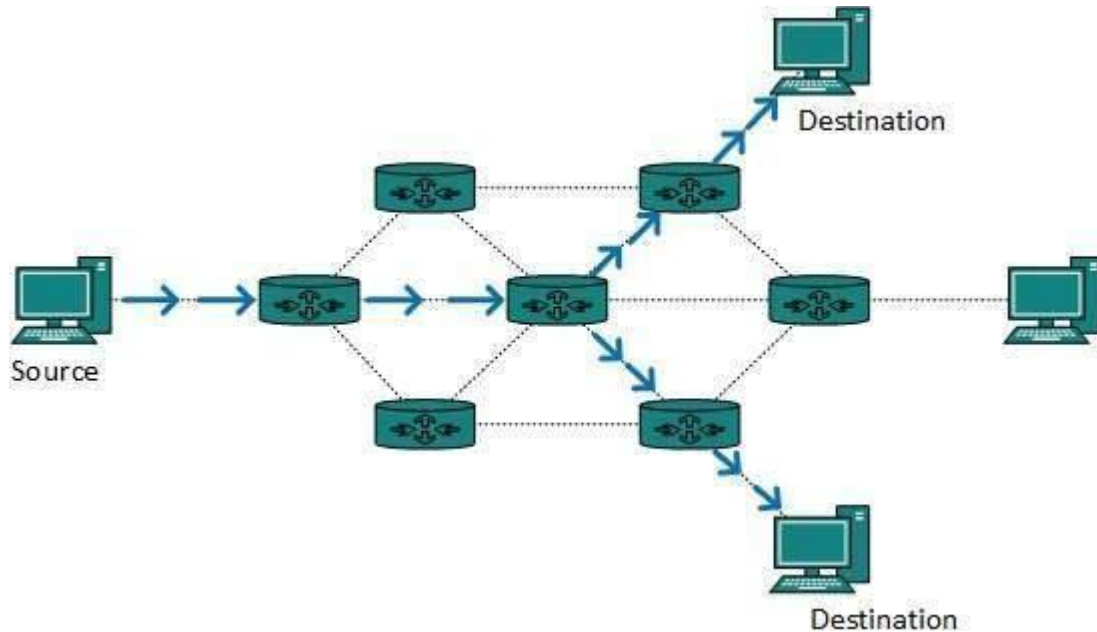
This method is easy on router's CPU but may cause the problem of duplicate packets received from peer routers.

Reverse path forwarding is a technique, in which router knows in advance about its predecessor from where it should receive broadcast. This technique is used to detect and discard duplicates.

Multicast Routing

Multicast routing is special case of broadcast routing with significance difference and challenges. In broadcast routing, packets are sent to all nodes even if they do not want it. But in Multicast routing, the data is sent to only nodes which wants to receive the packets.

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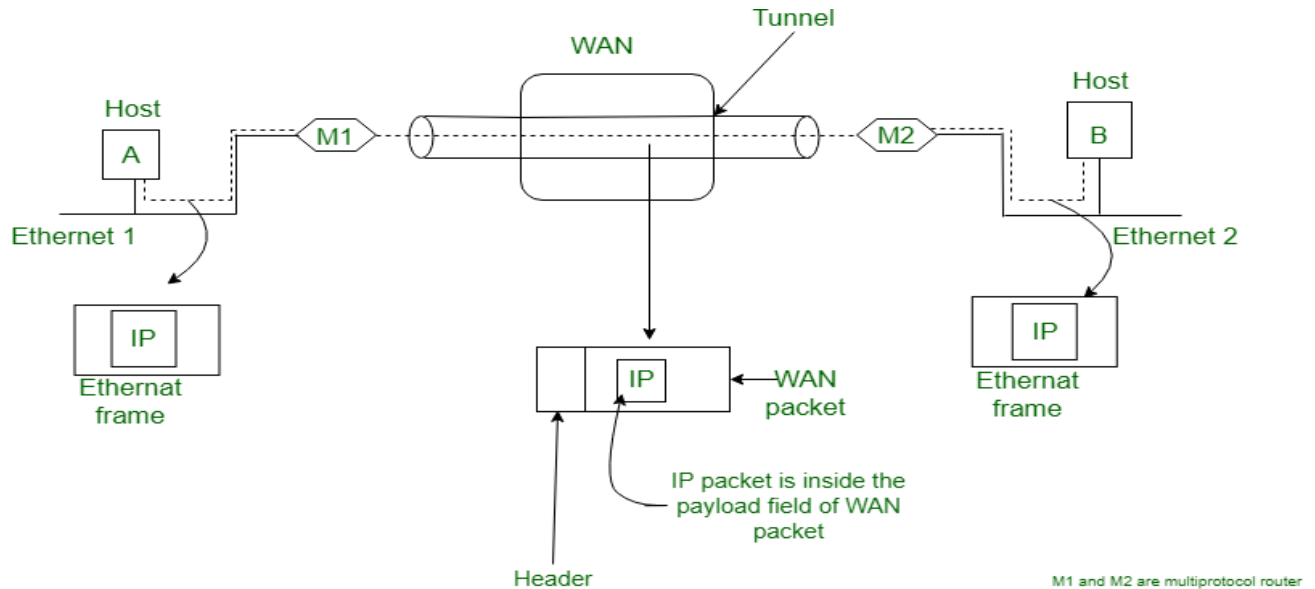
The router must know that there are nodes, which wish to receive multicast packets (or stream) then only it should forward. Multicast routing works spanning tree protocol to avoid looping.

Multicast routing also uses reverse path Forwarding technique, to detect and discard duplicates and loops.

5. Explain Tunneling concept with diagram. (3M)

A technique of internetworking called **Tunneling** is used when source and destination networks of same type are to be connected through a network of different type. For example, let us consider an Ethernet to be connected to another Ethernet through a WAN as:

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Tunneling

The task is sent on an IP packet from host A of Ethernet-1 to the host B of ethernet-2 via a WAN.

Sequence of events:

1. Host A constructs a packet which contains the IP address of Host B.
2. It then inserts this IP packet into an Ethernet frame and this frame is addressed to the multiprotocol router M1.
3. Host A then puts this frame on Ethernet.
4. When M1 receives this frame, it removes the IP packet, inserts it in the payload packet of the WAN network layer packet and addresses the WAN packet to M2. The multiprotocol router M2 removes the IP packet and sends it to host B in an Ethernet frame.

Why is this Technique called Tunneling?



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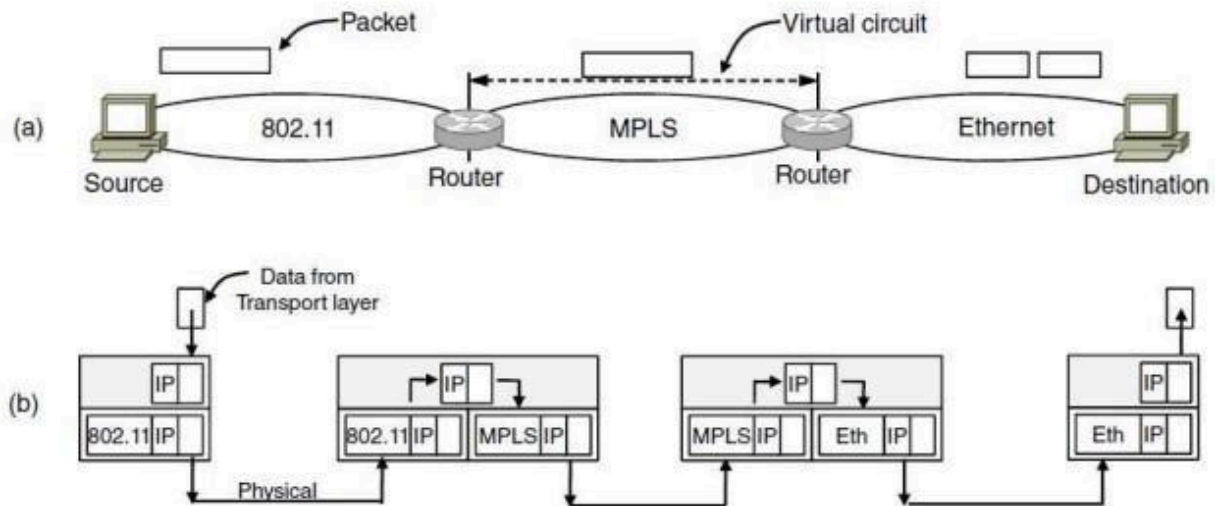
In this particular example, the IP packet does not have to deal with WAN, the host A and B also do not have to deal with the WAN. The multiprotocol routers M1 and M2 will have to understand about IP and WAN packets. Therefore, the WAN can be imagined to be equivalent to a big tunnel extending between multiprotocol routers M1 and M2 and the technique is called Tunneling.

Tunneling uses a layered protocol model such as those of the OSI or TCP/IP protocol suite. So, in other words, when data moves from host A to B it covers all the different level of the specified protocol (OSI, TCP/IP, etc.), while moving between different levels, data conversion (encapsulation) to suit different interfaces of the particular layer is called tunneling.

6. How networks can be connected. Explain with diagram.

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How Networks Can Be Connected



(a) A packet crossing different networks.

(b) Network and link layer protocol processing.

7. List out some ways networks can differ with each other.

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How Networks Differ

- Differences that can occur at network layer, which makes internetworking difficult:

Item	Some Possibilities
Service offered	Connection oriented versus connectionless
Protocols	IP, IPX, SNA, ATM, MPLS, AppleTalk, etc.
Addressing	Flat (802) versus hierarchical (IP)
Multicasting	Present or absent (also broadcasting)
Packet size	Every network has its own maximum
Quality of service	Present or absent; many different kinds
Error handling	Reliable, ordered, and unordered delivery
Flow control	Sliding window, rate control, other, or none
Congestion control	Leaky bucket, token bucket, RED, choke packets, etc.
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, by packet, by byte, or not at all

- It is impossible to resolve all differences, and the solution is to take a simple approach (as in the Internet)

8. Analyze how reliable and ordered delivery is achieved through TCP.

TCP is said to be reliable because when an application receives data, it knows that this is exactly the data that the sender sent. No part of the data was lost, reordered, etc. Error control is the part of TCP that deals with ensuring the reliability of connections. For example, if the sender sends two packets A and B and the receiver received only B, the TCP implementation on the receiver side will wait a little in case A took longer than B to arrive. If A finally arrives, the receiver will release A to the application before B. If A doesn't arrive after a while, the receiver will transmit a request to the sender to re-transmit A. (I am simplifying somewhat, see the specification for details.)

This is not related to congestion and flow control. They help the performance of the network, but they aren't needed or helpful to make a connection reliable.

TCP "guarantees" that a receiver will receive the reconstituted stream of bytes as it was originally sent by the sender. However, between the TCP send/receive endpoints (i.e., the physical network), the data can be received out of order, it can be fragmented, it



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can be corrupted, and it can even be lost. TCP accounts for these problems using a handshake mechanism that causes bad packets to be retransmitted. The TCP stack on the receiver places these packets in the order in which they were transmitted so that when you read from your TCP socket, you receive the data as it was originally sent.

9. Discuss about IP addresses.

An IP address is a unique address that identifies a device on the internet or a local network. IP stands for "Internet Protocol," which is the set of rules governing the format of data sent via the internet or local network.

In essence, IP addresses are the identifier that allows information to be sent between devices on a network: they contain location information and make devices accessible for communication. The internet needs a way to differentiate between different computers, routers, and websites. IP addresses provide a way of doing so and form an essential part of how the internet works.

An IP address is a string of numbers separated by periods. IP addresses are expressed as a set of four numbers — an example address might be 192.158.1.38. Each number in the set can range from 0 to 255. So, the full IP addressing range goes from 0.0.0.0 to 255.255.255.255.

IP addresses are not random. They are mathematically produced and allocated by the [Internet Assigned Numbers Authority](#) (IANA), a division of the [Internet Corporation for Assigned Names and Numbers](#) (ICANN). ICANN is a non-profit organization that was established in the United States in 1998 to help maintain the security of the internet and allow it to be usable by all. Each time anyone registers a domain on the internet, they go through a domain name registrar, who pays a small fee to ICANN to register the domain.

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Watch this video to learn what IP address is, why IP address is important and how to protect it from hackers:

10. How would you describe UDP and TCP protocols?

TCP is a connection-oriented protocol, whereas UDP is a connectionless protocol. A key difference between TCP and UDP is speed, as TCP is comparatively slower than UDP. Overall, UDP is a much faster, simpler, and efficient protocol, however, retransmission of lost data packets is only possible with TCP.

Both the protocols, i.e., TCP and UDP, are the transport layer protocol. It means that TCP requires connection prior to the communication, but the UDP does not require any connection.

Reliability

TCP is a reliable protocol as it provides assurance for the delivery of the data. It follows the acknowledgment mechanism. In this mechanism, the sender receives the acknowledgment from the receiver and checks whether the acknowledgment is positive or negative. If the ACK is positive means, the data has been received successfully. If ACK is negative, then TCP will resend the data. It also follows the flow and error control mechanism. UDP is an unreliable protocol as it does not ensure the delivery of the data.

FlowControl

TCP follows the flow control mechanism that ensures a large number of packets



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are not sent to the receiver at the same time, while UDP does not follow the flow control mechanism.

Ordering

TCP uses ordering and sequencing techniques to ensure that the data packets are received in the same order in which they are sent. On the other hand, UDP does not follow any ordering and sequencing technique; i.e., data can be sent in any sequence.

Speed

Since TCP establishes a connection between a sender and receiver, performs error checking, and also guarantees the delivery of data packets while UDP neither creates a connection nor it guarantees the delivery of data packets, so UDP is faster than TCP.

Flow of data

In TCP, data can flow in both directions means that it provides the full-duplex service. On the other hand, UDP is mainly suitable for the unidirectional flow of data.

○

1. Compare virtual circuit and data gram networks. (5M)

Virtual – circuit and datagram networks are categories of packet switching network. In virtual circuits, a virtual path is established between the source and the destination systems through which the data packets are transferred from the source to the destination. In the other hand, in datagram networks, each data packet called datagram contain adequate header information so that they can be individually routed by all intermediate network switching devices to the destination.

Computer networks that provide connection-oriented services are called Virtual Circuits while those providing connection-less services are called Datagram networks. For prior knowledge, the [Internet](#) which we use is actually based on a Datagram network (connection-less) at the network level as all packets from a source to a destination do not follow the same



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path.

Let us see what are the highlighting differences between.

Virtual Circuits:

1. It is connection-oriented, meaning that there is a reservation of resources like buffers, [CPU](#), [bandwidth](#), etc. for the time in which the newly setup VC is going to be used by a data transfer session.
2. The first sent packet reserves resources at each server along the path. Subsequent packets will follow the same path as the first sent packet for the connection time.
3. Since all the packets are going to follow the same path, a global header is required. Only the first packet of the connection requires a global header, the remaining packets generally don't require global headers.
4. Since all packets follow a specific path, packets are received in order at the destination.
5. Virtual Circuit Switching ensures that all packets successfully reach the Destination. No packet will be discarded due to the unavailability of resources.
6. From the above points, it can be concluded that [Virtual Circuits](#) are a highly reliable method of data transfer.
7. The issue with virtual circuits is that each time a new connection is set up, resources and extra information have to be reserved at every router along the path, which becomes problematic if many clients are trying to reserve a router's resources simultaneously.
8. It is used by the [ATM \(Asynchronous Transfer Mode\) Network](#), specifically for Telephone calls.

Datagram Networks :

1. It is a [connection-less service](#). There is no need for reservation of resources as there is no dedicated path for a connection session.
2. All packets are free to use any available path. As a result, intermediate routers calculate routes on the go due to dynamically changing routing tables on routers.
3. Since every packet is free to choose any path, all packets must be associated with a header with proper information about the source and the upper layer data.
4. The connection-less property makes data packets reach the destination in any order, which means that they can potentially be received out of order at the receiver's end.
5. Datagram networks are not as reliable as Virtual Circuits.

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The major drawback of Datagram Packet switching is that a packet can only be forwarded if resources such as the buffer, CPU, and bandwidth are available. Otherwise, the packet will be discarded.

But it is always easy and cost-efficient to implement datagram networks as there is no extra headache of reserving resources and making a dedicated each time an application has to communicate.

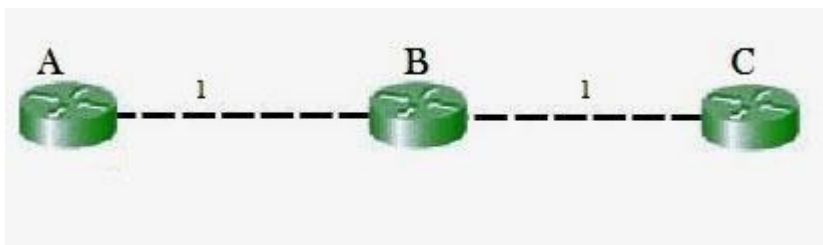
It is generally used by the IP network, which is used for Data services like the Internet.

2. What is count - to - infinity problem. Explain with an example.(5M)

The Count to Infinity problem **arises from the routing loop in this Distance Vector Routing(DVR) network**. Such Routing Loops usually occurs when 2 routers send an update together at the same time or when an interface goes down.

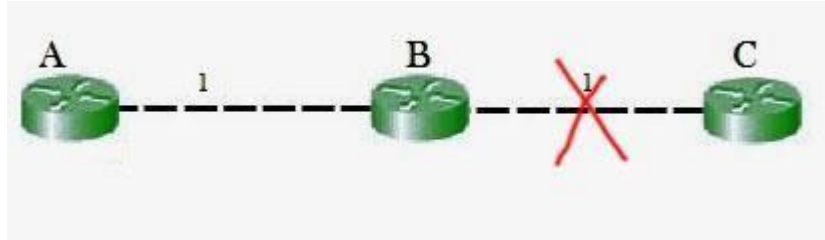
The main issue with Distance Vector Routing (DVR) protocols is Routing Loops since [Bellman-Ford Algorithm](#) cannot prevent loops. This routing loop in the DVR network causes the Count to Infinity Problem. Routing loops usually occur when an interface goes down or two routers send updates at the same time.

Counting to infinity problem:



So in this example, the Bellman-Ford algorithm will converge for each router, they will have entries for each other. B will know that it can get to C at a cost of 1, and A will know that it can get to C via B at a cost of 2.

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If the link between B and C is disconnected, then B will know that it can no longer get to C via that link and will remove it from its table. Before it can send any updates it's possible that it will receive an update from A which will be advertising that it can get to C at a cost of 2. B can get to A at a cost of 1, so it will update a route to C via A at a cost of 3. A will then receive updates from B later and update its cost to 4. They will then go on feeding each other bad information toward infinity which is called as **Count to Infinity problem**.

3. Discuss the following.i)Broadcast Routing ii)Multicast Routing (5M)

Broadcast Routing :

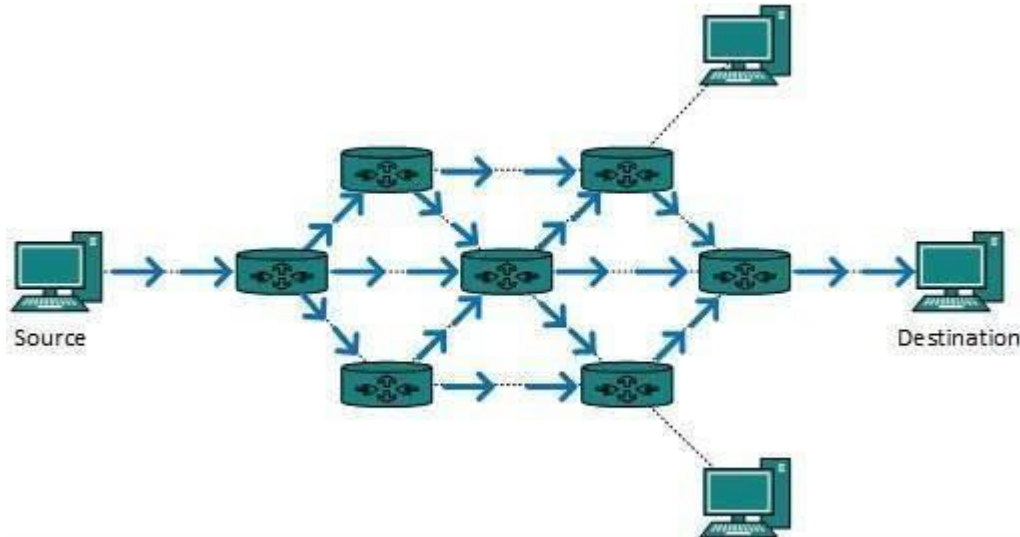
Broadcast routing

By default, the broadcast packets are not routed and forwarded by the routers on any network. Routers create broadcast domains. But it can be configured to forward broadcasts in some special cases. A broadcast message is destined to all network devices.

Broadcast routing can be done in two ways (algorithm):

- A router creates a data packet and then sends it to each host one by one. In this case, the router creates multiple copies of single data packet with different destination addresses. All packets are sent as unicast but because they are sent to all, it simulates as if router is broadcasting. This method consumes lots of bandwidth and router must destination address of each node.
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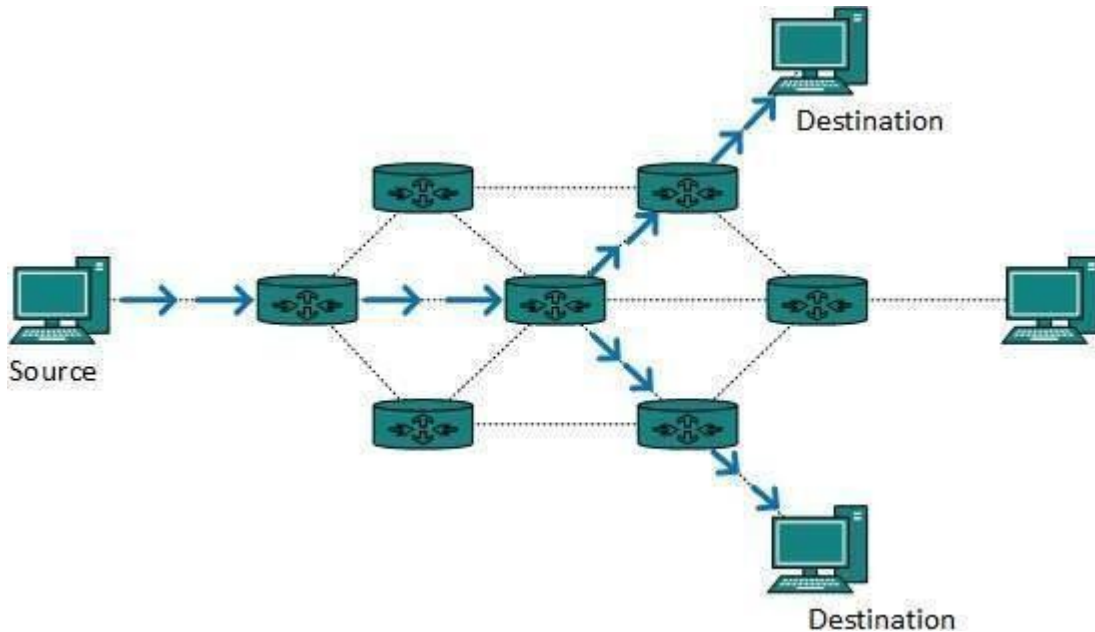


- This method is easy on router's CPU but may cause the problem of duplicate packets received from peer routers. Reverse path forwarding is a technique, in which router knows in advance about its predecessor from where it should receive broadcast. This technique is used to detect and discard duplicates.

Multicast Routing

Multicast routing is special case of broadcast routing with significance difference and challenges. In broadcast routing, packets are sent to all nodes even if they do not want it. But in Multicast routing, the data is sent to only nodes which wants to receive the packets.

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The router must know that there are nodes, which wish to receive multicast packets (or stream) then only it should forward. Multicast routing works spanning tree protocol to avoid looping.

Multicast routing also uses reverse path Forwarding technique, to detect and discard duplicates and loops.

4. With an example explain routing process in Hierarchical routing .(5M)

In hierarchical routing, the routers are divided into regions. Each router has complete details about how to route packets to destinations within its own region. But it does not have any idea about the internal structure of other regions.

As we know, in both LS and DV algorithms, every router needs to save some information about other routers. When network size is growing, the number of routers in the network will increase. Therefore, the size of routing table increases, then routers cannot handle network traffic as efficiently. To overcome this problem we are using hierarchical routing.

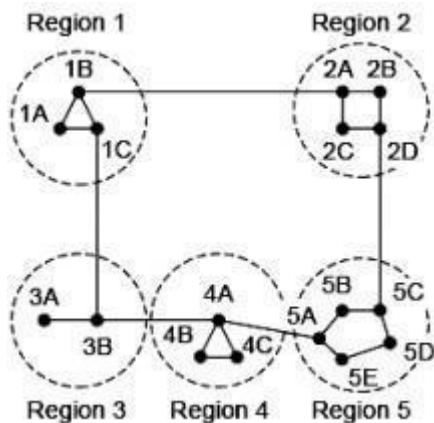
In hierarchical routing, routers are classified in groups called regions. Each router has information about the routers in its own region and it has no information about routers in other regions. So, routers save one record in their table for every other region.

For huge networks, a two-level hierarchy may be insufficient hence, it may be necessary to group the regions into clusters, the clusters into zones, the zones into groups and so on.

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Example

Consider an example of two-level hierarchy with five regions as shown in figure –



Let see the full routing table for router 1A which has 17 entries, as shown below –

Full Table for 1A

Dest.	Line	Hops
1A	-	-
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3

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Dest.	Line	Hops
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

When routing is done hierarchically then there will be only 7 entries as shown below –

Hierarchical Table for 1A

Dest.	Line	Hops
1A	-	-
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

Unfortunately, this reduction in table space comes with the increased path length.

Explanation

Step 1 – For example, the best path from 1A to 5C is via region 2, but hierarchical routing of all traffic to region 5 goes via region 3 as it is better for most of the other destinations of region 5.

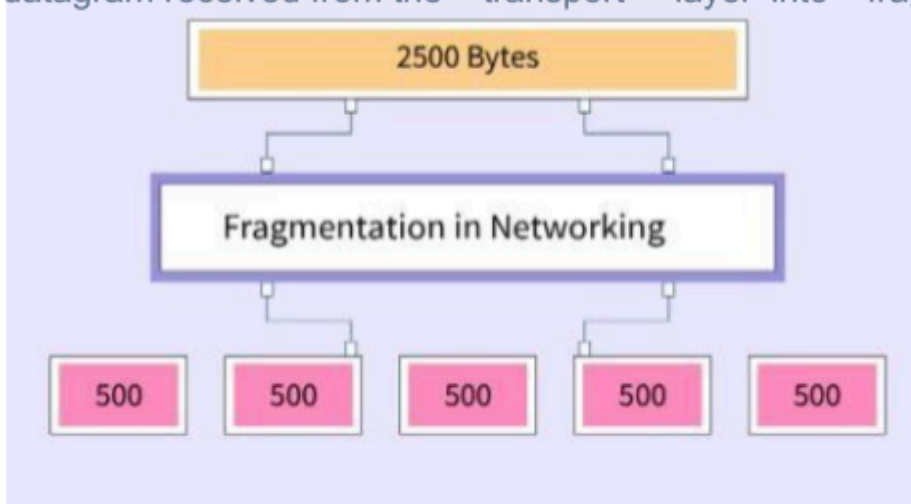
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Step 2 – Consider a subnet of 720 routers. If no hierarchy is used, each router will have 720 entries in its routing table.

Step 3 – Now if the subnet is partitioned into 24 regions of 30 routers each, then each router will require 30 local entries and 23 remote entries for a total of 53 entries.

5. Explain about packet fragmentation. (5M)

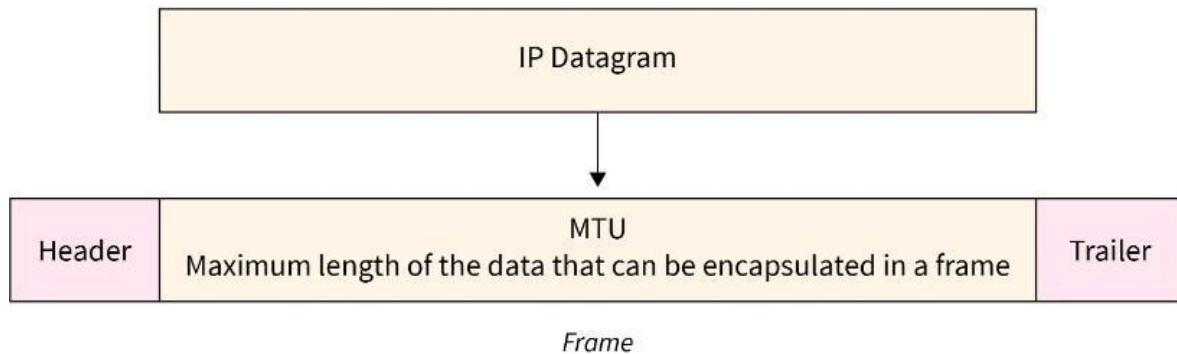
Fragmentation is a process that divides packets into smaller pieces (*fragments*) so that the resulting pieces can travel across a link with a smaller *maximum transmission unit (MTU)* than the original packet size. The network layer fragments data when the maximum size of a datagram exceeds the maximum size of data that can be retained in a frame, i.e., its *Maximum Transmission Unit (MTU)*. So that data flow is not impeded, the network layer separates the datagram received from the transport layer into fragments.



IP fragmentation is a process that divides packets into smaller pieces (*fragments*) so that the resulting pieces can travel across a link with a smaller **maximum transmission unit (MTU)** than the original packet size. The receiving host reassembles the fragments.

However, packets initially delivered across a network with one *MTU* may need to be routed over networks with a smaller *MTU* (*such as a WAN or VPN tunnel*). If the packet size in these circumstances is greater than the lower *MTU*, the data in the packet must be fragmented (*if possible*). This indicates that it is divided into fragments carried in brand-new packets (*fragments*) that are equal to or less than the lower *MTU*. This is known as fragmentation, and when the fragments arrive at their destination, the data is usually put back together.

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Some points related to the fragmentation:

- The maximum size of an IP datagram is $2^{16} - 1 = 65,535$ bytes, as the IP header has a total length of 16 bits.
- It is performed by the network layer at the destination side, typically at routers.
- Due to intelligent (*excellent*) segmentation by the transport layer, the source side does not require fragmentation. Specifically, the transport layer looks at the datagram and frame data limits and segments the data so that it can easily fit in a frame without the need for fragmentation.
- The receiver uses the identification (16 bits) field in the IP header to identify the packet. The identification number for each frame fragment is the same.
- The receiver uses the fragment offset (13 bits) field in the IP header to identify the series of frames.
- The extra header created by fragmentation results in overhead at the network layer.

Process of Fragmentation

- **RFC 791** specifies IP packet fragmentation, transmission, and reassembly mechanism.
- **RFC 815** specifies a streamlined reassembly algorithm. The Identification field in the IP header, along with the foreign and local internet addresses and the protocol ID, and the Fragment offset field in the IP header, coupled with the Don't Fragment and More Fragments flags, are used for fragmentation and reassembly of IP packets.
- If a receiving host receives a fragmented IP packet, it must put the packet back together and send it to the higher protocol layer. Reassembling is supposed to occur in the receiving host, but in reality, it might be carried out by an intermediate router. For instance, *network address translation (NAT)* can need to reassemble fragments to translate data streams.

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Step 4 – Some of local radio stations have helicopters flying around their cities to report on road congestion to make it possible for their mobile listeners to route their packets (cars) around hotspots. This is called traffic aware routing.

Step 5 – Sometimes it is not possible to increase capacity. The only way to reduce the congestion is to decrease the load. In a virtual circuit network, new connections can be refused if they would cause the network to become congested. This is called admission control.

Step 6 – Routers can monitor the average load, queueing delay, or packet loss. In all these cases, the rising number indicates growing congestion. The network is forced to discard packets that it cannot deliver. The general name for this is Load shedding. The better technique for choosing which packets to discard can help to prevent congestion collapse.

7. Illustrate in detail about UDP header format with neat sketch.(5M)

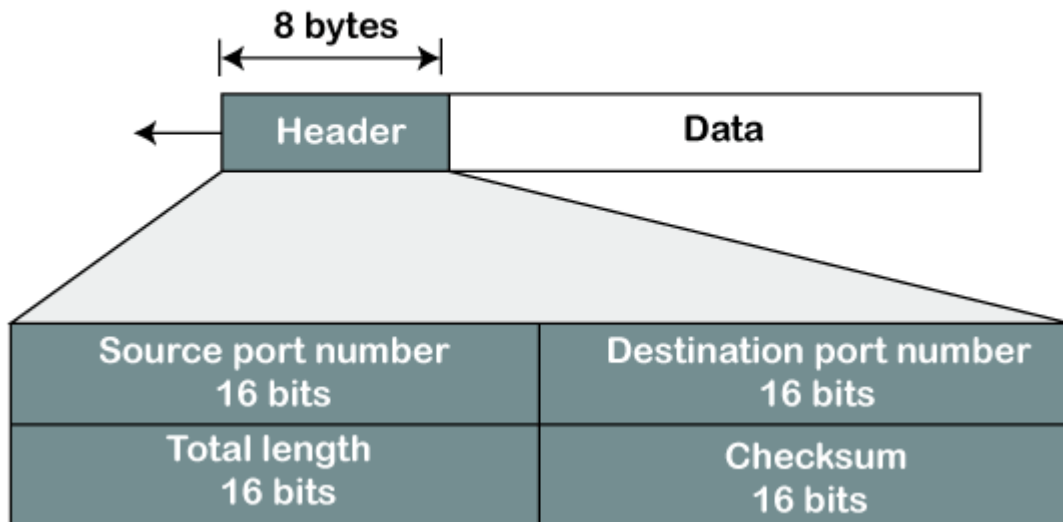
In computer networking, the UDP stands for User Datagram Protocol. The David P. Reed developed the UDP protocol in 1980. It is defined in RFC 768, and it is a part of the TCP/IP protocol, so it is a standard protocol over the internet. The UDP protocol allows the computer applications to send the messages in the form of datagrams from one machine to another machine over the Internet Protocol (IP) network. The UDP is an alternative communication protocol to the TCP protocol (transmission control protocol). Like TCP, UDP provides a set of rules that governs how the data should be exchanged over the internet. The UDP works by encapsulating the data into the packet and providing its own header information to the packet. Then, this UDP packet is encapsulated to the IP packet and sent off to its destination. Both the TCP and UDP protocols send the data over the internet protocol network, so it is also known as TCP/IP and UDP/IP. There are many differences between these two protocols. UDP enables the process to process communication, whereas the TCP provides host to host communication. Since UDP sends the messages in the form of datagrams, it is considered the best-effort mode of communication. TCP sends the individual packets, so it is a reliable transport medium. Another difference is that the TCP is a connection-oriented protocol whereas, the UDP is a connectionless protocol as it does not require any virtual circuit to transfer the data.

UDP also provides a different port number to distinguish different user requests and also provides the checksum capability to verify whether the complete data has arrived or not; the IP layer does not provide these two services.

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UDP Header Format

UDP Header Format



In UDP, the header size is 8 bytes, and the packet size is upto 65,535 bytes. But this packet size is not possible as the data needs to be encapsulated in the IP datagram, and an IP packet, the header size can be 20 bytes; therefore, the maximum of UDP would be 65,535 minus 20. The size of the data that the UDP packet can carry would be 65,535 minus 28 as 8 bytes for the header of the UDP packet and 20 bytes for IP header.

The UDP header contains four fields:

- o **Source port number:** It is 16-bit information that identifies which port is going to send the packet.
- o **Destination port number:** It identifies which port is going to accept the information. It is 16-bit information which is used to identify application-level service on the destination machine.
- o **Length:** It is 16-bit field that specifies the entire length of the UDP packet that includes the header also. The minimum value would be 8-byte as the size of the header is 8 bytes

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- o **Checksum:** It is a 16-bits field, and it is an optional field. This checksum field checks whether the information is accurate or not as there is the possibility that the information can be corrupted while transmission. It is an optional field, which means that it depends upon the application, whether it wants to write the checksum or not. If it does not want to write the checksum, then all the 16 bits are zero; otherwise, it writes the checksum. In UDP, the checksum field is applied to the entire packet, i.e., header as well as data part whereas, in IP, the checksum field is applied to only the header field.

8. With a diagram explain about TCP connection management. (5M)

In TCP, the connections are established using three way handshake technique.

TCP Connection Management includes TCP Connection Establishment & TCP Connection Release.

TCP Connection Establishment:

1. To establish a connection, one side, say, the server, passively waits for an incoming connection by executing the LISTEN and ACCEPT primitives, either specifying a specific source or nobody in particular.
2. The other side, say, the client, executes a CONNECT primitive, specifying the IP address and port to which it wants to connect, the maximum TCP segment size it is willing to accept, and optionally some user data (e.g., a password).
3. The CONNECT primitive sends a TCP segment with the SYN bit on and ACK bit off and waits for a response.
4. When this segment arrives at the destination, the TCP entity there checks to see if there is a process that has done a LISTEN on the port given in the Destination port field.
5. If not, it sends a reply with the (Reset) RST bit on to reject the connection.
6. If some process is listening to the port, that process is given the incoming TCP segment.
7. It can then either accept or reject the connection.
8. In the event that two hosts simultaneously attempt to establish a connection between the same two sockets, the sequence of events is as illustrated in Figure (b).



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9. The result of these events is that just one connection is established, not two be- cause connections are identified by their end points.

TCP Connection Release:

1. TCP connections are full duplex.
2. Each simplex connection is released independently of its sibling.
3. To release a connection, either party can send a TCP segment with the FIN bit set, which means that it has no more data to transmit.
4. When the FIN is acknowledged, that direction is shut down for new data.
5. Data may continue to flow indefinitely in the other direction, however.
6. When both directions have been shut down, the connection is released.
7. Normally, four TCP segments are needed to release a connection, one FIN and one ACK for each direction
8. However, it is possible for the first ACK and the second FIN to be contained in the same segment, reducing the total count to three.
9. To avoid the two-army problem, timers are used.
10. If a response to a FIN is not forthcoming within two maximum packet lifetimes, the sender of the FIN releases the connection.
11. The other side will eventually notice that nobody seems to be listening to it anymore and will time out as well.
12. While this solution is not perfect, given the fact that a perfect solution is theoretically impossible, it will have to do. In practice, problems rarely arise.
13. In each of the 11 states shown in above table, some specific events are legal.
14. Corresponding to every legal event some action may be taken, but if some other event happens, then error is reported.
15. Each Connection is always in the CLOSED State Initially.
16. It comes out of this state when it does either the passive open (LISTEN) or an active open (CONNECT).
17. A connection is established, if the other side does the opposite and the state be- comes ESTABLISHED.
18. When the both the sides initiate a connection release the connection is terminated and the state returns to CLOSED state.

Finite State Machine for Connection Establish & Release:

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State	Description
CLOSED	No Connection is active or pending.
LISTEN	The server is waiting for an incoming call.
SYN RCVD	A connection request has arrived; wait for ACK.
SYN SENT	The application has started to open a connection.
ESTABLISHED	The normal data transfer state.
FIN WAIT 1	The application has said it is finished.
FIN WAIT 2	The other side has agreed to release.
TIMED WAIT	Wait for all packets to die off.
CLOSING	Both sides have tried to close simultaneously.
CLOSE WAIT	The other side has initiated a release.
LAST ACK	Wait for ACK of FIN of last close.

19.

9. Formulate how would you differentiate TCP and UDP. (5M)

Differences between TCP and UDP

The following table highlights the major differences between TCP and UDP.

Key	TCP	UDP
Definition	It is a communications protocol, using which the data is transmitted between systems over the network. In this, the data is transmitted in the form of packets. It includes error-checking, guarantees the delivery and preserves the order of the data packets.	It is same as the TCP protocol except this doesn't guarantee the error-checking and data recovery. If you use this protocol, the data will be sent continuously, irrespective of the issues in the receiving end.
Design	TCP is a connection-oriented protocol.	UDP is a connectionless protocol.
Reliability	TCP is more reliable as it provides error checking support and also guarantees delivery of data to the destination router.	UDP, on the other hand, provides only basic error checking support using checksum. So, the delivery of data to the destination cannot

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		be guaranteed in UDP as in case of TCP.
Data transmission	In TCP, the data is transmitted in a particular sequence which means that packets arrive in-order at the receiver.	There is no sequencing of data in UDP in order to implement ordering it has to be managed by the application layer.
Performance	TCP is slower and less efficient in performance as compared to UDP. Also TCP is heavy-weight as compared to UDP.	UDP is faster and more efficient than TCP.
Retransmission	Retransmission of data packets is possible in TCP in case packet get lost or need to resend.	Retransmission of packets is not possible in UDP.
Sequencing	The Transmission Control Protocol has a function that allows data to be sequenced (TCP). This implies that packets arrive at the recipient in the sequence they were sent.	In UDP, there is no data sequencing. The application layer must control the order if it is necessary.
Header size	TCP uses a variable-length (20-60) bytes header.	UDP has a fixed-length header of 8 bytes.
Handshake	Handshakes such as SYN, ACK, and SYNACK are used.	It's a connectionless protocol, which means it doesn't require a handshake.
Broadcasting	Broadcasting is not supported by TCP.	Broadcasting is supported by UDP.
Examples	HTTP, HTTPS, FTP, SMTP, and Telnet use TCP.	DNS, DHCP, TFTP, SNMP, RIP, and VoIP use UDP.



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Conclusion

From the above discussion, we can conclude that TCP is the preferred protocol where reliability of data weighs more than transmission speed. UDP is connectionless and its header size is lightweight, which is why it is fast, but not as reliable as TCP.

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LECTURE NOTES

UNIT 5

Application Layer

- The layers below the application layer are there to provide transport services, but they do not do real work for users.

DNS:(Domain Name System)

- Way back in the ARPANET days, there was simply a file, hosts.txt, that listed all the computer names and their IP addresses.
- Every night, all the hosts would fetch it from the site at which it was maintained. For a network of a few hundred large timesharing machines, this approach worked reasonably well.
- However, well before many millions of PCs were connected to the Internet, everyone involved with it realized that this approach could not continue to work forever.
- For one thing, the size of the file would become too large. However, even more importantly, host name conflicts would occur constantly unless names were centrally managed, something unthinkable in a huge international network due to the load and latency.
- To solve these problems, DNS (Domain Name System) was invented in 1983.
- It has been a key part of the Internet ever since. The essence of DNS is the invention of a hierarchical, domain-based naming scheme and a distributed database system for implementing this naming scheme. It



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is primarily used for mapping host names to IP addresses but can also be used for other purposes.

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- DNS is defined in RFCs 1034, 1035, 2181, and further elaborated in many others.
 - Very briefly, the way DNS is used is as follows. To map a name onto an IP address, an application program calls a library procedure called the resolver, passing it the name as a parameter.
 - We saw an example of a resolver, `gethostbyname`, in Fig. 6-6. The resolver sends a query containing the name to a local DNS server, which looks up the name and returns a response containing the IP address to the resolver, which then returns it to the caller. The query and response messages are sent as UDP packets. Armed with the IP address, the program can then establish a TCP connection with the host or send it UDP packets.
 - For the Internet, the top of the naming hierarchy is managed by an organization called ICANN (Internet Corporation for Assigned Names and Numbers).
 - ICANN was created for this purpose in 1998, as part of the maturing of the Internet to a worldwide, economic concern.
 - Conceptually, the Internet is divided into SEC. 7.1 DNS—THE DOMAIN NAME SYSTEM 613 over 250 top-level domains, where each domain covers many hosts.
 - Each domain is partitioned into subdomains, and these are further partitioned, and so on.
 - All these domains can be represented by a tree, as shown in Fig. 7-1.
 - The leaves of the tree represent domains that have no subdomains (but do contain machines, of course).
 - A leaf domain may contain a single host, or it may represent a company and contain thousands of hosts.
 - eng cisco ieee acm eng
 - washington cs robot jack
 - jill ac co csl nec cs uwa keio edu oce vu cs law aero com gov org jp us edu mu- seum net au uk nl
 - Generic Countries
 - filts fluit
- Figure 7-1.
- A portion of the Internet domain name space. The top-level domains come in two flavors: generic and countries.
 - The generic domains, listed in Fig. 7-2, include original domains from the 1980s and domains introduced via applications to ICANN. Other generic top-level



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domains will be added in the future. The country domains include one entry for every country, as defined in ISO 3166.

- Internationalized country domain names that use non-Latin alphabets were introduced in 2010. These domains let people name hosts in Arabic, Cyrillic, Chinese, or other languages.
- Getting a second-level domain, such as name-of-company.com, is easy. The top-level domains are run by registrars appointed by ICANN.
- Getting a name merely requires going to a corresponding registrar (for com in this case) to check if the desired name is available and not somebody else's trademark.
- If there are no problems, the requester pays the registrar a small annual fee and gets the name.
- However, as the Internet has become more commercial and more international, it has also become more contentious, especially in matters related to naming. This controversy includes ICANN itself.
- Each domain is named by the path upward from it to the (unnamed) root. The components are separated by periods (pronounced "dot").
- Thus, the engineering department at Cisco might be eng.cisco.com., rather than a UNIX-style name such as /com/cisco/eng.
- Notice that this hierarchical naming means that eng.cisco.com. does not conflict with a potential use of eng in eng.washington.edu., which might be used by the English department at the University of Washington.
- Domain names can be either absolute or relative. An absolute domain name always ends with a period (e.g., eng.cisco.com.), whereas a relative one does not.
- Relative names have to be interpreted in some context to uniquely determine their true meaning. In both cases, a named domain refers to a specific node in the tree and all the nodes under it.
- Domain names are case-insensitive, so edu, Edu, and EDU mean the same thing. Component names can be up to 63 characters long, and full path names must not exceed 255 characters.



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- In principle, domains can be inserted into the tree in either generic or country domains.
- For example, cs.washington.edu could equally well be listed under the us coun- try domain as cs.washington.wa.us.
- Large companies often do so (e.g., sony.com, sony.net, and sony.nl). Each do- main controls how it allocates the domains under it.
- For example, Japan has domains ac.jp and co.jp that mirror edu and com. The Netherlands does not make this distinction and puts all organizations directly under nl.
- Thus, all three of the following are university computer science departments:
1. cs.washington.edu (University of Washington, in the U.S.). 2. cs.vu.nl (Vrije Universiteit, in The Netherlands). 3. cs.keio.ac.jp (Keio University, in Japan).
- To create a new domain, permission is required of the domain in which it will be included.
- For example, if a VLSI group is started at the University of Washington and wants to be known as vlsi.cs.washington.edu, it has to get permission from whoever manages cs.washington.edu.
- Similarly, if a new university is chartered, say, the University of Northern South Dakota, it must ask the manager of the edu domain to assign it unsd.edu (if that is still available).
- In this way, name conflicts are avoided and each domain can keep track of all its subdomains.
- Once a new domain has been created and registered, it can create subdomains, such as cs.unsd.edu, without getting permission from anybody higher up the tree.
- Thus, the primary function of DNS is to map domain names onto resource rec- ords. A resource record is a five-tuple.
- Although they are encoded in binary for efficiency, in most expositions resource records are presented as ASCII text, one line per resource record. The format we will use is as follows:

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- Domain name Time to live Class Type Value The Domain name tells the domain to which this record applies.
- Normally, many records exist for each domain and each copy of the database holds information about multiple domains. This field is thus the primary search key used to satisfy queries.
- The order of the records in the database is not significant. The Time to live field gives an indication of how stable the record is. Information that is highly stable is assigned a large value, such as 86400 (the number of seconds in 1 day).
- Information that is highly volatile is assigned a small value, such as 60 (1 minute).
- We will come back to this point later when we have discussed caching. The third field of every resource record is the Class. For Internet information, it is always IN.
- For non-Internet information, other codes can be used, but in practice these are rarely seen. The Type field tells what kind of record this is.
- There are many kinds of DNS records. The important types are listed in Fig. 7-3.
- An SOA record provides the name of the primary source of information about the name server's zone (described below), the email address of its administrator, a unique serial number, and various flags and timeouts.
- The most important record type is the A (Address) record. It holds a 32-bit IPv4 address of an interface for some host.
- The corresponding AAAA, or "quad A," record holds a 128-bit IPv6 address. Every Internet host must have at least one IP address so that other machines can communicate with it.
- Some hosts have two or more network interfaces, in which case they will have two or more type A or AAAA resource records.
- Consequently, DNS can return multiple addresses for a single name. A common record type is the MX record. It specifies the name of the host prepared to accept email for the specified domain.



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SMTP (Simple Mail Transfer Protocol) :

- Extensions Within the Internet, email is delivered by having the sending computer establish a TCP connection to port 25 of the receiving computer. Listening to this port is a mail server that speaks SMTP (Simple Mail Transfer Protocol).
- This server accepts incoming connections, subject to some security checks, and accepts messages for delivery.
- If a message cannot be delivered, an error report containing the first part of the undeliverable message is returned to the sender.
- SMTP is a simple ASCII protocol. This is not a weakness but a feature. Using ASCII text makes protocols easy to develop, test, and debug.
- They can be SEC. 7.2 ELECTRONIC MAIL 639 tested by sending commands manually, and records of the messages are easy to read.
- Most application-level Internet protocols now work this way (e.g., HTTP). We will walk through a simple message transfer between mail servers that delivers a message.
- After establishing the TCP connection to port 25, the sending machine, operating as the client, waits for the receiving machine, operating as the server, to talk first.
- The server starts by sending a line of text giving its identity and telling whether it is prepared to receive mail.
- If it is not, the client releases the connection and tries again later. If the server is willing to accept email, the client announces whom the email is coming from and whom it is going to.
- If such a recipient exists at the destination, the server gives the client the go-ahead to send the message.
- Then the client sends the message and the server acknowledges it. No checksums are needed because TCP provides a reliable byte stream.
- If there is more email, that is now sent. When all the email has been exchanged in both directions, the connection is released.

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- A sample dialog for sending the message of Fig. 7-14, including the numerical codes used by SMTP, is shown in Fig. 7-15.
- The lines sent by the client (i.e., the sender) are marked C:. Those sent by the server (i.e., the receiver) are marked S:.
- The first command from the client is indeed meant to be HELO. Of the various four-character abbreviations for HELLO, this one has numerous advantages over its biggest competitor.
- Why all the commands had to be four characters has been lost in the mists of time. In Fig. 7-15, the message is sent to only one recipient, so only one RCPT command is used.
- Such commands are allowed to send a single message to multiple receivers.
- Each one is individually acknowledged or rejected. Even if some recipients are rejected (because they do not exist at the destination), the message can be sent to the other ones.
- Finally, although the syntax of the four-character commands from the client is rigidly specified, the syntax of the replies is less rigid.
- Only the numerical code really counts. Each implementation can put whatever string it wants after the code.
- The basic SMTP works well, but it is limited in several respects. It does not include authentication.
- This means that the FROM command in the example could give any sender address that it pleases.
- This is quite useful for sending spam. Another limitation is that SMTP transfers ASCII messages, not binary data.
- This is why the base64 MIME content transfer encoding was needed. However, with that encoding the mail transmission uses bandwidth inefficiently, which is an issue for large messages.
- A third limitation is that SMTP sends messages in the clear. It has no encryption to provide a measure of privacy against prying eyes.
- To allow these and many other problems related to message processing to be addressed, SMTP was revised to have an extension mechanism.



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- This mechanism is a mandatory part of the RFC 5321 standard. The use of SMTP with extensions is called ESMTP (Extended SMTP).

World Wide Web :

World Wide Web, which is also known as a Web, is a collection of websites or web pages stored in web servers and connected to local computers through the internet. These websites contain text pages, digital images, audios, videos, etc. Users can access the content of these sites from any part of the world over the internet using their devices such as computers, laptops, cell phones, etc. The WWW, along with internet, enables the retrieval and display of text and media to your device.

The building blocks of the Web are web pages which are formatted in HTML and connected by links called "hypertext" or hyperlinks and accessed by HTTP. These links are electronic connections that link related pieces of information so that users can access the desired information quickly. Hypertext offers the advantage to select a word or phrase from text and thus to access other pages that provide additional information related to that word or phrase.

A web page is given an online address called a Uniform Resource Locator (URL). A particular collection of web pages that belong to a specific URL is called a website, e.g., *www.facebook.com*, *www.google.com*, etc. So, the World Wide Web is like a huge electronic book whose pages are stored on multiple servers across the world.

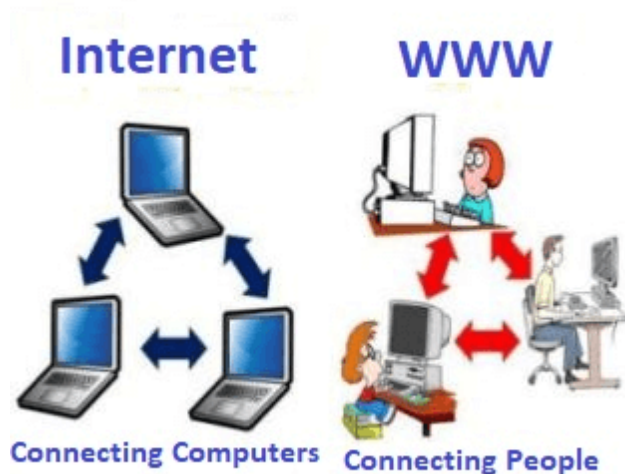
Small websites store all of their WebPages on a single server, but big websites or organizations place their WebPages on different servers in different countries so that when users of a country search their site they could get the information quickly from the nearest server.

Difference between World Wide Web and Internet:

Some people use the terms 'internet' and 'World Wide Web' interchangeably. They think they are the same thing, but it is not so. Internet is entirely different from WWW. It is a

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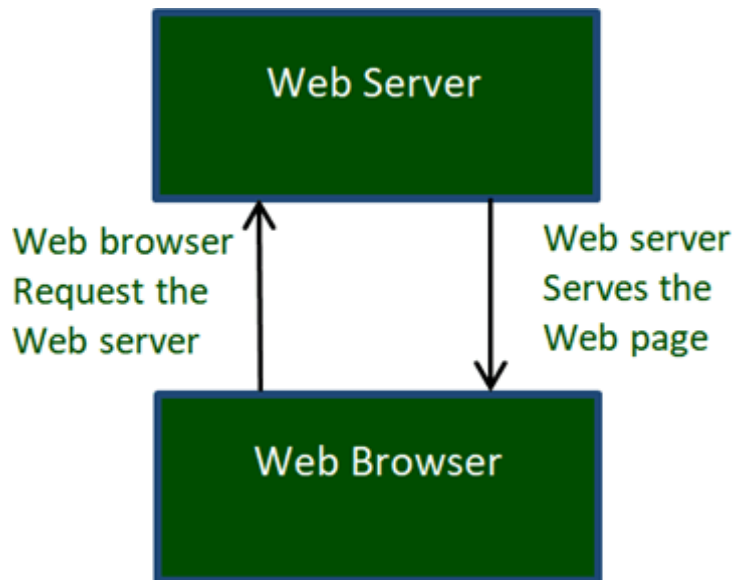
worldwide network of devices like computers, laptops, tablets, etc. It enables users to send emails to other users and chat with them online. For example, when you send an email or chatting with someone online, you are using the internet.



How the World Wide Web Works?

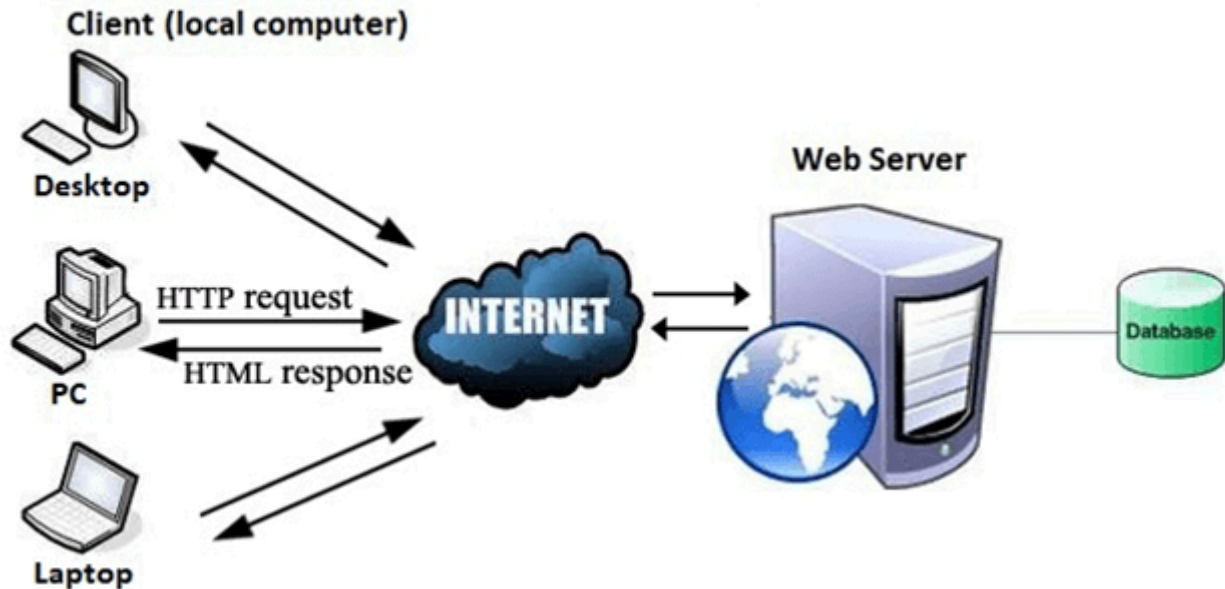
Now, we have understood that WWW is a collection of websites connected to the internet so that people can search and share information. Now, let us understand how it works!

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The Web works as per the internet's basic client-server format as shown in the following image. The servers store and transfer web pages or information to user's computers on the network when requested by the users. A web server is a software program which serves the web pages requested by web users using a browser. The computer of a user who requests documents from a server is known as a client. Browser, which is installed on the user' computer, allows users to view the retrieved documents.

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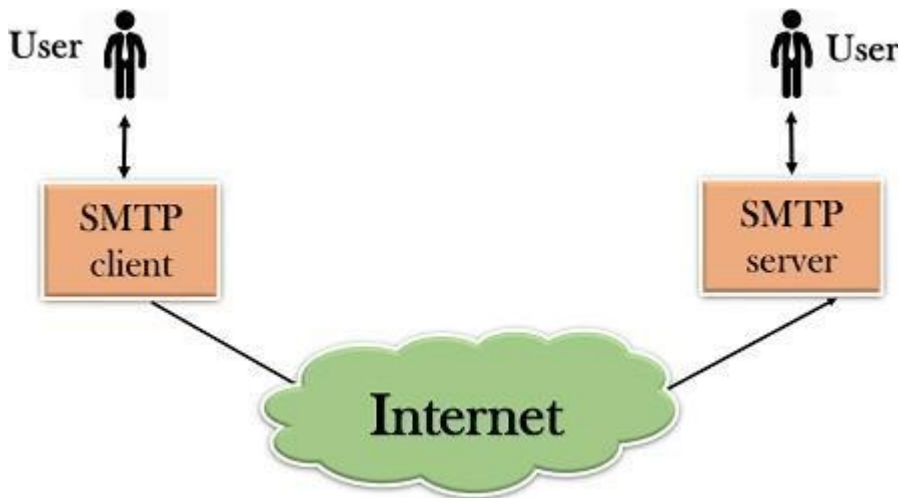
SMTP

- o SMTP stands for Simple Mail Transfer Protocol.
- o SMTP is a set of communication guidelines that allow software to transmit an electronic mail over the internet is called **Simple Mail Transfer Protocol**.
- o It is a program used for sending messages to other computer users based on e-mail addresses.
- o It provides a mail exchange between users on the same or different computers, and it also supports:
 - o It can send a single message to one or more recipients.
 - o Sending message can include text, voice, video or graphics.
 - o It can also send the messages on networks outside the internet.
- o The main purpose of SMTP is used to set up communication rules between servers. The servers have a way of identifying themselves and announcing what kind of

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communication they are trying to perform. They also have a way of handling the errors such as incorrect email address. For example, if the recipient address is wrong, then receiving server reply with an error message of some kind.

Components of SMTP



- o First, we will break the SMTP client and SMTP server into two components such as user agent (UA) and mail transfer agent (MTA). The user agent (UA) prepares the message, creates the envelope and then puts the message in the envelope. The mail transfer agent (MTA) transfers this mail across the internet.

Working of SMTP

1. **Composition of Mail:** A user sends an e-mail by composing an electronic mail message using a Mail User Agent (MUA). Mail User Agent is a program which is used to send and receive mail. The message contains two parts: body and header.

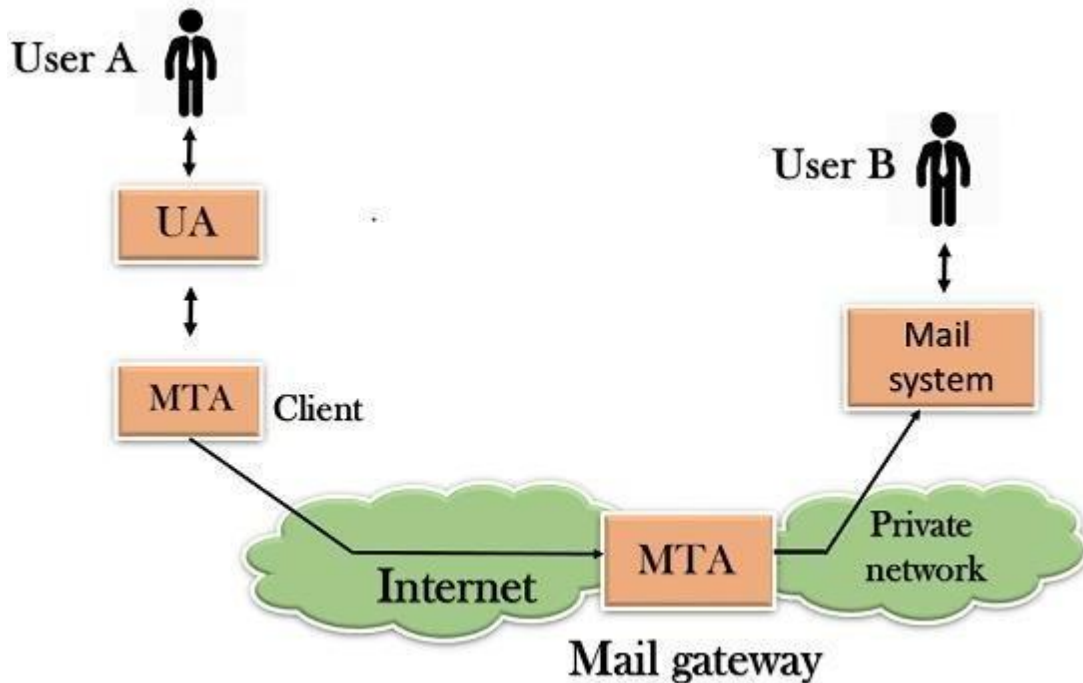


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The body is the main part of the message while the header includes information such as the sender and recipient address. The header also includes descriptive information such as the subject of the message. In this case, the message body is like a letter and header is like an envelope that contains the recipient's address.

2. **Submission of Mail:** After composing an email, the mail client then submits the completed e-mail to the SMTP server by using SMTP on TCP port 25.
3. **Delivery of Mail:** E-mail addresses contain two parts: username of the recipient and domain name. For example, vivek@gmail.com, where "vivek" is the username of the recipient and "gmail.com" is the domain name. If the domain name of the recipient's email address is different from the sender's domain name, then MSA will send the mail to the Mail Transfer Agent (MTA). To relay the email, the MTA will find the target domain. It checks the MX record from Domain Name System to obtain the target domain. The MX record contains the domain name and IP address of the recipient's domain. Once the record is located, MTA connects to the exchange server to relay the message.
4. **Receipt and Processing of Mail:** Once the incoming message is received, the exchange server delivers it to the incoming server (Mail Delivery Agent) which stores the e-mail where it waits for the user to retrieve it.
5. **Access and Retrieval of Mail:** The stored email in MDA can be retrieved by using MUA (Mail User Agent). MUA can be accessed by using login and password.

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Gate QB

Choose the best matching between Group 1 and

Group 2 Group –1

P. Data link layer

Q. Network layer

R. Transport
layer Group

– 2

1. Ensures reliable transport of data over a physical point-to-point

link 2. Encodes/decodes data for physical transmission

3. Allows end-to-end communication between two processes

4. Routes data from one network node to the next

(a) P – 1, Q – 4, R - 3

(b) P – 2, Q – 4, R - 1

(c) P – 2, Q – 3, R - 1

(d) P – 1, Q – 3, R - 2

Which one of the following is not a client server application?

(A) Internet chat

(B) Web browsing

(C) E-mail

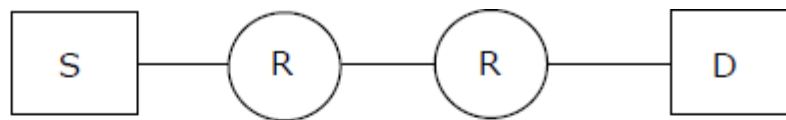
(D) Ping

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The protocol data unit (PDU) for the application layer in the Internet stack is

- (A) Segment
- (B) Datagram
- (C) Message
- (D) Frame

Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D.



- (A) Network layer – 4 times and Data link layer-4 times
- (B) Network layer – 4 times and Data link layer-3 times
- (C) Network layer – 4 times and Data link layer-6 times
- (D) Network layer – 2 times and Data link layer-6 times

In the following pairs of OSI protocol layer/sub-layer and its functionality, the INCORRECT pair is

- (A) Network layer and Routing.
- (B) Data Link Layer and Bit synchronization.
- (C) Transport layer and End-to-end process communication.
- (D) Medium Access Control sub-layer and Channel sharing.

Host A is sending data to host B over a full duplex link. A and B are using the sliding window protocol for flow control. The send and receive window sizes are 5 packets each. Data packets (sent only from A to B) are all 1000 bytes long and the transmission time for such a packet is 50

as



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Acknowledgement packets (sent only from B to A) are very small and require negligible transmission time. The propagation delay over the link is 200 μ s. What is the maximum achievable throughput in this communication?

- (A) 7.69×10^6 bps
- (B) 11.11×10^6 bps
- (C) 12.33×10^6 bps
- (D) 15.00×10^6 bps

In a packet switching network, packets are routed from source to destination along a single path having two intermediate nodes. If the message size is 24 bytes and each packet contains a header of 3 bytes, then the optimum packet size is:

- (A) 4
- (B) 6
- (C) 7
- (D) 9

Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?

- (A) 20
- (B) 40
- (C) 160
- (D) 320

Station A needs to send a message consisting of 9 packets to Station B using a sliding window (window size 3) and go-back-n error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no acks from B ever get lost),



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then what is the number of packets that A will transmit for sending the message to B?

- (A) 12
- (B) 14
- (C) 16
- (D) 18

The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The message that should be transmitted is:

- (A) 11001001000
- (B) 11001001011
- (C) 11001010
- (D) 110010010011

The distance between two stations M and N is L kilometers. All frames are K bits long. The propagation delay per kilometer is t seconds. Let R bits/second be the channel capacity. Assuming that processing delay is negligible, the minimum number of bits for the sequence number field in a frame for maximum utilization, when the sliding window protocol is used, is:

- (A) $\left\lceil \log_2 \frac{2LtR + 2K}{K} \right\rceil$
- (B) $\left\lceil \log_2 \frac{2LtR}{K} \right\rceil$
- (C) $\left\lceil \log_2 \frac{2LtR + K}{K} \right\rceil$
- (D) $\left\lceil \log_2 \frac{2LtR + K}{2K} \right\rceil$

Let G(x) be the generator polynomial used for CRC checking. What is the condition that should be satisfied by G(x) to detect odd number of bits in error?

- (A) G(x) contains more than two terms
- (B) G(x) does not divide $1+x^k$, for any k not exceeding the frame length



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(C) $1+x$ is a factor of $G(x)$

(D) $G(x)$ has an odd number of terms.

Frames of 1000 bits are sent over a 106 bps duplex link between two hosts. The propagation time is 25ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link). What is the minimum number of bits (l) that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.

(A) $l=2$

(B) $l=3$

(C) $l=4$

(D) $l=5$

Suppose that the sliding window protocol is used with the sender window size of 2^l , where l is the number of bits identified in the earlier part and acknowledgements are always piggy backed. After sending 2^l frames, what is the minimum time the sender will have to wait before starting transmission of the next frame? (Identify the closest choice ignoring the frame processing time.)

(A) 16ms

(B) 18ms

(C) 20ms

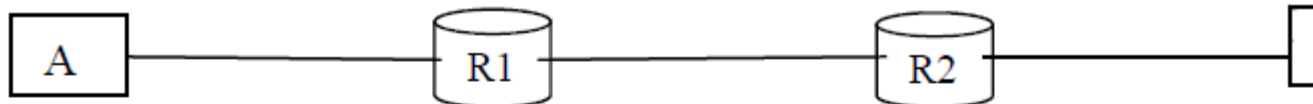
(D) 22ms

Consider the store and forward packet switched network given below. Assume that the bandwidth of each link is 10^6 bytes / sec. A user on host A sends a file of size 10^3 bytes to host B through routers R1 and R2 in three different ways. In the first case a single packet containing the complete file is transmitted from A to B. In the second case, the file is split into 10 equal parts, and these packets are transmitted from A to B. In the third case, the file is split into 20 equal parts and these packets are sent from A to B. Each packet contains 100 bytes of header information along with the user data.

Consider only transmission time and ignore processing, queuing and propagation delays. Also

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assume that there are no errors during transmission. Let T_1 , T_2 and T_3 be the times taken to transmit the file in the first, second and third case respectively. Which one of the following is CORRECT?



- (A) $T_1 < T_2 < T_3$
- (B) $T_1 > T_2 > T_3$
- (C) $T_2 = T_3, T_3 < T_1$
- (D) $T_1 = T_3, T_3 > T_2$

Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a

1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, the minimum number of bits required to represent the sequence number field is

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R13

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**B. Tech III Year I Semester
Examinations, March
- 2017 COMPUTER
NETWORKS**

(Common to CSE, IT)

Time: 3 hours

Max. Marks: 75

Note: This question paper contains two parts A and B.